

Temporal-State-Aware Compact Routing with Coherence-Budget Management in Quantum-Native Entanglement Overlay Networks

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Chapter 1

Introduction

1.1 Research Background

The Quantum Internet is envisioned as a future communication infrastructure that connects quantum processors, sensors, memories, and users through distributed quantum entanglement. Unlike the classical Internet, where information is transmitted as bits through relatively persistent communication links, quantum networks rely on the generation, storage, distribution, and manipulation of fragile quantum states.

Quantum entanglement is one of the central resources of quantum networking. It supports important applications such as quantum teleportation, distributed quantum computing, quantum sensing, and quantum key distribution. However, direct entanglement distribution over long distances is limited by optical loss, channel noise, imperfect operations, and the no-cloning theorem.

Quantum repeaters are introduced to overcome these limitations. A long communication path is divided into shorter elementary links, where entanglement is generated between neighboring nodes. Intermediate repeater nodes then perform entanglement swapping to establish end-to-end entanglement between remote users.

Although quantum repeaters improve communication distance, they introduce new routing and resource-management challenges. Entanglement generation is probabilistic, swapping may fail, purification consumes multiple entangled pairs, and quantum memories have limited coherence times. Therefore, a route that is topologically available may not necessarily be physically or temporally usable.

The quality of an entangled state is commonly measured using fidelity. A generated pair is useful only when its fidelity remains above an application-

specific threshold. However, fidelity decreases while entanglement is stored in quantum memory because of decoherence. This means that the usefulness of an entangled link depends not only on its existence, but also on its age, initial fidelity, coherence time, and expected delay before consumption.

Recent research has introduced age-aware concepts such as Fidelity-Age and Age of Entanglement. These studies demonstrate that temporal freshness affects delivery regularity, fairness, route reliability, and end-to-end quality. Age-aware scheduling and control policies can reduce starvation and improve the timeliness of usable entanglement deliveries.

In parallel, recent quantum-native network architectures have proposed artificial entanglement overlays, quantum addressing, Entanglement-Defined Controllers, compact routing tables, and bounded entangling stretch. In these architectures, selected nodes proactively maintain entanglement with remote nodes, creating an artificial topology above the physical network.

The Partial-Anchor and Full-Anchor compact-routing schemes provide scalable routing-table structures with sublinear memory requirements. For additive entangling-cost metrics, the schemes provide worst-case entangling-stretch bounds of five and three, respectively.

However, the temporal state of each stored artificial e-link is not explicitly represented in the compact routing table. Information such as generation time, current age, effective fidelity, remaining coherence budget, expiration risk, and regeneration urgency is not directly incorporated into route selection.

This creates a significant research problem. An artificial link may be structurally present but may become unusable before the required swapping sequence is completed. Similarly, a stale anchor link may reduce overlay reachability, increase regeneration delay, or invalidate the original entangling-stretch guarantee.

Therefore, scalable quantum-native routing requires more than static connectivity and cost information. It requires temporal-state awareness that can determine whether an artificial e-link will remain usable until route completion.

This thesis develops temporal compact routing for quantum-native entanglement overlays. It incorporates e-link age, effective fidelity, coherence budget, and regeneration cost into routing and maintenance decisions.

1.2 Research Motivation and Significance

The motivation for this research arises from the mismatch between the dynamic physical behavior of entanglement and the largely static assumptions

used in existing compact-routing formulations. Artificial e-links are created proactively, stored in finite-coherence quantum memories, consumed during entanglement swapping, and continuously regenerated. Therefore, their routing value changes over time.

A compact-routing protocol may find a valid overlay path whose e-links expire before route completion. This can occur when stored entanglement has already accumulated significant age, when the remaining coherence time is insufficient, or when regeneration delay is ignored during route selection.

This issue is particularly important in anchor-based routing architectures. In the Partial-Anchor Scheme, a small subset of anchor nodes maintains long-range artificial links. The failure or degradation of one such link may affect many source–destination pairs. In the Full-Anchor Scheme, more nodes maintain long-range links, which can improve route quality but also increases temporal-maintenance complexity.

Existing age-aware studies demonstrate that temporal freshness affects service reliability, fairness, and delivery quality. However, these studies mainly focus on scheduling, repeater chains, or local control. They do not analyze how temporal degradation affects compact-routing tables, anchor reliability, overlay reachability, or entangling stretch.

Conversely, existing compact-routing architectures provide strong scalability and stretch guarantees, but they do not explicitly model the time-dependent state of each stored artificial e-link. This separation motivates the integration of temporal entanglement management with scalable compact routing.

The significance of this research is both theoretical and practical. From a theoretical perspective, it extends static entangling-cost and stretch formulations into time-dependent routing models. It also investigates whether compact-routing guarantees remain meaningful when links age, decohere, and require regeneration.

From a practical perspective, temporal-state-aware routing may reduce the selection of stale or near-expiration links, improve end-to-end fidelity, reduce route failure, and prioritize regeneration resources more effectively. It may also help an Entanglement-Defined Controller decide whether to retain, regenerate, replace, or avoid a particular artificial link.

The proposed coherence-budget concept provides a direct estimate of whether an e-link is expected to remain usable until route completion. This allows routing decisions to account for both current quality and predicted future viability.

The research is also significant for large-scale quantum-network design because it aims to preserve the sublinear routing-table complexity of compact routing. A useful solution must improve temporal reliability without

requiring each node to maintain full global state.

Therefore, the central significance of this work lies in developing a routing framework that jointly addresses scalability, fidelity, temporal freshness, coherence limitations, and overlay maintenance.

1.3 Problem Statement

Existing quantum-native compact-routing architectures offer scalable tables, artificial overlays, and bounded entangling stretch. However, their decisions do not explicitly represent each stored e-link’s temporal condition.

This creates a fundamental inconsistency between routing abstraction and physical entanglement behavior. Artificial e-links are non-persistent resources. Their fidelity decreases during storage, they may expire before use, and they must be regenerated after consumption or degradation.

As a result, a route may be structurally valid but temporally infeasible. A selected path may contain one or more e-links that are too old, too close to decoherence, or unable to maintain the required fidelity until the expected completion of the swapping sequence.

The problem becomes more critical in anchor-based compact routing. In the Partial-Anchor Scheme, the degradation of one long-range anchor link may affect several source–destination paths. In the Full-Anchor Scheme, larger numbers of long-range links improve connectivity but increase maintenance and regeneration requirements.

Existing age-aware studies already model pair age, flow-level Fidelity-Age, end-to-end Age of Entanglement, discard, regeneration, and adaptive control. However, these approaches do not study quantum-native compact overlays, routing-table scalability, or entangling-stretch guarantees.

Therefore, the unresolved problem is not the absence of age-awareness or regeneration individually. The main problem is the absence of a compact-routing framework that explicitly incorporates per-e-link temporal state into route selection and overlay-maintenance decisions.

More specifically, existing work does not sufficiently answer the following questions:

- How should e-link age and effective fidelity be represented inside a compact routing table?
- How can the remaining coherence budget of an artificial e-link be estimated?
- How should a route be rejected when one or more e-links are expected to expire before completion?

- How should regeneration priorities be assigned across local, anchor, and tracked artificial links?
- How does temporal degradation affect the entangling-stretch guarantees of Partial-Anchor and Full-Anchor schemes?
- Can temporal-state awareness improve reliability without destroying sublinear routing-table scaling?

Accordingly, this thesis addresses the problem of designing a temporal-state-aware compact-routing framework that incorporates age, effective fidelity, coherence budget, and regeneration cost while preserving the scalability properties of quantum-native overlay routing.

1.4 Research Objectives

The primary objective of this research is to develop a temporal-state-aware compact-routing framework for quantum-native entanglement overlay networks.

The proposed framework aims to incorporate explicit temporal information into route selection and overlay-maintenance decisions while preserving the scalability advantages of compact routing.

The specific objectives are:

1. To define a temporal-state representation for each artificial e-link, including age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, and regeneration cost.
2. To develop a time-dependent extension of the static entangling-cost model.
3. To formulate a temporal route-feasibility condition that determines whether all e-links on a selected route are expected to remain usable until route completion.
4. To design coherence-budget-aware routing for Partial-Anchor and Full-Anchor overlays.
5. To develop an overlay-maintenance policy that determines when to retain, regenerate, replace, or avoid stale artificial e-links.
6. To define a temporal entangling-stretch metric for evaluating compact-routing performance under age-dependent fidelity degradation.

7. To analyze the impact of heterogeneous e-link ages, coherence times, fidelities, and regeneration delays on routing reliability.
8. To compare the method with static, fidelity-only, age-only, threshold-based, and regeneration-based routing.
9. To evaluate whether temporal-state awareness improves end-to-end fidelity, delivery success probability, route reliability, and overlay robustness without significantly increasing routing-table overhead.

The overall goal is to determine whether temporal-state information can be integrated into quantum-native compact routing in a way that improves physical reliability while preserving sublinear memory scaling.

1.5 Research Questions

This research is guided by the following questions:

1. How can explicit temporal state be incorporated into quantum-native compact routing-table entries without violating their sublinear scalability?
2. How should the age, effective fidelity, coherence time, and expected completion delay of an artificial e-link be combined into a remaining coherence-budget metric?
3. How can a routing protocol determine whether a structurally valid path is also temporally feasible?
4. How does temporal degradation affect the route quality and entangling-stretch guarantees of Partial-Anchor and Full-Anchor schemes?
5. How should regeneration be prioritized across local, anchor, and tracked links?
6. Can coherence-budget-aware routing reduce stale-link selection, route failure, and fidelity violation compared with static, fidelity-only, age-only, and fixed-threshold routing?
7. What trade-offs arise among end-to-end fidelity, delivery success probability, regeneration cost, routing-table overhead, and temporal entangling stretch?

1.6 Research Contributions

The expected contributions of this research are as follows:

1. A temporal-state extension of the quantum-native compact routing table that explicitly represents e-link age, effective fidelity, coherence time, remaining coherence budget, and regeneration cost.
2. A coherence-budget model that estimates whether an artificial e-link is expected to remain usable until the completion of an end-to-end entanglement route.
3. A time-dependent entangling-cost formulation that combines fidelity degradation, age, expiration risk, and regeneration delay.
4. A temporal route-feasibility criterion for rejecting structurally valid but temporally unreliable paths.
5. A coherence-budget-aware routing algorithm for Partial-Anchor and Full-Anchor entanglement overlays.
6. An overlay-maintenance policy for retaining, regenerating, replacing, or avoiding stale artificial e-links.
7. A temporal entangling-stretch formulation for compact routing under-heterogeneous age and decoherence conditions.
8. A comparative analysis of Partial-Anchor and Full-Anchor schemes under varying fidelity, coherence time, generation probability, and regeneration delay.
9. A simulation-based evaluation against static compact routing, fidelity-only routing, age-only routing, fixed-threshold routing, and basic regeneration heuristics.

These contributions are intended to connect temporal entanglement management with scalable quantum-native compact routing without claiming that age-awareness, fidelity-awareness, or regeneration are individually novel.

1.7 Research Methodology

This research follows a model-driven and simulation-based methodology to design and evaluate a temporal-state-aware routing framework for fidelity-preserving entanglement distribution in quantum networks. The methodology consists of the following interconnected stages.

First, the research problem is formulated by identifying the limitations of routing decisions based only on static topology or instantaneous link quality. Particular attention is given to the temporal degradation of stored entanglement, quantum-memory coherence limits, link fidelity, route-completion delay, and resource availability.

Second, a quantum-network system model is developed to represent physical quantum links, artificial entanglement links, quantum memories, entanglement generation, purification, swapping, and classical coordination. A time-dependent artificial entanglement overlay is then constructed to capture the currently available entanglement resources.

Third, temporal-state models are introduced for artificial entanglement links. These models describe link age, effective fidelity, predicted fidelity at consumption time, remaining coherence budget, staleness, and temporal usability. Based on these variables, explicit link-level and route-level feasibility conditions are formulated.

Fourth, a temporal-state-aware adaptive routing method is developed. Candidate routes are evaluated using predicted end-to-end fidelity, temporal feasibility, delay, resource requirements, and coherence-expiration risk. Stale or critical artificial links are excluded, repaired, or prioritized for regeneration according to their temporal condition and structural importance.

Fifth, the proposed method is implemented in a modular simulation environment. The implementation includes physical-link generation, temporal-state updates, route computation, artificial-link regeneration, topology generation, traffic generation, baseline algorithms, and performance measurement.

Sixth, controlled simulation experiments are conducted to compare the proposed method with selected baseline routing approaches. The evaluation considers request success ratio, end-to-end fidelity, routing delay, stale-link utilization, regeneration overhead, and computational scalability. Multi-seed experiments are used to reduce dependence on individual random realizations.

Finally, ablation studies, regeneration-budget experiments, and large-scale scalability tests are performed to examine the contribution of individual framework components and the behavior of the proposed method under different operating conditions. The resulting data are analyzed quantitatively to determine whether the proposed method satisfies the research objectives and answers the stated research questions.

The detailed system model is presented in Chapter 3, the proposed routing and regeneration method is described in Chapter 4, and the simulation methodology and experimental configuration are provided in Chapter 5.

1.8 Thesis Organization

This thesis is organized into six chapters.

Chapter 1 introduces the research background, motivation, problem statement, objectives, research questions, and expected contributions. It explains why temporal-state awareness is necessary for scalable quantum-native compact routing.

Chapter 2 reviews the relevant literature on quantum networking, entanglement distribution, routing, noise, decoherence, fidelity degradation, temporal entanglement metrics, age-aware control, quantum-native addressing, artificial entanglement overlays, and compact routing. It also identifies the gap between temporal entanglement management and scalable compact-routing architectures.

Chapter 3 presents the system model and problem formulation. It defines the physical and artificial network topologies, e-link temporal state, fidelity-decay model, coherence budget, route-feasibility conditions, time-dependent entangling cost, and temporal entangling stretch.

Chapter 4 introduces the proposed temporal-state-aware compact-routing and overlay-maintenance algorithms. It describes the routing-table extension, coherence-budget-aware path selection, stale-link handling, regeneration prioritization, and the adaptations required for Partial-Anchor and Full-Anchor schemes.

Chapter 5 describes the simulation environment, experimental design, baseline methods, parameter settings, and evaluation metrics. It presents and analyzes the results in terms of fidelity, delivery success probability, route failure, regeneration cost, memory overhead, and temporal entangling stretch.

Chapter 6 concludes the thesis by summarizing the main findings, discussing the limitations of the proposed framework, and outlining directions for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the theoretical and technical foundations required for temporal-state-aware compact routing in quantum networks. The discussion begins with the basic principles of quantum communication, including qubits, entanglement, quantum memories, entanglement swapping, purification, and probabilistic link generation. These concepts are essential because routing in a quantum network differs fundamentally from routing in a classical packet-switched network.

The review then examines existing approaches to entanglement distribution and quantum network routing. Particular attention is given to fidelity-aware routing, decoherence-aware control, adaptive scheduling, and resource-constrained repeater networks. These research areas establish that route quality depends not only on connectivity, but also on the physical quality and temporal condition of the entangled resources used along a route.

Recent studies have introduced explicit temporal metrics for quantum networks, including pair age, Fidelity-Age, and Age of Entanglement [1, 2]. These metrics capture the fact that stored entanglement degrades over time and that delayed delivery may reduce application-level usefulness. However, most age-aware studies focus on scheduling, repeater chains, or local control policies rather than scalable quantum-native compact routing.

In parallel, quantum-native routing architectures have introduced artificial entanglement overlays, quantum addressing, compact routing tables, and bounded entangling stretch. The proposed architecture in [3] demonstrates how Partial-Anchor and Full-Anchor schemes can achieve sublinear routing-table growth while preserving constant stretch. Nevertheless, the temporal state of individual artificial e-links is not explicitly represented in the routing

table or in the stretch analysis.

Therefore, this literature review focuses on the intersection of two previously separate research directions: temporal entanglement management and scalable compact routing. The objective is to identify whether age, effective fidelity, remaining coherence budget, and regeneration delay can be incorporated into a quantum-native compact routing framework without destroying its scalability properties.

2.2 Fundamentals of Quantum Networks

Quantum networks extend classical communication infrastructures by enabling the generation, storage, distribution, and manipulation of quantum states across spatially separated nodes. Their operation relies on physical principles that have no direct classical counterpart, including quantum superposition, entanglement, measurement-induced state collapse, and the no-cloning theorem. These principles fundamentally constrain how information can be transmitted and how network resources must be managed. [4, 5] A qubit is the basic unit of quantum information and can be represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (2.1)$$

where α and β are complex probability amplitudes satisfying

$$|\alpha|^2 + |\beta|^2 = 1. \quad (2.2)$$

Unlike a classical bit, which takes either the value 0 or 1, a qubit may exist in a coherent superposition of both basis states until measurement.

Quantum entanglement is a non-classical correlation shared between two or more quantum systems. A widely used example is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle). \quad (2.3)$$

The two qubits in this state cannot be described independently, even when they are stored at distant nodes. This property makes entanglement a fundamental communication resource for quantum teleportation, distributed quantum computing, quantum sensing, and quantum key distribution [4, 5]. Direct entanglement distribution over long distances is strongly limited by photon loss, channel noise, and imperfect hardware [6, 7] Quantum repeaters are therefore introduced to divide a long communication path into shorter elementary links. Entanglement is first generated between neighboring nodes and then extended over multiple links through entanglement swapping. If two

adjacent entangled pairs are available, an intermediate repeater can perform a Bell-state measurement to establish entanglement between the remote end nodes.

The quality of an entangled state is commonly measured using fidelity [6, 8]. For a quantum state described by density matrix ρ and a target pure state $|\psi_{\text{tar}}\rangle$, fidelity is defined as

$$F = \langle \psi_{\text{tar}} | \rho | \psi_{\text{tar}} \rangle. \quad (2.4)$$

A high fidelity value indicates that the generated state is close to the intended target state. In practical quantum networks, fidelity decreases due to channel noise, imperfect gates, measurement errors, and storage-induced decoherence.

Quantum memories temporarily store entangled qubits until other links or operations become available [9, 5]. However, stored states are not persistent resources. Their quality deteriorates over time because of decoherence, and an entangled pair may eventually become unusable if its fidelity falls below an application-specific threshold. Consequently, quantum-network control must account not only for whether an entangled pair exists, but also for its current physical and temporal condition.

Entanglement purification can improve fidelity by consuming multiple low-quality entangled pairs to probabilistically produce a smaller number of higher-quality pairs. This creates an important trade-off between fidelity improvement and resource consumption. Similarly, entanglement swapping extends communication distance but may reduce end-to-end fidelity and introduces additional probabilistic failure.

These characteristics make quantum networking fundamentally stateful. Routing and resource-allocation decisions must consider link-generation probability, memory occupancy, fidelity, decoherence, swapping success, purification cost, and the age of stored entanglement. This statefulness motivates the temporal-state-aware routing framework developed in this thesis.

2.3 Entanglement Distribution in Quantum Repeater Networks

Entanglement distribution is the process of establishing shared entangled states between spatially separated quantum nodes [6, 5]. In a practical quantum network, direct end-to-end entanglement generation becomes increasingly difficult as the communication distance grows because optical loss, im-

perfect photon transmission, detector inefficiency, and hardware noise reduce the probability of successful link establishment.

To overcome these limitations, a long communication path is divided into a sequence of shorter elementary links connected through quantum repeaters [6, 7]. Entanglement is first generated independently over each elementary link. After adjacent links have successfully established entangled pairs, intermediate repeater nodes perform entanglement swapping to extend the entanglement over a longer distance.

Let the quantum network be represented by an undirected graph [7, 10, 11, 12, 13].

$$G = (V, E), \quad (2.5)$$

where V denotes the set of quantum nodes and E denotes the set of physical quantum links. For an elementary link $e \in E$, the outcome of an entanglement-generation attempt in time slot t may be modeled as

$$X_t(e) \sim \text{Bernoulli}(p_e), \quad (2.6)$$

where p_e is the probability of successfully generating an entangled pair over link e .

When multiple transmission modes are available, the probability that at least one entangled pair is successfully generated over a link can be expressed as

$$p_{\text{link}}(e) = 1 - (1 - p_e)^{S(e)}, \quad (2.7)$$

where $S(e)$ denotes the number of parallel generation attempts or transmission modes allocated to the link.

Consider an end-to-end path

$$\pi = (e_1, e_2, \dots, e_k), \quad (2.8)$$

consisting of k elementary links. If each elementary link is available and each intermediate Bell-state measurement succeeds independently with probability q , the approximate end-to-end entanglement-success probability is

$$P_{\text{e2e}}(\pi) = \left(\prod_{j=1}^k p_{\text{link}}(e_j) \right) q^{k-1}. \quad (2.9)$$

This expression illustrates why longer paths are generally less reliable. As the number of links increases, more elementary-link generations and swapping operations must succeed simultaneously.

Entanglement swapping is essential for extending the communication distance. Suppose node B shares an entangled pair with node A and another entangled pair with node C . By performing a Bell-state measurement on its local qubits, node B can establish entanglement between A and C . The original elementary pairs are consumed during this operation, demonstrating that entanglement is a non-persistent network resource.

The quality of the resulting end-to-end entangled state depends on the fidelities of all elementary pairs and the errors introduced by swapping operations. For Werner-state approximations, a transformed parameter may be used to compose link fidelities along a path. If

$$p_{W,e} = \frac{4F_e - 1}{3}, \quad (2.10)$$

then the end-to-end Werner parameter for a path can be approximated by

$$p_{W,\text{end}} = \prod_{e \in \pi} p_{W,e}, \quad (2.11)$$

and the resulting end-to-end fidelity becomes

$$F_{\text{end}} = \frac{1 + 3p_{W,\text{end}}}{4}. \quad (2.12)$$

A delivered entangled state is considered usable only when

$$F_{\text{end}} \geq F_{\min}, \quad (2.13)$$

where $F_{\min}$ is an application-specific fidelity threshold.

Quantum memories allow successfully generated elementary pairs to be stored while the network waits for other links to become available. However, storage introduces a temporal dependency because the fidelity of stored entanglement decreases due to decoherence. Therefore, two links that are simultaneously available may have different ages and different effective fidelities.

Entanglement purification may be applied when the fidelity of available pairs is insufficient. Purification consumes two or more imperfect entangled pairs and probabilistically produces a smaller number of higher-fidelity pairs. Although purification improves quality, it also increases resource consumption, memory occupancy, and delivery delay.

Consequently, entanglement distribution involves several coupled decisions, including which links should attempt generation, which pairs should be stored or discarded, where purification should be performed, when swapping should occur, and which end-to-end path should be selected. These

decisions directly influence throughput, fidelity, latency, resource consumption, and entanglement freshness.

In quantum-native overlay architectures, proactively established entanglement creates artificial links between nodes that may not be physically adjacent. These artificial links can reduce distribution delay and routing complexity, but they must be continuously monitored, replenished, and re-configured because they are consumed during use and degraded by decoherence [3]. This motivates the need for routing methods that consider both the structural role and temporal condition of each entangled link.

2.4 Quantum Network Routing

Quantum network routing determines how entanglement should be established between remote source and destination nodes through a sequence of physical or artificial links. Unlike classical routing, the objective is not merely to forward packets through a connected topology. Instead, the routing process must identify and coordinate a sequence of entanglement-generation, storage, purification, and swapping operations capable of producing an end-to-end entangled state of sufficient quality [7, 10, 8]. [7, 10, 8] A routing path between source node s and destination node d may be represented as

$$\pi_{s,d} = (v_0 = s, v_1, \dots, v_k = d), \quad (2.14)$$

where consecutive nodes share physical or pre-established entanglement links.

In classical networks, routing cost is commonly based on hop count, delay, bandwidth, or packet-loss probability. In quantum networks, routing decisions must additionally consider quantum-specific resources [8, 14].

- elementary-link generation probability,
- quantum-memory availability,
- storage time and decoherence,
- link and end-to-end fidelity,
- entanglement-swapping reliability,
- purification requirements,
- resource consumption, and
- the age of stored entanglement.

A generic quantum-routing problem may be expressed as

$$\pi_{s,d}^* = \arg \min_{\pi \in \mathcal{P}_{s,d}} C(\pi), \quad (2.15)$$

where $\mathcal{P}_{s,d}$ is the set of feasible paths between s and d , and $C(\pi)$ is a quantum-specific route-cost function [7, 8].

Early quantum-routing approaches adapted classical shortest-path algorithms by assigning quantum-aware weights to physical links. A simple additive path cost can be written as

$$C(\pi) = \sum_{e \in \pi} w(e), \quad (2.16)$$

where $w(e)$ may reflect distance, generation delay, memory consumption, fidelity loss, or another link-level property. Although this formulation is computationally convenient, it may be insufficient when route quality depends on multiplicative success probabilities or non-additive fidelity composition.

For example, if link-generation and swapping events are independent, an end-to-end success-oriented route may be selected by maximizing

$$P_{\text{e2e}}(\pi) = \left(\prod_{e \in \pi} p_{\text{link}}(e) \right) \left(\prod_{r \in \pi} q_r \right), \quad (2.17)$$

where q_r denotes the swapping-success probability at repeater r . By applying a negative logarithm, this multiplicative objective can be transformed into an additive routing cost.

Fidelity-aware routing explicitly selects paths according to the quality of the expected end-to-end entangled state. A path is feasible only if

$$F_{\text{end}}(\pi) \geq F_{\text{min}}. \quad (2.18)$$

Among feasible paths, a routing protocol may maximize expected fidelity, minimize purification cost, maximize delivery rate, or balance multiple objectives. However, a route with high initial fidelity may become unsuitable if the required entangled pairs remain in memory for too long [14, 8, 15].

Reactive routing computes or updates a path after a demand arrives, whereas proactive routing creates and maintains entanglement resources before requests are received. Reactive approaches may reduce idle resource consumption but introduce setup delay. Proactive approaches can reduce service latency by maintaining pre-established entanglement, but they require continuous memory allocation, monitoring, and regeneration [16, 17].

Adaptive routing responds to changes in link state, memory availability, noise, fidelity, and request load. Recent approaches formulate the routing

process as a Markov decision process, partially observable Markov decision process, or reinforcement-learning problem. In these models, the controller selects actions based on the current or estimated network state rather than relying on a fixed link metric [11, 12, 13].

Routing becomes especially challenging when multiple source–destination requests compete for the same stored entangled pairs. Since each pair can be consumed by at most one swapping or purification operation, path selection must be coordinated with scheduling and resource allocation. Therefore, routing and scheduling cannot always be treated as independent layers [18, 19, 20].

Quantum-native routing introduces a further conceptual shift by operating over an artificial topology formed by pre-established entanglement. In this setting, an artificial link may connect physically remote nodes and can be consumed through entanglement swapping. The routing table therefore stores information about currently available entangled resources rather than only physical next hops [3, 10].

The quantum-native compact-routing architecture in [3] defines a general entangling-cost metric $w(i, j)$ and uses it to select e-neighbors, anchors, and artificial entanglement paths. The framework supports a wide class of possible metrics, including fidelity, memory usage, and decoherence-related cost. Nevertheless, the proposed routing-table structure does not explicitly track the temporal state of each stored e-link.

This limitation is important because the same artificial link may have different routing value at different times. A recently generated link with high effective fidelity and sufficient remaining coherence may be preferable to an older link with lower generation cost. Consequently, routing decisions should consider not only static structural connectivity but also time-dependent entanglement usability. Temporal-state-aware routing therefore requires joint consideration of fidelity evolution, entanglement age, memory coherence, and future delivery reliability [1, 2, 21, 22].

2.5 Noise, Decoherence, and Fidelity Degradation

Noise and decoherence are fundamental limitations in practical quantum networks. Unlike classical information, quantum states are highly sensitive to environmental interaction, imperfect control operations, photon loss, detector errors, and finite memory coherence. These effects reduce the quality and reliability of distributed entanglement and therefore directly influence

routing and resource-allocation decisions [6, 5].

Quantum noise can arise from several physical mechanisms. Common models include depolarizing noise, dephasing, amplitude damping, gate errors, and measurement errors [23, 9]. Although these models differ physically, their common consequence is a reduction in the fidelity of the stored or transmitted quantum state.

For an ideal target state $|\psi_{\text{tar}}\rangle$ and an actual density matrix ρ , fidelity is defined as

$$F = \langle \psi_{\text{tar}} | \rho | \psi_{\text{tar}} \rangle. \quad (2.19)$$

The value of F lies between zero and one, with larger values indicating that the actual state is closer to the target state. In entanglement-distribution systems, fidelity is commonly used as the main quality metric for determining whether an entangled pair is suitable for communication tasks [7, 8].

A practical application usually imposes a minimum acceptable fidelity threshold:

$$F \geq F_{\min}. \quad (2.20)$$

An entangled pair whose fidelity falls below $F_{\min}$ is treated as unusable, even if the pair still physically exists in quantum memory.

One important source of degradation is memory decoherence. When an entangled pair is stored, its quality decreases as a function of storage duration. Quantum memories therefore introduce a temporal dependency into entanglement management because stored resources gradually lose their usefulness over time [9, 1, 24, 22]. A simplified exponential decay model may be written as

$$F_{ij}(t) = F_{ij}^0 \exp\left(-\frac{A_{ij}(t)}{T_{2,ij}}\right), \quad (2.21)$$

where F_{ij}^0 is the initial fidelity of the pair shared by nodes i and j , $A_{ij}(t)$ is its age at time t , and $T_{2,ij}$ is the corresponding coherence time.

This model captures the basic principle that older entangled pairs generally have lower effective fidelity. More detailed physical models may include dephasing, amplitude damping, correlated noise, or hardware-specific memory behavior.

For Werner-state approximations, the fidelity may instead be represented through the Werner parameter

$$p_W = \frac{4F - 1}{3}. \quad (2.22)$$

Under storage-induced decoherence, this parameter may evolve according to

$$p_W(t + \Delta) = p_W(t) \exp\left(-\frac{\Delta}{T_2}\right). \quad (2.23)$$

The resulting fidelity can then be recovered using

$$F(t) = \frac{1 + 3p_W(t)}{4}. \quad (2.24)$$

Noise also accumulates during entanglement swapping. If multiple elementary pairs are combined along a path, the final end-to-end state depends on the fidelity of every stored pair and the reliability of each Bell-state measurement. Therefore, a path that is feasible at the beginning of a time slot may become infeasible after waiting, additional generation delay, or repeated swapping attempts [6, 7].

Purification can partially compensate for fidelity degradation. However, purification consumes multiple entangled pairs and succeeds only probabilistically. As a result, it introduces a trade-off between fidelity improvement, memory usage, throughput, and delay [6].

The temporal effect of decoherence is particularly important for proactive entanglement overlays. Artificial e-links may be generated in advance and stored until they are needed. During this waiting period, their effective fidelity decreases. Consequently, the routing value of an e-link is not static but changes with its temporal condition [3, 1, 21, 19].

A more realistic time-dependent link state can therefore be represented as

$$\mathcal{L}_{ij}(t) = [F_{ij}^0, A_{ij}(t), F_{ij}^{\text{eff}}(t), T_{2,ij}]. \quad (2.25)$$

The effective fidelity may be modeled as

$$F_{ij}^{\text{eff}}(t) = f(F_{ij}^0, A_{ij}(t), T_{2,ij}, \eta_{ij}, \epsilon_{ij}), \quad (2.26)$$

where η_{ij} and ϵ_{ij} represent channel and operational noise parameters.

Existing quantum-native compact-routing work allows the entangling-cost metric to represent fidelity degradation, memory use, or decoherence abstractly [3]. However, it does not provide an explicit temporal-state model or derive how age-dependent fidelity affects route feasibility and entangling stretch.

This gap motivates the development of a routing framework in which each artificial e-link is evaluated according to its current effective fidelity, age, and remaining coherence budget rather than by static connectivity alone [1, 2, 22, 21].

2.6 Temporal State of Entanglement

The temporal state of entanglement describes how the usefulness of a stored entangled resource changes over time. Unlike classical links, an entangled link is not a persistent resource. Its fidelity may decrease during storage, it may expire after a finite coherence interval, and it is consumed when used in swapping, purification, or application-level delivery.

For an entangled pair generated between nodes i and j at time t_{ij}^{gen} , its age at time t can be defined as

$$A_{ij}(t) = t - t_{ij}^{\text{gen}}. \quad (2.27)$$

A newly generated pair has a small age, whereas a pair that has remained in quantum memory for several slots has a larger age. Since decoherence accumulates during storage, age provides an indirect indicator of the physical quality of the pair [1, 2].

However, age and fidelity are not identical. Two pairs with the same age may have different fidelities because of differences in initial fidelity, coherence time, channel noise, or hardware quality. Therefore, a temporal-state model should combine age with physical parameters rather than using age alone.

A basic temporal link state may be expressed as

$$\mathcal{L}_{ij}(t) = [A_{ij}(t), F_{ij}^0, F_{ij}^{\text{eff}}(t), T_{2,ij}], \quad (2.28)$$

where F_{ij}^0 is the initial fidelity, $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity, and $T_{2,ij}$ is the memory coherence time.

A stored pair remains usable only while its effective fidelity satisfies

$$F_{ij}^{\text{eff}}(t) \geq F_{\min}. \quad (2.29)$$

This condition induces a finite usability interval. Once the pair is expected to fall below the required threshold, it should be discarded, purified, regenerated, or excluded from candidate routes [7, 8].

A useful quantity for routing is the remaining coherence budget. It may be approximated as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.30)$$

where $\widehat{D}_{ij}$ denotes the expected additional delay required before the entangled link is consumed in an end-to-end operation.

The value $B_{ij}(t)$ estimates whether the stored resource is likely to remain usable until route completion. A negative or very small coherence budget indicates a high risk of expiration before the required swapping or delivery operations are completed. Such temporal constraints motivate routing strategies that jointly consider resource age, fidelity, and future operation delay [1, 2, 19].

Temporal freshness has also been studied at the flow level. Fidelity-Age measures the time elapsed since the most recent delivery of an entangled pair whose fidelity exceeds a prescribed threshold. For a source–destination request (s, d) , it can be represented as

$$A_{s,d}(t) = t - \max \left\{ t_n^{(s,d)} \leq t \right\}, \quad (2.31)$$

where $t_n^{(s,d)}$ denotes a successful usable-delivery epoch.

This flow-level metric differs from pair age. Pair age describes the current storage duration of an individual entangled resource, whereas Fidelity-Age describes how long a flow has waited since its last usable delivery [1].

Another related concept is the Age of Entanglement of an end-to-end link. In repeater-chain models, the age may reset after successful swapping according to the accumulated ages of the elementary links used to create the end-to-end state. This captures the fact that a newly established end-to-end pair may already contain storage-induced degradation inherited from its component links [2]. These temporal quantities can therefore be classified into three levels:

- **pair age**, describing the age of an individual stored Bell pair;
- **flow-level age**, describing the time since the last usable delivery for a request; and
- **end-to-end entanglement age**, describing the freshness of the most recently generated remote entangled state.

These different temporal metrics provide complementary views of entanglement freshness and have recently been considered for adaptive scheduling and resource management in quantum networks [1, 2, 19, 21].

For compact routing over artificial entanglement overlays, pair-level or e-link-level temporal state is particularly important. The routing protocol must determine whether each stored artificial link will remain usable throughout the expected route-completion interval.

A temporally feasible route π may therefore be required to satisfy

$$B_{ij}(t) > 0, \quad \forall (i, j) \in \pi. \quad (2.32)$$

Alternatively, feasibility may be expressed as

$$F_{ij}^{\text{eff}}(t + \widehat{D}_\pi) \geq F_{\min}, \quad \forall (i, j) \in \pi, \quad (2.33)$$

where $\widehat{D}_\pi$ is the predicted completion delay of the selected route.

This perspective changes routing from a static connectivity problem into a time-dependent resource-selection problem. A route may be topologically available but temporally infeasible because one or more e-links are too old or too close to expiration. Therefore, future route quality depends not only on current link availability but also on the expected evolution of quantum resources during execution [7, 8].

The quantum-native compact-routing architecture in [3] recognizes the stateful and ephemeral nature of entanglement and discusses continuous overlay regeneration. Nevertheless, its routing-table entries do not explicitly include age, current fidelity, generation time, or remaining coherence budget.

This thesis extends compact-routing entries with temporal state and examines how degraded artificial links affect feasibility, fidelity, and entangling stretch [1, 2].

2.7 Age-Aware Scheduling and Control

Age-aware scheduling has recently emerged as an important research direction in quantum networking because entangled resources are both probabilistic and time-sensitive. A scheduler must decide which flow, link, or swapping operation should receive limited network resources while accounting for fidelity degradation, memory constraints, and service regularity [1, 2].

One recent approach introduces the Fidelity-Age metric, which measures the time elapsed since the most recent successful delivery of an entangled pair whose fidelity satisfies a prescribed threshold [1]. For a source–destination request (s, d) , the Fidelity-Age process can be written as

$$A_{s,d}(t) = t - \max \left\{ t_n^{(s,d)} \leq t \right\}, \quad (2.34)$$

where $t_n^{(s,d)}$ denotes the time of the most recent usable entanglement delivery.

The age increases when no usable delivery occurs and resets when a newly delivered end-to-end pair satisfies

$$F_{\text{end}} \geq F_{\text{min}}. \quad (2.35)$$

This formulation captures both delivery regularity and fidelity-based usability. A flow may have high throughput on average but still experience long periods without a usable entangled pair. Therefore, mean throughput alone is insufficient for evaluating temporal service quality in quantum networks [1].

The corresponding long-run average Fidelity-Age can be expressed as

$$\bar{A}_{s,d} = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T A_{s,d}(t). \quad (2.36)$$

The system objective is then to minimize a weighted sum of average ages:

$$\min_{\phi \in \Phi} \sum_{(s,d) \in \mathcal{P}} w_{s,d} \bar{A}_{s,d}, \quad (2.37)$$

subject to attempt-budget, memory, link-exclusivity, swapping, purification, and fidelity constraints.

The Fidelity-Age study proposes lightweight scheduling methods, including threshold-based and index-based policies. A threshold policy prioritizes flows whose age exceeds a predefined value, whereas an index policy ranks flows using a combination of instantaneous utility and current age [1].

These scheduling concepts are closely related to adaptive routing because routing decisions should consider not only the current availability of entangled resources but also their expected future quality and freshness. Therefore, age-aware scheduling provides an important foundation for routing frameworks that explicitly model temporal link states [8, 1, 21, 19]. A generic age-aware scheduling score may be represented as

$$I_{s,d}(t) = \frac{U_{s,d}(t)}{1 + \beta A_{s,d}(t)}, \quad (2.38)$$

where $U_{s,d}(t)$ represents an instantaneous utility, such as expected success probability or fidelity, and β controls the influence of age.

The simulations reported in [1] show that age-aware scheduling can substantially reduce extreme-age events, starvation, and unfairness while preserving throughput. However, the study focuses primarily on flow scheduling over a repeater grid rather than on compact routing over a quantum-native artificial overlay.

A related approach introduces the Age of Entanglement for satellite-assisted repeater chains with intermittent availability [2]. In this model,

the network state includes satellite visibility, elementary-link ages, and the end-to-end entanglement age:

$$s_t = (v_{12}^t, v_{23}^t, m_{12}^t, m_{23}^t, \Delta_t^e). \quad (2.39)$$

The action space includes elementary-link generation, waiting, and entanglement swapping:

$$a_t = (a_{12}^t, a_{23}^t, a_{\text{sw}}^t). \quad (2.40)$$

A successful swapping operation resets the end-to-end Age of Entanglement according to the ages of the two elementary links:

$$\Delta_{t+1}^e = m_{12}^t + m_{23}^t. \quad (2.41)$$

If swapping does not succeed, the age increases:

$$\Delta_{t+1}^e = \Delta_t^e + 1. \quad (2.42)$$

The problem is formulated as an average-cost Markov decision process, where the controller jointly decides whether to generate, wait, discard, regenerate, or swap entanglement. Relative Value Iteration is then used to obtain an optimal stationary policy [2].

This work is important because it demonstrates that generation, storage, discard, regeneration, and swapping decisions can be optimized jointly under temporal degradation. Therefore, regeneration or age-aware control alone cannot be considered a sufficient research contribution.

Nevertheless, the satellite-based Age of Entanglement model is limited to a three-node linear repeater chain. It does not consider general graph routing, multiple competing routes, quantum addressing, compact routing tables, artificial ESP overlays, Partial-Anchor or Full-Anchor structures, or entangling-stretch guarantees.

The distinction between scheduling and routing is therefore essential. Age-aware scheduling determines which request or operation should receive service, while compact routing determines which sequence of artificial e-links should be selected to establish end-to-end entanglement.

Existing age-aware studies provide strong temporal-control mechanisms, but they do not examine how stale e-links affect the structural guarantees of a compact routing architecture. In particular, they do not analyze whether aging may invalidate an anchor link, increase effective entangling stretch, reduce overlay reachability, or require proactive replacement of specific artificial links.

This motivates a joint perspective in which temporal state influences both route selection and overlay maintenance. The proposed thesis therefore extends existing age-aware control by embedding temporal e-link information into a scalable quantum-native compact-routing framework.

2.8 Quantum-Native Addressing and Entanglement Overlay Networks

Quantum-native network architecture departs from the classical assumption that network nodes only forward information over persistent physical links. In the Quantum Internet, entanglement itself acts as the primary communication resource, and network nodes must generate, store, monitor, manipulate, and replenish entangled states [3, 5].

The architecture proposed in [3] introduces a two-tier hierarchy. The upper tier is composed of Entanglement Service Providers, referred to as ESPs, while the lower tier contains quantum edge nodes that consume entanglement resources for applications such as teleportation, distributed quantum computing, sensing, and quantum key distribution.

The ESPs form an entangled backbone by proactively generating and maintaining entanglement with selected remote ESPs. This creates an artificial topology over the underlying physical network.

Let the physical quantum network be represented as

$$G = (V, E), \quad (2.43)$$

where V is the set of network nodes and E is the set of physical quantum links.

The artificial entanglement overlay may be represented as

$$G_E(t) = (V_E, E_E(t)), \quad (2.44)$$

where V_E denotes the set of ESPs and $E_E(t)$ denotes the set of artificial entanglement links available at time t .

Unlike the physical topology, the artificial topology is dynamic. An artificial e-link may be created through elementary-link generation and entanglement swapping, consumed by an application, degraded by decoherence, or removed after failure. Therefore, the state of an artificial overlay depends not only on connectivity but also on the temporal condition of the underlying entanglement resources [3, 1].

The architecture includes an Entanglement-Defined Controller, or EDC, which performs three main functions:

- monitoring the availability and fidelity of entangled resources;
- reconfiguring the artificial topology; and
- enforcing routing, recovery, and resource-allocation policies.

The EDC is conceptually similar to a software-defined networking controller, but its role is broader because it must manage non-persistent and stateful quantum resources.

The proposed architecture also separates the control plane from the data plane. The control plane coordinates routing and entanglement-resource management, whereas the data plane performs entanglement generation, purification, swapping, forwarding, and overlay repair.

A key feature of the architecture is continuous overlay maintenance. Artificial links are consumed during end-to-end entanglement establishment and must therefore be regenerated. This means that overlay refresh, link regeneration, and resource replenishment are already recognized as necessary architectural functions [3].

Thus, overlay maintenance or regeneration alone is not a novel contribution. The research opportunity lies in determining how maintenance decisions should be made under explicit temporal-state information, including link age, effective fidelity, and remaining coherence time.

The same architecture introduces a quantum addressing scheme in which each node is assigned a quantum address represented by a computational-basis state. For a network containing n nodes, the address length is

$$N = \lceil \log_2 n \rceil. \quad (2.45)$$

A quantum address can be represented as

$$|v_i\rangle \in \{|x\rangle \mid x \in \{0, 1\}^N\}. \quad (2.46)$$

Unlike a classical address, a quantum address may also appear in a superposition representing a set of nodes. For example,

$$|A\rangle = \frac{1}{\sqrt{M}} \sum_{i=1}^M |v_i\rangle \quad (2.47)$$

may represent a set of M quantum nodes.

This allows routing information about multiple ESPs to be encoded compactly inside a quantum routing table [3].

The quantum routing table does not store only physical next hops. Instead, it contains information about locally stored e-bits, the quantum identities of the remote ESPs sharing those e-bits, and superposed addresses representing e-neighborhood information.

A conceptual routing-table entry can be written as

$$\mathcal{R}_{ij} = [\text{e-bits}, |v_j\rangle, |\mathcal{N}(v_j)\rangle, \text{anchor status}]. \quad (2.48)$$

This structure enables routing decisions over the artificial entanglement topology rather than only over the physical graph [3].

However, the routing-table design in [3] does not explicitly include

- entanglement age,
- generation time,
- current effective fidelity,
- remaining coherence time,
- expiration probability, or
- regeneration urgency.

This omission is significant because two e-links with the same structural role may have very different temporal usability due to different storage durations, coherence properties, and degradation rates [1, 2].

A temporally extended routing-table entry may therefore be represented as

$$\mathcal{R}_{ij}(t) = [\text{e-bits}, |v_j\rangle, |\mathcal{N}(v_j)\rangle, A_{ij}(t), F_{ij}^{\text{eff}}(t), B_{ij}(t)], \quad (2.49)$$

where $A_{ij}(t)$ is the e-link age, $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity, and $B_{ij}(t)$ is the remaining coherence budget.

The integration of this temporal information creates a direct connection between quantum-native addressing, compact routing, and age-dependent entanglement management. This connection forms the central research direction of the present thesis.

2.9 Compact Quantum Routing

Compact routing aims to reduce routing-table size while preserving bounded route quality. In a large-scale quantum network, maintaining entanglement between every pair of ESPs would require excessive quantum memory, control signaling, and continuous regeneration. Therefore, a scalable routing protocol must operate over a sparse artificial entanglement overlay [3].

The quantum-native routing architecture in [3] introduces compact routing based on two design objectives:

- sublinear routing-table size; and
- constant entangling stretch.

For a network with n_e Entanglement Service Providers, each ESP maintains approximately

$$\tilde{O}(\sqrt{n_e}) \quad (2.50)$$

routing-table entries, where the soft- O notation suppresses logarithmic factors.

The routing framework defines a general link-entanglement cost

$$w(i, j), \quad (2.51)$$

which represents the cost of establishing entanglement between ESPs v_i and v_j [3].

The cost metric is assumed to support an order relation and a composition operator $\oplus$. For a multi-hop artificial route

$$R = (v_i, v_{r_1}, \dots, v_{r_k}, v_j), \quad (2.52)$$

the accumulated entangling cost is represented as

$$w_R(i, j) = w(i, r_1) \oplus w(r_1, r_2) \oplus \dots \oplus w(r_k, j). \quad (2.53)$$

The optimal entangling cost between two ESPs is

$$w^*(i, j) = \min_{R \in \mathcal{R}_{i,j}} w_R(i, j), \quad (2.54)$$

where $\mathcal{R}_{i,j}$ is the set of feasible entanglement routes between v_i and v_j . The entangling stretch of a routing protocol is defined as

$$ES(i, j) = \frac{w_{R_{QR}}(i, j)}{w^*(i, j)}, \quad (2.55)$$

where R_{QR} is the route selected by the quantum-routing protocol.
The worst-case entangling stretch is

$$ES = \max_{v_i, v_j \in V_2} ES(i, j). \quad (2.56)$$

A routing protocol is optimal when

$$ES = 1. \quad (2.57)$$

A value greater than one indicates that the protocol may choose a route with higher entangling cost than the globally optimal route.

The compact-routing architecture proposes two variants: the Partial-Anchor Scheme and the Full-Anchor Scheme [3].

2.9.1 Partial-Anchor Scheme

In the Partial-Anchor Scheme, each ESP proactively maintains low-cost artificial links with the nodes in its e-neighborhood. A smaller subset of ESPs, called anchor nodes, maintains additional long-range artificial links [3].

The e-neighborhood of ESP v_i is denoted by

$$\mathcal{N}(v_i), \quad (2.58)$$

and contains the ESPs with the lowest entangling cost relative to v_i .

The number of e-neighbors is selected approximately as

$$k = (1 + m)\sqrt{n_e} \log n_e, \quad (2.59)$$

where $m > 1$ is a design parameter.

The anchor set is denoted by

$$\mathcal{T} \subseteq V_2, \quad (2.60)$$

with cardinality approximately

$$|\mathcal{T}| = \sqrt{n_e}. \quad (2.61)$$

Each anchor proactively establishes and maintains entanglement with the other anchors. This creates a sparse long-range backbone over the local e-neighborhood structure [3].

Under this design, any ESP can reach another ESP through at most three artificial entanglement hops with high probability.

For additive entangling-cost metrics, the Partial-Anchor Scheme guarantees

$$ES \leq 5. \quad (2.62)$$

The scheme provides low memory overhead but may produce larger routing cost because only a subset of ESPs maintains long-range artificial links.

However, this architecture assumes that the entanglement resources forming the artificial overlay remain sufficiently available and does not explicitly model the temporal evolution of individual e-links, including aging, fidelity degradation, and remaining coherence budget.

2.9.2 Full-Anchor Scheme

In the Full-Anchor Scheme, every ESP maintains both local e-neighborhood links and selected long-range artificial links [3].

The ESP set is divided into tracked subsets, and each ESP proactively maintains entanglement with the nodes belonging to one assigned tracked set.

This design improves overlay connectivity while preserving sublinear table size.

Under the Full-Anchor Scheme, any ESP can reach another ESP through at most two artificial entanglement hops with high probability.

For additive entangling-cost metrics, the Full-Anchor Scheme guarantees

$$ES \leq 3. \quad (2.63)$$

Therefore, the Full-Anchor Scheme provides lower worst-case entangling stretch than the Partial-Anchor Scheme, but it requires more ESPs to maintain long-range entanglement resources.

The main trade-off can be summarized as

$$\text{lower overlay-maintenance complexity} \quad \longleftrightarrow \quad \text{lower entangling stretch.} \quad (2.64)$$

The existing stretch guarantees are primarily structural. They assume that the artificial links required by the routing scheme are available and usable when selected [3].

However, an artificial link may become stale, lose fidelity, expire, or require regeneration before it can be used. Consequently, the static stretch bound may not accurately reflect the actual routing cost under temporal degradation. This limitation motivates the integration of temporal resource information into compact quantum routing.

To address this issue, a time-dependent entangling cost can be introduced:

$$w_t(i, j) = \alpha C_F(i, j, t) + \beta C_A(i, j, t) + \gamma C_B(i, j, t) + \delta C_R(i, j, t), \quad (2.65)$$

where

- $C_F(i, j, t)$ represents fidelity-related cost;
- $C_A(i, j, t)$ represents entanglement-age cost;
- $C_B(i, j, t)$ represents remaining-coherence-budget risk; and
- $C_R(i, j, t)$ represents regeneration cost or delay.

The corresponding temporal entangling stretch may be defined as

$$ES_t(i, j) = \frac{w_t(R_{QR}(i, j))}{w_t(R_t^*(i, j))}, \quad (2.66)$$

where $R_t^*(i, j)$ is the optimal temporally feasible route at time t .

This extension allows compact-routing performance to be evaluated when artificial links have heterogeneous ages, coherence times, fidelities, and regeneration delays.

The central research question is whether temporal-state awareness can improve fidelity and route reliability while preserving the original compact-routing property

Table 2.1: Comparison of Closest Prior Work

Research Line	Age	Fidelity	Decoherence	Routing	Compact Overlay	Stretch
Quantum-native compact routing	No explicit age	Abstractly supported	Abstractly supported	Yes	Yes	Yes
Fidelity-aware routing	No explicit age	Yes	Partially considered	Yes	No	No
Fidelity-Age scheduling	Flow-level age	Yes	Yes	Scheduling-oriented	No	No
Satellite Age of Entanglement	Pair and end-to-end age	Age-based quality	Yes	Chain control	No	No
Adaptive routing and resource control	Partial state information	Yes	Yes	Yes	Limited	No
Belief-state adaptive routing	Pair-level state	Yes	Yes	Yes	No	No
Proposed thesis	Explicit e-link age	Predicted effective fidelity	Temporal degradation model	Temporal-state-aware routing	Yes	Temporal entangling stretch

The quantum-native architecture in [3] provides artificial entanglement overlays, quantum addressing, compact routing tables, Partial-Anchor and Full-Anchor schemes, and bounded entangling stretch. However, it does not explicitly represent e-link age, generation time, remaining coherence budget, or time-varying fidelity inside the routing table.

The Fidelity-Age framework in [1] explicitly combines delivery freshness with a minimum fidelity threshold. It also considers finite memory, probabilistic generation, purification, swapping, and scheduling fairness. Nevertheless, its primary focus is flow scheduling rather than compact route construction over a quantum-native overlay.

The satellite-assisted Age of Entanglement model in [2] jointly controls generation, waiting, swapping, discard, and regeneration under intermittent visibility and memory degradation. However, it is limited to a three-node repeater chain and does not study general graph routing, artificial overlays, compact routing tables, or entangling stretch.

Existing adaptive routing approaches also investigate dynamic quantum-resource states and uncertainty-aware decision making, but they do not combine temporal e-link modeling with scalable compact overlay routing [11, 12, 13].

The comparison shows that age-awareness, fidelity-awareness, decoherence modeling, regeneration, and adaptive control already exist in the literature. Therefore, none of these components can be claimed as individually novel.

The remaining research opportunity lies in their integration with scalable quantum-native compact routing. More specifically, existing work does not sufficiently characterize how stale artificial e-links affect route feasibility, overlay reachability, anchor reliability, memory overhead, and entangling stretch.

This motivates a temporal-state-aware routing framework that extends compact quantum routing by incorporating dynamic e-link age, effective fidelity, and coherence-aware route selection.

2.10 Limitations of Existing Literature

Although recent research has significantly advanced quantum routing, entanglement scheduling, fidelity management, and age-aware control, several limitations remain at the intersection of temporal entanglement management and scalable compact routing.

2.10.1 Absence of Explicit Temporal State in Compact Routing Tables

Existing quantum-native compact-routing tables primarily store entangled resources, e-hop identities, e-neighborhood information, and anchor-related metadata [3]. However, they do not explicitly include the generation time,

age, current effective fidelity, remaining coherence budget, or expiration risk of each stored e-link.

This omission prevents the routing protocol from distinguishing between two structurally equivalent artificial links that have different temporal usability.

2.10.2 Static Entangling-Stretch Guarantees

The Partial-Anchor and Full-Anchor schemes provide constant worst-case entangling-stretch bounds. For additive metrics, these bounds are

$$ES \leq 5 \tag{2.67}$$

for the Partial-Anchor Scheme and

$$ES \leq 3 \tag{2.68}$$

for the Full-Anchor Scheme.

However, these guarantees are derived under the assumption that the required artificial links are available and usable when selected [3, 14, 8]. They do not characterize the effect of stale links, fidelity decay, regeneration delay, or temporal route infeasibility.

2.10.3 Architectural Limits of Age-Aware Studies

Recent age-aware studies explicitly model pair age, Fidelity-Age, end-to-end Age of Entanglement, discard, regeneration, and adaptive control [1, 2]. Nevertheless, these works mainly focus on scheduling, small repeater chains, or flow-level control.

They do not investigate quantum addressing, compact routing tables, artificial ESP overlays, anchor structures, or entangling-stretch guarantees.

2.10.4 Regeneration without Overlay-Aware Prioritization

Existing work already recognizes that depleted or stale entangled resources must be discarded or regenerated [1, 2]. Therefore, regeneration alone is not a new contribution.

The unresolved issue is how regeneration resources should be prioritized across a compact artificial overlay. Regenerating one anchor link may restore global reachability, while regenerating a local e-neighbor link may improve

only a small part of the network. Existing studies do not sufficiently quantify this structural difference.

2.10.5 No Joint Analysis of Temporal and Scalability Trade-offs

Current research does not jointly characterize the relationship among

$$\begin{aligned} &\text{fidelity, age, coherence budget,} \\ &\text{regeneration cost, entangling stretch,} \\ &\text{routing-table overhead.} \end{aligned} \tag{2.69}$$

This trade-off is central to large-scale quantum-native routing because improved temporal freshness may require additional memory, control updates, or regeneration attempts.

2.10.6 Lack of Temporal Route-Feasibility Analysis

A route may be structurally available but temporally infeasible. For example, one or more artificial links may be expected to decohere below the required fidelity threshold before the end-to-end swapping sequence is completed.

Existing compact-routing schemes lack an explicit temporal-feasibility condition:

$$F_{ij}^{\text{eff}}(t + \widehat{D}_\pi) \geq F_{\min}, \quad \forall (i, j) \in \pi. \tag{2.70}$$

Without this condition, a selected route may fail despite being topologically valid.

Therefore, a key open problem is how to integrate temporal-state information into compact quantum routing while preserving scalability, bounded routing-table complexity, and low entangling stretch.

2.10.7 Limited Evaluation of Heterogeneous Temporal States

Artificial e-links may differ in initial fidelity, coherence time, age, generation probability, and regeneration delay. Existing compact-routing analysis largely abstracts these differences through a general cost metric [3].

A detailed evaluation is still required to determine how Partial-Anchor and Full-Anchor schemes behave when e-links have heterogeneous temporal states, including different ages, fidelities, and remaining coherence times.

These limitations indicate that the main unresolved problem is not the absence of age-awareness or regeneration individually. Rather, it is the absence of an explicit temporal-state framework integrated with scalable quantum-native compact routing and its analytical guarantees.

2.11 Identified Research Gap

Prior work has advanced fidelity-aware routing, age metrics, adaptive control, and quantum-native compact routing. It already models pair age, Fidelity-Age, discard, regeneration, and adaptive control [1, 2, 14, 3]. Thus, these individual concepts are not claimed as novel.

At the same time, quantum-native compact-routing architectures provide scalable routing tables, artificial entanglement overlays, quantum addressing, Partial-Anchor and Full-Anchor schemes, and bounded entangling stretch [3]. However, their routing tables do not explicitly represent the temporal state of each stored artificial e-link.

In particular, the following e-link attributes are not directly incorporated into the compact routing-table structure:

- generation time;
- current age;
- current effective fidelity;
- remaining coherence budget;
- expiration probability;
- predicted route-completion risk; and
- regeneration urgency.

Although the general entangling-cost metric can abstractly represent fidelity, memory use, or decoherence, the existing framework does not define an explicit time-dependent cost model.

Consequently, the effect of temporal degradation on compact-routing guarantees remains insufficiently characterized. A route may satisfy the structural conditions of the Partial-Anchor or Full-Anchor scheme but fail because one or more required e-links become unusable before completion.

This leads to the central research gap:

Existing studies do not sufficiently integrate per-e-link temporal state with scalable quantum-native compact routing [1, 2, 3]. The effects of fidelity decay, coherence budget, and regeneration delay on route feasibility, overlay reliability, and entangling stretch remain unclear.

The gap lies at the intersection of two established research directions:

$$\begin{aligned} & \text{temporal entanglement management} \\ & + \text{ quantum-native compact routing.} \end{aligned} \quad (2.71)$$

Addressing this gap requires a framework that extends compact routing-table entries with temporal state while preserving sublinear routing-table scaling and bounded routing overhead.

A time-dependent routing entry may be defined as

$$\mathcal{R}_{ij}(t) = [|v_j\rangle, A_{ij}(t), F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), C_{ij}^{\text{regen}}]. \quad (2.72)$$

The remaining coherence budget may be represented as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.73)$$

where $\widehat{D}_{ij}$ is the expected additional delay before the e-link is consumed.

This temporal state can then support route rejection, rerouting, selective regeneration, anchor-link replacement, and overlay repair.

The research gap is therefore not the absence of temporal concepts in quantum networking. Rather, it is the lack of a rigorous integration of temporal e-link state into compact overlay routing and stretch analysis [3, 1, 2].

2.12 Proposed Research Direction

Based on the identified gap, this thesis proposes a temporal-state-aware extension of quantum-native compact routing for artificial entanglement overlay networks [3, 1, 2]. The main objective is to incorporate explicit e-link temporal information into routing and maintenance decisions while preserving the scalability advantages of compact routing.

The proposed research title is

**Temporal-State-Aware Quantum Routing for
Fidelity-Preserving Entanglement Distribution
in Quantum Networks**

Unlike approaches that study fidelity, age, or compact routing separately, the proposed framework integrates temporal state into routing decisions. It evaluates whether age, fidelity evolution, coherence budget, and regeneration cost improve reliability while preserving compact-routing scalability.

$$\mathcal{L}_{ij}(t) = [A_{ij}(t), F_{ij}^0, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), C_{ij}^{\text{regen}}], \quad (2.74)$$

where

- $A_{ij}(t)$ is the age of the e-link;
- F_{ij}^0 is the initial fidelity;
- $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity;
- $T_{2,ij}$ is the coherence time;
- $B_{ij}(t)$ is the remaining coherence budget; and
- C_{ij}^{regen} is the regeneration cost or delay.

The remaining coherence budget is defined as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.75)$$

where $\widehat{D}_{ij}$ denotes the predicted additional delay before the link is consumed.

A route is considered temporally feasible only if every e-link remains usable until route completion. This can be expressed as

$$F_{ij}^{\text{eff}}(t + \widehat{D}_{\pi}) \geq F_{\min}, \quad \forall (i, j) \in \pi. \quad (2.76)$$

The static entangling cost is extended into a time-dependent cost:

$$w_t(i, j) = \alpha C_F(i, j, t) + \beta C_A(i, j, t) + \gamma C_B(i, j, t) + \delta C_R(i, j, t), \quad (2.77)$$

where the terms represent fidelity cost, age cost, coherence-budget risk, and regeneration cost, respectively.

The corresponding temporal entangling stretch is defined as

$$ES_t(i, j) = \frac{w_t(R_{\text{QR}}(i, j))}{w_t(R_t^*(i, j))}. \quad (2.78)$$

The framework supports the following temporal management actions:

- retain a usable e-link;
- reject a temporally infeasible route;
- reroute through fresher links;
- regenerate a stale e-link;
- replace an unreliable anchor link; and
- repair the artificial overlay.

The main research questions are:

1. How can explicit e-link temporal state be incorporated into a quantum-native compact routing table while preserving sublinear memory scaling?
2. How do age, fidelity degradation, coherence budget, and regeneration delay affect route feasibility in Partial-Anchor and Full-Anchor overlays?
3. How does temporal degradation affect entangling stretch and end-to-end reliability?
4. Can coherence-budget-aware routing outperform static compact routing, fidelity-only routing, age-only routing, and fixed-threshold approaches?

The expected contributions of this thesis are:

1. a temporal-state-aware extension of compact quantum routing tables for artificial entanglement overlays;
2. a coherence-budget model that captures the remaining usability of temporal e-links;
3. a time-dependent entangling cost based on fidelity evolution, age, and coherence degradation;
4. a temporal route-feasibility criterion for evaluating future link usability;
5. a coherence-budget-aware routing and overlay-maintenance algorithm;
6. a temporal entangling-stretch formulation for measuring degradation-aware routing performance; and

7. a comprehensive evaluation of Partial-Anchor and Full-Anchor overlay schemes under heterogeneous temporal conditions.

The evaluation will compare the proposed method against static compact routing, fidelity-only routing, age-only routing, fixed-cutoff routing, and regeneration heuristics. Performance will be measured using end-to-end fidelity, delivery success probability, throughput, route failure rate, memory overhead, regeneration cost, and temporal entangling stretch.

This research direction preserves the key advantage of quantum-native compact routing, namely scalable overlay-based communication, while addressing its limitation in representing time-dependent entanglement-resource states.

Chapter 3

System Model and Problem Formulation

3.1 Introduction

This chapter presents the system model and formal problem formulation for temporal-state-aware compact routing in quantum-native entanglement overlay networks. The objective is to extend the existing quantum-native compact-routing architecture by incorporating explicit temporal information associated with each artificial entanglement link (e-link).

The proposed model distinguishes between the underlying physical quantum network and the artificial entanglement overlay. The physical network represents the actual quantum communication infrastructure, including quantum nodes, repeaters, quantum memories, and physical quantum channels. In contrast, the artificial overlay represents proactively established entangled connections among Entanglement Service Providers (ESPs), enabling routing over an abstract entanglement topology rather than only over physical connectivity.

Unlike classical network links, artificial e-links are dynamic and time-dependent resources [3, 1, 2]. Their availability and quality are affected by probabilistic entanglement generation, finite quantum-memory coherence, decoherence-induced fidelity degradation, entanglement consumption during swapping operations, and regeneration processes. Therefore, the routing cost of an artificial e-link cannot be considered static and must be evaluated according to its current temporal state.

To capture this dynamic behavior, each artificial e-link is associated with a temporal state containing its generation time, age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, and regeneration-

related cost. These temporal attributes enable the routing protocol to determine whether a structurally available route can remain physically usable throughout the expected end-to-end entanglement establishment process.

This chapter further defines the time-dependent entangling-cost model and introduces the concept of temporal entangling stretch. These extensions provide a framework for evaluating compact-routing schemes, including the Partial-Anchor and Full-Anchor architectures, under heterogeneous conditions involving different link ages, fidelity levels, coherence characteristics, and regeneration requirements.

Finally, routing is formulated as temporal resource selection. The objective is to minimize time-dependent entangling cost subject to fidelity, coherence, memory, and compact-routing constraints.

3.2 Network Architecture

The proposed system follows a two-tier quantum-network architecture composed of quantum edge nodes and Entanglement Service Providers (ESPs). The edge nodes generate application-level entanglement requests, while the ESPs maintain, manage, and route shared entanglement resources across the quantum-network core [3, 10].

Figure 3.1 illustrates the overall layered architecture considered in this thesis.

The complete quantum network is represented as

$$G = (V, E), \quad (3.1)$$

where V denotes the set of quantum nodes and E represents the set of physical quantum communication links.

The node set is partitioned into two disjoint subsets:

$$V = V_1 \cup V_2, \quad V_1 \cap V_2 = \emptyset, \quad (3.2)$$

where V_1 represents quantum edge nodes and V_2 represents Entanglement Service Providers (ESPs).

The numbers of edge nodes and ESPs are defined as

$$|V_1| = n_u, \quad |V_2| = n_e, \quad (3.3)$$

such that

$$|V| = n = n_u + n_e. \quad (3.4)$$

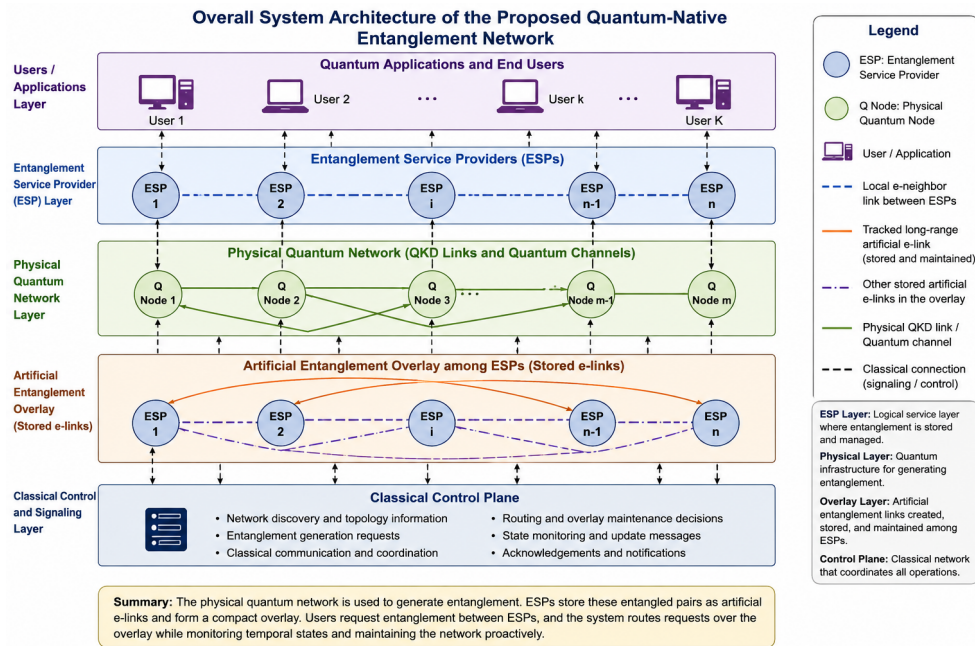


Figure 3.1: Overall architecture of the quantum-native entanglement network, including the user layer, Entanglement Service Provider layer, physical quantum network, artificial entanglement overlay, and classical control plane.

Each edge node is connected to a serving ESP through a local quantum connection. When an application requests end-to-end entanglement, the corresponding ESP coordinates entanglement generation, routing, and resource allocation operations over the ESP-level network.

The ESPs constitute the quantum-network core. Each ESP is assumed to contain quantum memories, entanglement-generation interfaces, Bell-state measurement capabilities, and classical communication channels required for entanglement establishment, heralding, synchronization, and control operations.

An Entanglement-Defined Controller (EDC), denoted by $\mathcal{C}_{\text{EDC}}$, manages the control plane of the artificial entanglement overlay [3, 25]. The controller maintains information about artificial e-link availability, routing-table states, temporal link conditions, regeneration status, and network constraints.

The main functions of the EDC include:

- monitoring artificial e-link availability and temporal state;
- constructing and updating e-neighborhood information;
- assigning anchor and tracked-link responsibilities;
- coordinating route-selection decisions;
- prioritizing stale-link regeneration;
- repairing depleted artificial-overlay resources; and
- enforcing fidelity, memory, and coherence constraints.

The architecture separates the physical quantum network from the artificial entanglement overlay. The physical network represents the available quantum communication infrastructure, while the artificial overlay represents dynamically maintained entanglement resources established through entanglement generation and swapping operations [3, 10]

The physical ESP-level network is represented as

$$G_P = (V_2, E_P), \quad (3.5)$$

where E_P denotes the set of physical quantum links connecting ESPs. The artificial entanglement overlay at time t is represented as

$$G_A(t) = (V_2, E_A(t)), \quad (3.6)$$

where $E_A(t)$ represents the set of currently available artificial e-links.

An artificial e-link is defined as

$$e_{ij}^A(t) \in E_A(t), \quad (3.7)$$

and represents an entangled resource shared between ESPs v_i and v_j . The two ESPs may be physically adjacent or may be connected through multi-hop entanglement generation and swapping operations.

Because artificial e-links are generated, consumed, degraded, and regenerated dynamically, the artificial overlay is time varying:

$$E_A(t+1) \neq E_A(t). \quad (3.8)$$

The proposed framework considers both Partial-Anchor and Full-Anchor compact-routing schemes. In the Partial-Anchor Scheme, every ESP maintains local e-neighbor links, while only selected anchor ESPs maintain long-range artificial links. In the Full-Anchor Scheme, every ESP maintains local e-neighbor links together with additional long-range tracked links.

Routing decisions are performed using local compact routing-table information combined with temporal-state updates provided by the EDC. Therefore, each ESP does not require complete global network knowledge, which enables scalable routing for large quantum networks.

This architecture provides the structural foundation for defining the physical network model, artificial entanglement overlay, temporal e-link state, and coherence-budget-aware routing problem in the following sections.

3.3 Physical Quantum Network Model

The physical quantum network represents the underlying hardware-level infrastructure that enables entanglement generation and distribution among Entanglement Service Providers. It includes quantum communication channels, quantum memories, entanglement-generation devices, Bell-state measurement modules, and classical communication links required for synchronization and coordination.

The ESP-level physical quantum network is modeled as an undirected graph

$$G_P = (V_2, E_P), \quad (3.9)$$

where V_2 denotes the set of Entanglement Service Providers and E_P represents the set of physical quantum links connecting them.

A physical quantum link between ESPs v_i and v_j is defined as

$$e_{ij}^P = (v_i, v_j) \in E_P. \quad (3.10)$$

Each physical link is characterized by a parameter vector

$$\Theta_{ij}^P = [L_{ij}, p_{ij}^{\text{gen}}, F_{ij}^0, T_{2,ij}, q_{ij}, M_{ij}, D_{ij}^{\text{gen}}], \quad (3.11)$$

where

- L_{ij} represents the physical distance between ESPs v_i and v_j ;
- p_{ij}^{gen} denotes the probability of successful elementary-entanglement generation;
- F_{ij}^0 represents the initial fidelity of a successfully generated entangled pair;
- $T_{2,ij}$ denotes the coherence time of the quantum memory;
- q_{ij} represents the success probability of local operations, including Bell-state measurements;
- M_{ij} denotes the available quantum-memory capacity; and
- D_{ij}^{gen} represents the expected elementary-link generation delay.

The network evolution is considered in discrete time slots:

$$t \in \mathbb{N}. \quad (3.12)$$

For physical link e_{ij}^P , the result of an elementary entanglement-generation attempt at slot t is modeled as a Bernoulli random variable:

$$X_{ij}(t) \sim \text{Bernoulli}(p_{ij}^{\text{gen}}). \quad (3.13)$$

Therefore,

$$X_{ij}(t) = \begin{cases} 1, & \text{if elementary entanglement generation succeeds,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.14)$$

When multiple parallel generation modes are available, denoted by S_{ij} , the probability of obtaining at least one successful elementary pair is

$$p_{ij}^{\text{link}} = 1 - (1 - p_{ij}^{\text{gen}})^{S_{ij}}. \quad (3.15)$$

The elementary-generation probability is affected by channel loss and can be approximated using an exponential attenuation model:

$$p_{ij}^{\text{gen}} = \eta_{ij} = \exp(-\alpha L_{ij}), \quad (3.16)$$

where α is the channel attenuation coefficient.

Each successfully generated elementary entangled pair occupies quantum-memory resources at both endpoint ESPs. Let

$$m_i(t) \quad (3.17)$$

denote the number of occupied memory units at ESP v_i during time slot t . The memory constraint is given by

$$m_i(t) \leq M_i, \quad \forall v_i \in V_2. \quad (3.18)$$

A physical entanglement path between ESPs v_i and v_j is represented as

$$P_{ij} = (v_i, v_{r_1}, \dots, v_{r_k}, v_j). \quad (3.19)$$

If all elementary links along the path are successfully generated and intermediate Bell-state measurements occur independently, the approximate end-to-end establishment probability is

$$P_{\text{succ}}(P_{ij}) = \left(\prod_{e_{uv}^P \in P_{ij}} p_{uv}^{\text{link}} \right) \left(\prod_{v_r \in P_{ij} \setminus \{v_i, v_j\}} q_r \right). \quad (3.20)$$

The expected physical-route establishment delay is denoted by

$$D^{\text{phys}}(P_{ij}). \quad (3.21)$$

This delay includes link generation, classical heralding, memory waiting, purification, and entanglement swapping.

A physical route is considered feasible only when sufficient quantum-memory resources are available and the resulting entangled state satisfies the required fidelity condition:

$$F_{\text{end}}(P_{ij}) \geq F_{\text{min}}. \quad (3.22)$$

The physical quantum network provides the substrate for constructing artificial entanglement links. An artificial e-link between physically non-adjacent ESPs is established through a feasible physical path and stored at the corresponding ESP endpoints.

Therefore, the quality of an artificial e-link depends on the physical path used for its creation, including generation delay, success probability, initial fidelity, memory consumption, and accumulated operational errors.

3.4 Artificial Entanglement Overlay Model

The artificial overlay contains proactive entangled connections among ESPs. Unlike a physical link, an artificial e-link may connect nonadjacent ESPs through multihop generation and swapping.

At time slot t , the artificial entanglement overlay is represented as

$$G_A(t) = (V_2, E_A(t)), \quad (3.23)$$

where V_2 denotes the set of ESPs and $E_A(t)$ represents the set of available artificial e-links at time t .

An artificial e-link between ESPs v_i and v_j is defined as

$$e_{ij}^A(t) = (v_i, v_j) \in E_A(t). \quad (3.24)$$

The availability state of an artificial e-link is represented by

The storage-availability state of an artificial e-link is represented by

$$x_{ij}(t) = \begin{cases} 1, & \text{if the artificial e-link is stored and unconsumed at time } t, \\ 0, & \text{otherwise.} \end{cases} \quad (3.25)$$

An artificial e-link is considered usable for routing only when it is available in quantum memory and its effective fidelity satisfies the link-level fidelity requirement:

$$x_{ij}(t) = 1, \quad F_{ij}^{\text{eff}}(t) \geq F_{\min}^{\text{link}}. \quad (3.26)$$

Each artificial e-link is created through an underlying physical path. Let

$$P_{ij}^{\text{gen}} = (v_i, v_{k_1}, \dots, v_{k_m}, v_j) \quad (3.27)$$

denote the physical route that creates the e-link between v_i and v_j . Intermediate ESPs perform the required swapping operations.

The generation success probability of the artificial e-link is determined by the physical path:

$$P_{ij}^A = P_{\text{succ}}(P_{ij}^{\text{gen}}). \quad (3.28)$$

The corresponding expected generation delay is

$$D_{ij}^A = D^{\text{phys}}(P_{ij}^{\text{gen}}). \quad (3.29)$$

The initial fidelity of the generated artificial e-link is denoted by

$$F_{ij}^{A,0}. \quad (3.30)$$

This initial fidelity depends on the quality of elementary entangled pairs, the number of entanglement-swapping operations, purification procedures, and accumulated hardware-induced errors during link establishment.

Because artificial e-links are temporary, the overlay evolves through generation, consumption, degradation, expiration, and regeneration. The evolution of the artificial overlay is expressed as

$$E_A(t+1) = (E_A(t) \setminus E_{\text{cons}}(t) \setminus E_{\text{drop}}(t)) \cup E_{\text{gen}}(t), \quad (3.31)$$

where

- $E_{\text{cons}}(t)$ denotes the set of artificial e-links consumed during routing, swapping, or application operations;
- $E_{\text{drop}}(t)$ represents expired, stale, or unusable e-links; and
- $E_{\text{gen}}(t)$ represents newly generated or regenerated artificial e-links.

Each artificial e-link requires quantum-memory resources at both endpoint ESPs. Let

$$n_{ij}^{\text{ebit}}(t) \quad (3.32)$$

denote the number of stored e-bits associated with artificial link (i, j) .

The memory constraint at ESP v_i is therefore

$$\sum_{j: e_{ij}^A(t) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i. \quad (3.33)$$

A routing path over the artificial overlay between source ESP v_s and destination ESP v_d is represented as

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d). \quad (3.34)$$

A feasible overlay route requires that every consecutive ESP pair along the path has an available artificial e-link:

$$e_{uv}^A(t) \in E_A(t), \quad \forall (u, v) \in R_{s,d}(t). \quad (3.35)$$

Establishing end-to-end entanglement along the route requires entanglement swapping at intermediate ESPs. Therefore, a route containing $h+1$ artificial e-links requires h swapping operations.

Assuming independent swapping success probabilities, the route-level establishment probability is approximated as

$$P_{\text{route}}(R_{s,d}(t)) = \left(\prod_{(i,j) \in R_{s,d}(t)} z_{ij}(t) \right) \left(\prod_{v_r \in R_{s,d}(t) \setminus \{v_s, v_d\}} q_r \right). \quad (3.36)$$

The artificial overlay follows compact-routing structures based on local e-neighborhood maintenance and long-range connectivity mechanisms. In this thesis, both Partial-Anchor and Full-Anchor schemes are considered as baseline overlay architectures for evaluating temporal-state-aware routing performance.

3.4.1 Partial-Anchor Overlay

In the Partial-Anchor Scheme, each ESP maintains artificial e-links with ESPs belonging to its local e-neighborhood

$$\mathcal{N}(v_i). \quad (3.37)$$

A smaller subset of ESPs, referred to as anchor nodes, maintains additional long-range artificial links. The anchor set is defined as

$$\mathcal{T} \subseteq V_2. \quad (3.38)$$

The artificial-link set in the Partial-Anchor overlay can be represented as

$$E_A^{\text{PA}}(t) = E_{\text{local}}(t) \cup E_{\text{anchor}}(t). \quad (3.39)$$

3.4.2 Full-Anchor Overlay

In the Full-Anchor Scheme, every ESP maintains local e-neighbor links together with selected long-range tracked links.

The artificial-link set is represented as

$$E_A^{\text{FA}}(t) = E_{\text{local}}(t) \cup E_{\text{tracked}}(t). \quad (3.40)$$

Compared with the Partial-Anchor Scheme, the Full-Anchor architecture provides stronger overlay connectivity and shorter artificial routes, but requires additional quantum-memory resources and long-range entanglement maintenance.

In both compact-routing architectures, structural availability alone is insufficient for reliable routing. An artificial e-link may remain present in the

routing table while its fidelity has degraded, its coherence budget has been exhausted, or its remaining lifetime is insufficient for completing an end-to-end entanglement operation.

Therefore, the following section introduces an explicit temporal-state model for artificial e-links.

3.5 Temporal State of Artificial E-Links

The temporal state of an artificial e-link describes how its routing usability evolves after link establishment. Although an e-link may remain physically stored in quantum memory, decoherence gradually reduces its fidelity, increases expiration risk, and changes its suitability for future routing operations.

For an artificial e-link shared between ESPs v_i and v_j , let

$$t_{ij}^{\text{gen}} \quad (3.41)$$

denote the time slot when the e-link was successfully generated or regenerated.

The age of the artificial e-link at time t is defined as

$$A_{ij}(t) = t - t_{ij}^{\text{gen}}. \quad (3.42)$$

For a newly generated e-link,

$$A_{ij}(t) = 0. \quad (3.43)$$

If the e-link remains stored without consumption or regeneration, its age evolves as

$$A_{ij}(t+1) = A_{ij}(t) + 1. \quad (3.44)$$

The activity state of an artificial e-link is represented by

$$x_{ij}(t) \in \{0, 1\}, \quad (3.45)$$

where

$$x_{ij}(t) = \begin{cases} 1, & \text{if the e-link is stored and available,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.46)$$

The complete temporal state vector of an artificial e-link is defined as

$$\mathcal{L}_{ij}(t) = [x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.47)$$

The components represent:

- $x_{ij}(t)$: current availability state of the artificial e-link;
- $A_{ij}(t)$: elapsed storage time after generation or regeneration;
- $F_{ij}^{A,0}$: initial fidelity after establishment;
- $F_{ij}^{\text{eff}}(t)$: current effective fidelity;
- $T_{2,ij}$: coherence time of the quantum memory;
- $B_{ij}(t)$: remaining coherence budget;
- D_{ij}^{regen} : expected regeneration delay; and
- C_{ij}^{regen} : regeneration resource cost.

The temporal state is updated at the beginning of each time slot. Depending on generation, aging, consumption, or expiration events, the state evolves according to

$$\mathcal{L}_{ij}(t+1) = \begin{cases} \mathcal{L}_{ij}^{\text{new}}, & \text{if generation succeeds,} \\ \mathcal{L}_{ij}^{\text{age}}, & \text{if the e-link remains stored,} \\ \mathcal{L}_{ij}^{\text{inactive}}, & \text{if consumed, discarded, or expired.} \end{cases} \quad (3.48)$$

For a newly generated e-link,

$$\mathcal{L}_{ij}^{\text{new}} = [1, 0, F_{ij}^{A,0}, F_{ij}^{A,0}, T_{2,ij}, B_{ij}^0, D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.49)$$

For an aged but still usable e-link,

$$\mathcal{L}_{ij}^{\text{age}} = [1, A_{ij}(t) + 1, F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t+1), T_{2,ij}, B_{ij}(t+1), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.50)$$

An artificial e-link is classified as stale when its temporal quality becomes insufficient. This occurs if

$$F_{ij}^{\text{eff}}(t) < F_{\min}^{\text{link}}, \quad (3.51)$$

or

$$B_{ij}(t) \leq 0, \quad (3.52)$$

or

$$A_{ij}(t) \geq A_{ij}^{\max}. \quad (3.53)$$

The stale-link indicator is defined as

$$s_{ij}(t) = \begin{cases} 1, & \text{if the e-link is stale,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.54)$$

A stale e-link may still physically exist in quantum memory; however, it should not participate in temporally feasible routing decisions.

The temporally usable-link indicator is therefore defined as

$$u_{ij}(t) = x_{ij}(t) (1 - s_{ij}(t)). \quad (3.55)$$

The temporally usable artificial overlay is given by

$$G_U(t) = (V_2, E_U(t)), \quad (3.56)$$

where

$$E_U(t) = \{e_{ij}^A(t) \in E_A(t) : u_{ij}(t) = 1\}. \quad (3.57)$$

This distinction between stored and temporally usable artificial links is fundamental for the proposed routing framework. A routing table may contain an e-link entry that is structurally valid but temporally unreliable. Therefore, route selection is performed over the temporally usable overlay $G_U(t)$ rather than the complete stored overlay $G_A(t)$.

3.6 Fidelity and Decoherence Model

The fidelity and decoherence model describes the evolution of artificial e-link quality during storage, generation, purification, and entanglement-swapping operations. Since the proposed routing framework relies on stored entanglement resources, the temporal degradation of fidelity must be explicitly considered in routing decisions.

For an artificial e-link shared between ESPs v_i and v_j , let

$$F_{ij}^{A,0} \quad (3.58)$$

denote the initial fidelity immediately after successful e-link generation or regeneration.

The initial fidelity depends on several physical factors, including the quality of elementary entangled pairs, the number of entanglement-swapping operations, purification performance, and accumulated hardware imperfections.

The current effective fidelity of the stored artificial e-link at time t is represented as

$$F_{ij}^{\text{eff}}(t). \quad (3.59)$$

A general temporal fidelity model can be expressed as

$$F_{ij}^{\text{eff}}(t) = f_{\text{dec}} \left(F_{ij}^{\text{A},0}, A_{ij}(t), T_{2,ij}, \epsilon_{ij}^{\text{op}} \right), \quad (3.60)$$

where $\epsilon_{ij}^{\text{op}}$ represents accumulated operational errors caused by imperfect quantum operations.

The analysis and simulation use an exponential decoherence model:

$$F_{ij}^{\text{eff}}(t) = F_{\infty} + \left(F_{ij}^{\text{A},0} - F_{\infty} \right) \exp \left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}} \right), \quad (3.61)$$

where Δ denotes the duration of one time slot, $T_{2,ij}$ is the coherence time of the quantum memory, and F_{∞} represents the fidelity limit under the selected noise model.

For a two-qubit Werner-state approximation, the fully decohered state approaches the maximally mixed state, whose fidelity is

$$F_{\infty} = \frac{1}{4}. \quad (3.62)$$

Therefore, the storage-induced fidelity degradation model becomes

$$F_{ij}^{\text{eff}}(t) = \frac{1}{4} + \left(F_{ij}^{\text{A},0} - \frac{1}{4} \right) \exp \left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}} \right). \quad (3.63)$$

This formulation ensures that the fidelity gradually approaches the physical lower bound of the Werner-state model rather than incorrectly decaying to zero.

An equivalent representation can be obtained using the Werner parameter:

$$p_{ij}^W(t) = \frac{4F_{ij}^{\text{eff}}(t) - 1}{3}. \quad (3.64)$$

The initial Werner parameter is defined as

$$p_{ij}^{W,0} = \frac{4F_{ij}^{A,0} - 1}{3}. \quad (3.65)$$

Under exponential storage decoherence, the Werner parameter evolves as

$$p_{ij}^W(t) = p_{ij}^{W,0} \exp\left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}}\right). \quad (3.66)$$

The corresponding effective fidelity is recovered by

$$F_{ij}^{\text{eff}}(t) = \frac{1 + 3p_{ij}^W(t)}{4}. \quad (3.67)$$

For an artificial-overlay route

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d), \quad (3.68)$$

the end-to-end fidelity depends on the effective fidelity of all artificial e-links along the selected path and the fidelity-retention factors of intermediate swapping operations.

Let the set of artificial e-links contained in route $R_{s,d}(t)$ be denoted by

$$E_A(R_{s,d}(t)). \quad (3.69)$$

Under the Werner-state approximation, the route parameter before swapping imperfections is

$$p_{\text{route}}^W(t) = \prod_{(i,j) \in E_A(R_{s,d}(t))} p_{ij}^W(t). \quad (3.70)$$

To model imperfect entanglement-swapping operations, let

$$\lambda_r \in [0, 1] \quad (3.71)$$

denote the fidelity-retention factor of the Bell-state measurement operation performed at intermediate ESP v_r .

The end-to-end Werner parameter after considering swapping imperfections is approximated as

$$p_{s,d}^{W,\text{end}}(t) = \left(\prod_{(i,j) \in E_A(R_{s,d}(t))} p_{ij}^W(t) \right) \left(\prod_{v_r \in R_{s,d}(t) \setminus \{v_s, v_d\}} \lambda_r \right). \quad (3.72)$$

The resulting end-to-end fidelity is obtained as

$$F_{s,d}^{\text{end}}(t) = \frac{1 + 3p_{s,d}^{W,\text{end}}(t)}{4}. \quad (3.73)$$

A selected route is considered fidelity-feasible at the current time only if

$$F_{s,d}^{\text{end}}(t) \geq F_{\min}^{\text{end}}. \quad (3.74)$$

However, current-time fidelity alone is insufficient for temporal routing decisions. In practice, additional waiting time may occur due to classical communication, resource scheduling, and intermediate swapping operations. During this interval, the stored e-links continue to experience decoherence.

Let

$$\widehat{D}_{s,d}^{\text{route}}(t) \quad (3.75)$$

denote the predicted completion delay of the selected route.

For an artificial e-link (i, j) , the predicted fidelity at consumption time is given by

$$\widehat{F}_{ij}^{\text{cons}}(t) = \frac{1}{4} + \left(F_{ij}^{\text{eff}}(t) - \frac{1}{4} \right) \exp \left(-\frac{\widehat{D}_{ij}^{\text{use}}(t)}{T_{2,ij}} \right), \quad (3.76)$$

where $\widehat{D}_{ij}^{\text{use}}(t)$ represents the predicted waiting time before the e-link is consumed during route establishment.

Using these predicted link fidelities, the predicted end-to-end Werner parameter is defined as

$$\widehat{p}_{s,d}^{W,\text{end}}(t) = \left(\prod_{(i,j) \in E_A(R_{s,d}(t))} \frac{4\widehat{F}_{ij}^{\text{cons}}(t) - 1}{3} \right) \left(\prod_{v_r \in R_{s,d}(t) \setminus \{v_s, v_d\}} \lambda_r \right). \quad (3.77)$$

The end-to-end fidelity at route completion can then be estimated as

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) = \frac{1 + 3\widehat{p}_{s,d}^{W,\text{end}}(t)}{4}. \quad (3.78)$$

A route is considered temporally fidelity-feasible only if

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (3.79)$$

In some cases, purification may be applied when the stored fidelity is insufficient but additional entangled pairs are available. The output fidelity of a purification procedure can be represented generally as

$$F_{ij}^{\text{pur}} = \Phi_{\text{pur}} \left(F_{ij}^{(1)}, F_{ij}^{(2)} \right). \quad (3.80)$$

Purification is performed only when the expected fidelity improvement justifies the additional memory consumption, processing delay, and probabilistic success cost.

Therefore, the proposed routing framework distinguishes among:

- current stored-link fidelity;
- predicted fidelity at consumption time;
- end-to-end route fidelity after swapping; and
- optional purification-enhanced fidelity.

This temporal fidelity evaluation prevents the routing algorithm from selecting an artificial e-link that appears feasible at the decision time but becomes unusable before completion of the end-to-end entanglement establishment process.

3.7 Remaining Coherence-Budget Model

The remaining coherence budget estimates whether an artificial e-link can remain usable until it is consumed by an end-to-end entanglement-routing operation. Unlike simple age-based metrics, the coherence budget jointly considers the current storage duration, memory coherence limit, and the expected future delay before utilization.

For artificial e-link (i, j) , let

$$T_{2,ij} \quad (3.81)$$

denote the coherence time of the corresponding quantum memory, and let

$$A_{ij}(t) \quad (3.82)$$

represent the current age of the e-link at time slot t .

A basic estimate of the remaining coherence lifetime is

$$R_{ij}^{\text{coh}}(t) = T_{2,ij} - A_{ij}(t)\Delta, \quad (3.83)$$

where Δ is the duration of one time slot.

However, remaining coherence lifetime alone does not guarantee successful utilization because additional delay may occur before the e-link participates in an end-to-end entanglement operation.

Let

$$\widehat{D}_{ij}^{\text{use}}(t) \quad (3.84)$$

denote the predicted additional delay before artificial e-link (i, j) is consumed.

The remaining coherence budget is therefore defined as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t)\Delta - \widehat{D}_{ij}^{\text{use}}(t). \quad (3.85)$$

A positive coherence budget indicates that the e-link is expected to remain within its coherence interval until consumption:

$$B_{ij}(t) > 0. \quad (3.86)$$

A non-positive value indicates that the e-link is likely to expire before it can be used:

$$B_{ij}(t) \leq 0. \quad (3.87)$$

The predicted utilization delay can include several components:

$$\widehat{D}_{ij}^{\text{use}}(t) = \widehat{D}_{ij}^{\text{queue}}(t) + \widehat{D}_{ij}^{\text{coord}}(t) + \widehat{D}_{ij}^{\text{pur}}(t) + \widehat{D}_{ij}^{\text{swap}}(t), \quad (3.88)$$

where

- $\widehat{D}_{ij}^{\text{queue}}(t)$ represents scheduling and resource-waiting delay;
- $\widehat{D}_{ij}^{\text{coord}}(t)$ represents classical coordination delay;
- $\widehat{D}_{ij}^{\text{pur}}(t)$ represents purification-related delay; and
- $\widehat{D}_{ij}^{\text{swap}}(t)$ represents the expected delay before entanglement swapping.

For a route

$$R_{s,d}(t), \quad (3.89)$$

the route-level coherence budget is determined by the bottleneck e-link:

$$B_{s,d}^{\text{route}}(t) = \min_{(i,j) \in E_A(R_{s,d}(t))} B_{ij}(t). \quad (3.90)$$

This formulation reflects the fact that a single expired artificial e-link can invalidate the complete end-to-end route.

Therefore, a route is coherence-feasible only when

$$B_{s,d}^{\text{route}}(t) > 0. \quad (3.91)$$

To compensate for uncertainty in delay prediction, a safety margin can be introduced:

$$\delta_{\text{safe}} > 0. \quad (3.92)$$

The robust feasibility condition becomes

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in E_A(R_{s,d}(t)). \quad (3.93)$$

A normalized coherence-budget ratio is defined as

$$\tilde{B}_{ij}(t) = \frac{B_{ij}(t)}{T_{2,ij}}. \quad (3.94)$$

This normalization enables comparison among e-links with different coherence times. The normalized budget satisfies

$$\tilde{B}_{ij}(t) \in (-\infty, 1]. \quad (3.95)$$

A larger positive value indicates a temporally reliable e-link, while a value close to zero indicates a high expiration risk.

The coherence-budget risk cost is defined as

$$C_B(i, j, t) = \begin{cases} \frac{1}{\tilde{B}_{ij}(t) + \varepsilon}, & \tilde{B}_{ij}(t) > 0, \\ +\infty, & \tilde{B}_{ij}(t) \leq 0, \end{cases} \quad (3.96)$$

where $\varepsilon > 0$ avoids numerical instability near zero.

This cost increases sharply when an e-link approaches expiration, discouraging the selection of temporally unreliable resources.

The coherence-budget state can also be classified using predefined thresholds:

$$\text{State}_{ij}^B(t) = \begin{cases} \text{Safe}, & B_{ij}(t) \geq B_{\text{high}}, \\ \text{Warning}, & B_{\text{low}} \leq B_{ij}(t) < B_{\text{high}}, \\ \text{Critical}, & 0 < B_{ij}(t) < B_{\text{low}}, \\ \text{Expired}, & B_{ij}(t) \leq 0. \end{cases} \quad (3.97)$$

Safe links can be used normally, warning links require monitoring, critical links may trigger rerouting or regeneration, and expired links are removed from the temporally usable overlay.

The regeneration urgency of artificial e-link (i, j) is defined as

$$U_{ij}^{\text{regen}}(t) = \omega_1 C_B(i, j, t) + \omega_2 C_F(i, j, t) + \omega_3 I_{ij}^{\text{struct}}, \quad (3.98)$$

where

- $C_B(i, j, t)$ represents coherence-budget risk;
- $C_F(i, j, t)$ represents fidelity-degradation risk; and
- I_{ij}^{struct} represents the structural importance of the e-link in the compact overlay.

The structural-importance term enables anchor links and frequently used artificial links to receive higher regeneration priority than links with limited routing influence.

Therefore, the coherence-budget model establishes a direct relationship among quantum-memory lifetime, route feasibility, temporal routing decisions, and artificial-overlay maintenance.

3.8 Temporal Route-Feasibility Condition

A route over the artificial entanglement overlay may be structurally valid but temporally infeasible. Structural validity only indicates that all required artificial e-links are currently available in the overlay. Temporal feasibility additionally requires that these e-links remain usable until the completion of the end-to-end entanglement-establishment process.

Let

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d) \quad (3.99)$$

denote a candidate artificial-overlay route between source ESP v_s and destination ESP v_d at time t .

The artificial e-link set contained in the route is denoted by

$$E_A(R_{s,d}(t)). \quad (3.100)$$

The route is structurally feasible if every required artificial e-link exists and is currently stored:

$$x_{ij}(t) = 1, \quad \forall (i, j) \in E_A(R_{s,d}(t)). \quad (3.101)$$

However, structural availability alone does not guarantee successful route completion. Each artificial e-link must satisfy temporal-state, fidelity, delay, and memory constraints.

The first requirement is the remaining coherence-budget constraint:

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in E_A(R_{s,d}(t)), \quad (3.102)$$

where δ_{safe} represents the required safety margin against delay-estimation uncertainty.

The second requirement is the predicted link-level fidelity constraint:

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}}, \quad \forall (i, j) \in E_A(R_{s,d}(t)). \quad (3.103)$$

The third requirement considers the final end-to-end fidelity after all swapping operations:

$$\hat{F}_{s,d}^{\text{end}}(t + \hat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (3.104)$$

The fourth requirement limits the route-completion delay:

$$\hat{D}_{s,d}^{\text{route}}(t) \leq D_{\max}, \quad (3.105)$$

where $D_{\max}$ is the maximum acceptable establishment delay.

The fifth requirement ensures that sufficient quantum-memory resources are available along the route:

$$\sum_{j: e_{ij}^A \in R_{s,d}(t)} n_{ij}^{\text{ebit}}(t) \leq M_i, \quad \forall v_i \in R_{s,d}(t). \quad (3.106)$$

A binary temporal route-feasibility indicator is defined as

$$\chi_{s,d}(R_{s,d}(t)) \in \{0, 1\}. \quad (3.107)$$

For compactness, define the active-edge set of the selected route as

$$\mathcal{E}_{s,d}(t) = E_A(R_{s,d}(t)). \quad (3.108)$$

The indicator takes the value one only when all temporal-routing requirements are satisfied:

$$\begin{aligned}
\chi_{s,d}(R_{s,d}(t)) = & \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} x_{ij}(t) \\
& \times \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} \mathbf{1}[B_{ij}(t) \geq \delta_{\text{safe}}] \\
& \times \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} \mathbf{1}[\widehat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}}] \\
& \times \mathbf{1}[\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}] \\
& \times \mathbf{1}[\widehat{D}_{s,d}^{\text{route}}(t) \leq D_{\max}] \\
& \times \prod_{v_i \in R_{s,d}(t)} \mathbf{1} \left[\sum_{j: e_{ij}^A \in R_{s,d}(t)} n_{ij}^{\text{ebit}}(t) \leq M_i \right].
\end{aligned} \tag{3.109}$$

If $\chi_{s,d}(R_{s,d}(t)) = 0$, the route must be rejected, regenerated, repaired, or replaced by an alternative temporally feasible path.

The set of all temporally feasible routes between ESPs v_s and v_d is defined as

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \{R_{s,d}(t) : \chi_{s,d}(R_{s,d}(t)) = 1\}. \tag{3.110}$$

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \tag{3.111}$$

then no currently available artificial-overlay route satisfies all temporal, fidelity, delay, and memory constraints.

In this situation, the Entanglement-Defined Controller may perform one of the following recovery actions:

- regenerate one or more stale artificial e-links;
- replace critical anchor or tracked links;
- wait for additional entanglement-generation opportunities;
- apply purification when sufficient entangled resources are available; or
- declare the request temporarily infeasible.

Therefore, the proposed framework classifies each candidate route into three temporal states:

$$\text{RouteState} = \begin{cases} \text{Feasible,} & \text{if all constraints hold,} \\ \text{Repairable,} & \text{if maintenance restores feasibility,} \\ \text{Infeasible,} & \text{if no valid recovery exists.} \end{cases} \quad (3.112)$$

A repairable route contains one or more critical artificial e-links whose regeneration delay and resource requirements remain within predefined limits.

Let

$$\mathcal{E}_{\text{crit}}(R_{s,d}(t)) \quad (3.113)$$

denote the set of critical artificial e-links contained in route $R_{s,d}(t)$:

$$\mathcal{E}_{\text{crit}}(R_{s,d}(t)) = \left\{ (i, j) \in E_A(R_{s,d}(t)) : B_{ij}(t) < \delta_{\text{safe}} \vee \hat{F}_{ij}^{\text{cons}}(t) < F_{\text{min}}^{\text{link}} \right\}. \quad (3.114)$$

The predicted route-repair delay is calculated as

$$\hat{D}_{s,d}^{\text{repair}}(t) = \sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R_{s,d}(t))} D_{ij}^{\text{regen}}. \quad (3.115)$$

A route is considered repairable only if the required regeneration operations satisfy both delay and resource constraints:

$$\hat{D}_{s,d}^{\text{repair}}(t) \leq D_{\text{max}}^{\text{repair}}, \quad (3.116)$$

and

$$\sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R_{s,d}(t))} C_{ij}^{\text{regen}} \leq C_{\text{max}}^{\text{repair}}. \quad (3.117)$$

If both conditions are satisfied, the controller may regenerate the critical links and restore route feasibility. Otherwise, an alternative route must be selected or the request must wait for future resource availability.

This temporal route-feasibility model prevents the compact-routing protocol from selecting artificial e-links solely because they remain structurally present in the routing table. Instead, routing decisions are based on predicted future usability, ensuring that selected routes remain reliable throughout the complete entanglement-establishment process.

3.9 Time-Dependent Entangling-Cost Metric

The time-dependent entangling-cost metric quantifies the cost of selecting an artificial e-link according to its current physical and temporal state. Unlike static routing metrics, where an e-link has a fixed cost, the proposed metric dynamically changes with fidelity degradation, entanglement age, remaining coherence budget, regeneration overhead, and structural characteristics.

For an artificial e-link (i, j) at time t , the proposed time-dependent cost is defined as

$$w_t(i, j) = \alpha_F C_F(i, j, t) + \alpha_A C_A(i, j, t) + \alpha_B C_B(i, j, t) + \alpha_R C_R(i, j, t) + \alpha_S C_S(i, j, t). \quad (3.118)$$

where

- $C_F(i, j, t)$ represents fidelity degradation cost;
- $C_A(i, j, t)$ represents entanglement-age cost;
- $C_B(i, j, t)$ represents coherence-budget risk;
- $C_R(i, j, t)$ represents regeneration overhead; and
- $C_S(i, j)$ represents structural routing cost.

The weighting coefficients satisfy

$$\alpha_F, \alpha_A, \alpha_B, \alpha_R, \alpha_S \geq 0, \quad (3.119)$$

with

$$\alpha_F + \alpha_A + \alpha_B + \alpha_R + \alpha_S = 1. \quad (3.120)$$

The fidelity cost is based on the predicted fidelity at consumption:

$$C_F(i, j, t) = 1 - \hat{F}_{ij}^{\text{cons}}(t). \quad (3.121)$$

Therefore, artificial e-links with higher predicted fidelity obtain lower routing cost.

The age-related cost is defined as

$$C_A(i, j, t) = \min \left\{ \frac{A_{ij}(t)}{A_{ij}^{\max}}, 1 \right\}. \quad (3.122)$$

This cost increases as the stored e-link approaches its maximum allowable age.

The coherence-budget risk cost is defined as

$$C_B(i, j, t) = \begin{cases} \frac{1}{\tilde{B}_{ij}(t) + \varepsilon}, & \tilde{B}_{ij}(t) > 0, \\ +\infty, & \tilde{B}_{ij}(t) \leq 0, \end{cases} \quad (3.123)$$

where

$$\tilde{B}_{ij}(t) = \frac{B_{ij}(t)}{T_{2,ij}}. \quad (3.124)$$

A smaller remaining coherence budget therefore produces a larger routing penalty.

The regeneration-related cost is defined as

$$C_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\max}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\max}^{\text{regen}}}, \quad (3.125)$$

where

$$\lambda_D + \lambda_C = 1. \quad (3.126)$$

The first term represents regeneration delay, while the second term represents regeneration resource consumption.

The structural routing cost models the role of the artificial e-link inside the compact overlay:

$$C_S(i, j) = \begin{cases} c_{\text{local}}, & \text{if } (i, j) \text{ is an e-neighbor link,} \\ c_{\text{anchor}}, & \text{if } (i, j) \text{ is an anchor link,} \\ c_{\text{tracked}}, & \text{if } (i, j) \text{ is a tracked long-range link.} \end{cases} \quad (3.127)$$

The structural cost and structural importance are treated separately because they serve different purposes.

The structural routing cost

$$C_S(i, j) \quad (3.128)$$

is used during route selection, whereas the structural importance

$$I_{ij}^{\text{struct}} \quad (3.129)$$

is used during overlay maintenance and regeneration prioritization.

The structural importance may depend on

- the number of source–destination pairs using the e-link;
- whether the link connects anchor ESPs;
- whether link removal affects overlay connectivity;
- the betweenness of the link in the artificial topology; and
- the availability of alternative temporally feasible paths.

For a complete route

$$R_{s,d}(t), \quad (3.130)$$

the total time-dependent entangling cost is

$$W_t(R_{s,d}(t)) = \sum_{(i,j) \in E_A(R_{s,d}(t))} w_t(i,j). \quad (3.131)$$

The routing objective is therefore formulated as selecting the temporally feasible route with minimum total cost:

$$R_{s,d}^*(t) = \arg \min_{R_{s,d}(t) \in \mathcal{R}_{s,d}^{\text{TF}}(t)} W_t(R_{s,d}(t)). \quad (3.132)$$

This formulation converts static compact routing into temporal resource selection. Route quality depends on both topology and evolving entanglement states.

A normalized structural-importance score is defined as

$$I_{ij}^{\text{struct}} = \mu_1 I_{ij}^{\text{anchor}} + \mu_2 I_{ij}^{\text{betweenness}} + \mu_3 I_{ij}^{\text{demand}} + \mu_4 I_{ij}^{\text{scarcity}}, \quad (3.133)$$

where

$$\sum_{k=1}^4 \mu_k = 1, \quad (3.134)$$

and the coefficients μ_k determine the relative contribution of each structural factor.

For a candidate artificial-overlay route

$$R_{s,d}(t), \quad (3.135)$$

the accumulated time-dependent entangling cost is represented using the general composition operator:

$$w_t(R_{s,d}(t)) = \bigoplus_{(i,j) \in E_A(R_{s,d}(t))} w_t(i,j). \quad (3.136)$$

For the additive cost case used in this thesis, the route cost becomes

$$w_t(R_{s,d}(t)) = \sum_{(i,j) \in E_A(R_{s,d}(t))} w_t(i,j). \quad (3.137)$$

The optimal temporally feasible route is selected as

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R), \quad (3.138)$$

where $\mathcal{R}_{s,d}^{\text{TF}}(t)$ denotes the set of routes satisfying all temporal feasibility constraints.

If no fully feasible route exists, the controller evaluates repairable routes. The repair-aware route cost is defined as

$$w_t^{\text{repair}}(R) = w_t(R) + \rho_D \widehat{D}_{s,d}^{\text{repair}}(t) + \rho_C C_{s,d}^{\text{repair}}(t), \quad (3.139)$$

where the additional terms represent regeneration delay and regeneration resource consumption.

The total repair cost is given by

$$C_{s,d}^{\text{repair}}(t) = \sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R)} C_{ij}^{\text{regen}}. \quad (3.140)$$

The minimum-cost repairable route is therefore

$$R_{s,d}^{\text{repair},*}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{repair}}(t)} w_t^{\text{repair}}(R). \quad (3.141)$$

For routing convergence, the proposed metric preserves the non-negative ordering property required by additive shortest-path algorithms. Specifically, every finite link cost satisfies

$$w_t(i,j) \geq 0. \quad (3.142)$$

Artificial e-links that violate temporal feasibility conditions are excluded by assigning

$$w_t(i,j) = +\infty. \quad (3.143)$$

Therefore, stale or expired artificial e-links cannot be selected during temporally feasible route optimization.

The weighting parameters are not assumed to be universally optimal. Instead, their influence is evaluated through sensitivity analysis to determine how fidelity, age, coherence budget, regeneration overhead, and structural factors affect routing performance.

3.10 Temporal Entangling Stretch

Temporal entangling stretch evaluates routing under time-varying artificial entanglement resources. Unlike static stretch, which assumes that all artificial links remain available, temporal entangling stretch considers fidelity degradation, coherence limitation, and regeneration-related effects.

For source ESP v_s and destination ESP v_d , let

$$R_{QR}(s, d, t) \quad (3.144)$$

denote the route selected by the proposed compact quantum-routing protocol at time t .

Let

$$R_t^*(s, d) \quad (3.145)$$

denote the minimum-cost temporally feasible route between v_s and v_d under the same temporal network state.

The temporal entangling stretch is defined as

$$ES_t(s, d) = \frac{w_t(R_{QR}(s, d, t))}{w_t(R_t^*(s, d))}. \quad (3.146)$$

When both routes are temporally feasible and have finite cost,

$$ES_t(s, d) \geq 1. \quad (3.147)$$

The optimal value is achieved when

$$ES_t(s, d) = 1, \quad (3.148)$$

which indicates that the compact-routing protocol selects the minimum-cost temporally feasible route.

A value greater than one indicates that the selected route has higher temporal entangling cost than the optimal feasible route. This additional cost may result from compact-routing restrictions, limited local visibility, anchor constraints, or avoidance of temporally degraded e-links.

The worst-case temporal entangling stretch at time t is defined as

$$ES_t^{\max} = \max_{v_s, v_d \in V_2, v_s \neq v_d} ES_t(s, d). \quad (3.149)$$

The mean temporal stretch over evaluated source–destination pairs is

$$\overline{ES}_t = \frac{1}{|\mathcal{D}(t)|} \sum_{(s,d) \in \mathcal{D}(t)} ES_t(s, d), \quad (3.150)$$

where $\mathcal{D}(t)$ represents the set of source–destination demands with at least one temporally feasible route at time t .

Over a simulation period of T time slots, the time-average temporal stretch is

$$\overline{ES} = \frac{1}{T} \sum_{t=1}^T \overline{ES}_t. \quad (3.151)$$

To evaluate worst-case temporal degradation, the tail stretch is measured using the 95th percentile:

$$ES_{95} = \inf \{x : \Pr [ES_t(s, d) \leq x] \geq 0.95\}. \quad (3.152)$$

The percentile metric captures rare but important cases where temporal degradation causes significant routing overhead.

A structurally reachable route may still be temporally infeasible because one or more artificial e-links may expire before route completion. Therefore, temporal stretch is evaluated only when at least one temporally feasible route exists.

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \quad (3.153)$$

the corresponding demand is classified as temporally disconnected rather than being assigned an infinite stretch value.

This distinction separates routing inefficiency from resource unavailability and provides a more accurate evaluation of temporal-state-aware quantum routing performance. The temporal disconnection rate measures the fraction of source–destination requests for which no temporally feasible route exists during the evaluation period:

$$P_{\text{disc}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} \mathbf{1} [\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset]}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.154)$$

A lower value of P_{disc} indicates that the artificial overlay maintains higher temporal availability under dynamic entanglement conditions.

To quantify the impact of temporal degradation, the proposed temporal stretch is compared with the original static entangling stretch.

Let

$$ES_{\text{static}}(s, d) \quad (3.155)$$

denote the stretch obtained using the original static entangling-cost metric without temporal-state information.

The temporal stretch inflation is defined as

$$\Delta ES_t(s, d) = ES_t(s, d) - ES_{\text{static}}(s, d). \quad (3.156)$$

A positive value indicates that aging, fidelity degradation, coherence limitation, or regeneration constraints increase the effective routing cost compared with the static case.

The normalized temporal-stretch inflation is defined as

$$\Delta ES_t^{\text{rel}}(s, d) = \frac{ES_t(s, d) - ES_{\text{static}}(s, d)}{ES_{\text{static}}(s, d)}. \quad (3.157)$$

The temporal-stretch performance is evaluated separately for Partial-Anchor and Full-Anchor artificial overlays.

For the Partial-Anchor Scheme, the original static compact-routing guarantee is

$$ES_{\text{static}}^{\text{PA}} \leq 5. \quad (3.158)$$

For the Full-Anchor Scheme, the corresponding static guarantee is

$$ES_{\text{static}}^{\text{FA}} \leq 3. \quad (3.159)$$

These structural guarantees assume that the required artificial links remain available and usable. Under temporal degradation, however, the original route may become infeasible due to fidelity loss, coherence expiration, or regeneration delay.

Therefore, the following conditions are not assumed:

$$ES_t^{\text{PA}} \leq 5, \quad (3.160)$$

and

$$ES_t^{\text{FA}} \leq 3. \quad (3.161)$$

Instead, this thesis evaluates the degradation of compact-routing guarantees under realistic temporal conditions by measuring:

- the frequency of static-bound violations;
- the temporal characteristics of links causing violations;
- the recovery capability after e-link regeneration;
- the additional memory and control overhead required for temporal maintenance; and
- the robustness difference between Partial-Anchor and Full-Anchor overlays.

The violation indicators are defined as

$$I_{\text{viol}}^{\text{PA}}(s, d, t) = \mathbf{1} \left[ES_t^{\text{PA}}(s, d) > 5 \right], \quad (3.162)$$

and

$$I_{\text{viol}}^{\text{FA}}(s, d, t) = \mathbf{1} \left[ES_t^{\text{FA}}(s, d) > 3 \right]. \quad (3.163)$$

The average violation probability is then calculated as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.164)$$

This metric quantifies how frequently temporal degradation causes the compact-routing stretch guarantees to be violated. The stretch-bound violation rate is defined as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.165)$$

This metric directly quantifies whether temporal degradation weakens the structural guarantees of compact routing.

The proposed temporal entangling-stretch framework therefore evaluates not only route efficiency, but also the robustness of compact-routing guarantees under realistic entanglement aging, decoherence, fidelity degradation, and regeneration delay.

3.11 Problem Formulation

The routing problem is formulated as the joint selection of temporally feasible artificial-overlay routes and regeneration decisions. The objective is to minimize the time-dependent entangling cost while satisfying fidelity, coherence-budget, memory, exclusivity, regeneration, and compact-routing constraints.

The stretch-bound violation rate is defined as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.166)$$

This metric quantifies whether temporal degradation weakens the structural guarantees of compact routing.

The proposed temporal entangling-stretch framework therefore evaluates not only route efficiency but also the robustness of compact-routing guarantees under realistic entanglement aging, decoherence, and regeneration delay.

Let

$$\mathcal{D}(t) \quad (3.167)$$

denote the set of source–destination entanglement requests arriving at time slot t .

For each request

$$k = (s, d) \in \mathcal{D}(t), \quad (3.168)$$

the routing controller selects an artificial-overlay route

$$R_{s,d}(t) \in \mathcal{R}_{s,d}(t), \quad (3.169)$$

where $\mathcal{R}_{s,d}(t)$ is the set of candidate routes between source ESP v_s and destination ESP v_d .

A binary route-selection variable is defined as

$$y_{s,d}^R(t) = \begin{cases} 1, & \text{if route } R \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.170)$$

At most one route is selected for each demand:

$$\sum_{R \in \mathcal{R}_{s,d}(t)} y_{s,d}^R(t) \leq 1, \quad \forall (s, d) \in \mathcal{D}(t). \quad (3.171)$$

A selected route must satisfy temporal feasibility:

$$y_{s,d}^R(t) \leq \chi_{s,d}(R), \quad (3.172)$$

where $\chi_{s,d}(R)$ is the temporal route-feasibility indicator.

The route-selection objective is

$$\min \sum_{(s,d) \in \mathcal{D}(t)} \sum_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} y_{s,d}^R(t) w_t(R). \quad (3.173)$$

To include route repair and rejection decisions, the objective is extended as

$$\min \left[\sum_{(s,d) \in \mathcal{D}(t)} \sum_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} y_{s,d}^R(t) w_t(R) + \lambda_{\text{rep}} C_{\text{repair}}(t) + \lambda_{\text{rej}} N_{\text{reject}}(t) \right], \quad (3.174)$$

where

- $C_{\text{repair}}(t)$ represents regeneration and overlay repair cost;
- $N_{\text{reject}}(t)$ represents rejected requests;
- λ_{rep} controls repair-cost importance; and
- λ_{rej} controls rejection penalty.

The regeneration cost is

$$C_{\text{repair}}(t) = \sum_{(i,j) \in E_A(t)} r_{ij}(t) C_{ij}^{\text{regen}}, \quad (3.175)$$

where

$$r_{ij}(t) \in \{0, 1\} \quad (3.176)$$

is the regeneration decision variable:

$$r_{ij}(t) = \begin{cases} 1, & \text{if e-link } (i, j) \text{ is regenerated,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.177)$$

The regeneration budget constraint is

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\text{max}}. \quad (3.178)$$

The memory constraint at ESP v_i is

$$\sum_{j: e_{ij}^A(t) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i. \quad (3.179)$$

The end-to-end fidelity constraint is

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\text{min}}^{\text{end}}. \quad (3.180)$$

The coherence-budget constraint is

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall(i, j) \in R_{s,d}(t). \quad (3.181)$$

The route completion delay constraint is

$$\widehat{D}_{s,d}^{\text{route}}(t) \leq D_{\text{max}}. \quad (3.182)$$

The e-link exclusivity constraint ensures that available stored entangled resources are not allocated simultaneously to multiple requests:

$$\sum_{(s,d) \in \mathcal{D}(t)} \sum_{R:(i,j) \in R} y_{s,d}^R(t) \leq n_{ij}^{\text{ebit}}(t). \quad (3.183)$$

The compact-routing constraint requires sublinear routing-table size:

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (3.184)$$

where $\mathcal{T}_i(t)$ represents the routing table of ESP v_i .

The complete optimization problem is summarized as

$$\min_{\mathbf{y}(t), \mathbf{r}(t)} C_{\text{route}}(t) + \lambda_{\text{rep}} C_{\text{repair}}(t) + \lambda_{\text{rej}} N_{\text{reject}}(t) \quad (3.185)$$

subject to

$$\text{route-selection constraints,} \quad (3.186)$$

$$\text{temporal-feasibility constraints,} \quad (3.187)$$

$$\text{fidelity constraints,} \quad (3.188)$$

$$\text{coherence-budget constraints,} \quad (3.189)$$

$$\text{memory constraints,} \quad (3.190)$$

$$\text{e-link exclusivity constraints,} \quad (3.191)$$

$$\text{regeneration-budget constraints,} \quad (3.192)$$

$$\text{compact-routing-table constraints.} \quad (3.193)$$

The complete optimization problem becomes computationally expensive for large networks because route selection, regeneration, resource allocation, and temporal-state evolution are strongly coupled.

Therefore, the following chapter develops a practical heuristic algorithm that preserves compact-routing scalability while using temporal e-link information for adaptive path selection and overlay maintenance.

3.12 Chapter Summary

This chapter presented the system model and formal problem formulation for temporal-state-aware compact routing in quantum-native entanglement overlay networks.

The model separated the physical quantum network from the artificial entanglement overlay. The physical layer represents hardware connectivity, elementary-link generation, quantum memories, swapping operations, and generation delay, while the artificial overlay represents pre-established entangled resources maintained among Entanglement Service Providers.

Each artificial e-link was extended with an explicit temporal state containing availability, age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, regeneration delay, and regeneration cost.

A fidelity-degradation model was introduced to estimate both current and predicted future e-link quality. The coherence-budget model determined whether an artificial link could remain usable until route completion.

A temporal route-feasibility condition was developed by combining link availability, fidelity constraints, coherence requirements, memory limitations, and completion delay.

A time-dependent entangling-cost metric was introduced by combining fidelity cost, age cost, coherence-budget risk, regeneration overhead, and structural routing cost.

Temporal entangling stretch, temporal disconnection rate, stretch inflation, and stretch-bound violation rate were defined to evaluate the robustness of compact-routing schemes under dynamic entanglement conditions.

Finally, the routing and overlay-maintenance problem was formulated as a joint optimization problem involving route selection and regeneration decisions under fidelity, coherence, memory, exclusivity, regeneration-budget, and compact-routing constraints.

The next chapter presents the proposed temporal-state-aware compact-routing algorithm, including temporal-state evaluation, adaptive route selection, coherence-budget-aware regeneration, and overlay maintenance procedures.

Chapter 4

Proposed Temporal-State-Aware Compact Routing Method

4.1 Introduction

This chapter presents the proposed temporal-state-aware compact routing method for quantum-native entanglement overlay networks. The method extends compact quantum routing by incorporating the time-dependent state of stored artificial e-links into route selection and overlay-maintenance decisions.

The proposed framework is motivated by the observation that an artificial e-link may remain structurally present in the routing table while its effective routing utility deteriorates because of aging, decoherence, fidelity degradation, memory waiting time, and delayed consumption. Therefore, a routing decision based only on topological reachability or static entangling cost may select a route that is unlikely to satisfy the required end-to-end fidelity at completion time.

The proposed method addresses this limitation through five coordinated operations:

1. monitoring the temporal state of each stored artificial e-link;
2. constructing compact-routing candidate paths;
3. filtering candidates using temporal feasibility conditions;
4. selecting the minimum-cost temporally feasible route; and

5. regenerating or replacing critical artificial e-links when necessary.

The method operates on the artificial entanglement overlay while using physical-network information to estimate generation delay, regeneration cost, fidelity evolution, and memory consumption.

At each time slot, the controller updates the state of every artificial e-link according to

$$\mathcal{L}_{ij}(t) = [x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}, I_{ij}^{\text{struct}}]. \quad (4.1)$$

Based on this temporal state, the controller classifies each artificial e-link as

$$\text{State}_{ij}(t) \in \{\text{Safe}, \text{Warning}, \text{Critical}, \text{Expired}\}. \quad (4.2)$$

Only safe or admissible warning-state links are directly considered during route construction. Critical links may trigger route repair or proactive regeneration, whereas expired links are removed from the temporally usable overlay.

The proposed routing process is expressed as

$$\text{Monitor} \rightarrow \text{Construct} \rightarrow \text{Filter} \rightarrow \text{Select} \rightarrow \text{Maintain}. \quad (4.3)$$

The method preserves the compact-routing principle by limiting route discovery to local e-neighbors, anchor nodes, and tracked long-range artificial links. It does not require every ESP to maintain complete global routing information.

The main design challenge is therefore to improve temporal reliability without converting the compact-routing architecture into a fully centralized global-state routing system.

This chapter first defines the design objectives and overall framework. It then describes temporal-state monitoring, candidate-route construction, temporal-feasibility filtering, time-dependent route selection, proactive regeneration, overlay maintenance, algorithmic steps, and computational complexity.

4.2 Design Objectives

The proposed method is designed to improve the reliability of compact quantum routing by incorporating temporal information of artificial e-links while preserving the scalability advantages of compact routing. The method does

not require every Entanglement Service Provider to maintain complete global knowledge of the network state.

The main design objectives are described as follows.

4.2.1 Temporal Reliability

The first objective is to prevent the routing protocol from selecting artificial e-links that are structurally available but expected to become unusable before route completion.

For every selected route $R_{s,d}(t)$, the proposed method aims to satisfy the coherence-budget constraint

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in R_{s,d}(t), \quad (4.4)$$

where $B_{ij}(t)$ denotes the remaining coherence budget of artificial e-link (i, j) and δ_{safe} represents the required safety margin.

In addition, the selected route must satisfy the predicted end-to-end fidelity requirement at the expected completion time:

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (4.5)$$

These constraints ensure that routing decisions consider future link usability rather than only the current availability state.

4.2.2 Fidelity Preservation

The second objective is to preserve the quality of distributed entanglement by considering both the current fidelity and the predicted fidelity at the expected consumption time.

The routing method therefore avoids selecting an artificial e-link solely because its instantaneous fidelity satisfies the threshold. Instead, it evaluates whether the link quality remains sufficient after additional waiting, coordination, purification, and swapping operations.

The fidelity-preservation objective can be expressed as

$$\max \widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)). \quad (4.6)$$

This objective is embedded in the proposed time-dependent entangling-cost metric instead of being optimized separately. Therefore, high-fidelity and temporally stable routes receive lower routing costs during route selection.

4.2.3 Compact Routing-State Preservation

The proposed method must preserve the compact-routing property of the underlying quantum-native architecture.

Let $\mathcal{T}_i(t)$ denote the routing table maintained by ESP v_i . The routing-state constraint is defined as

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (4.7)$$

where $n_e = |V_2|$ represents the number of ESPs.

The method therefore relies on local e-neighbors, anchor information, and selected tracked e-links instead of maintaining complete all-pairs routing information.

4.2.4 Low Route-Selection Cost

The method aims to select a temporally feasible route with minimum time-dependent entangling cost.

For each demand (s, d) , the route-selection objective is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.8)$$

The metric $w_t(R)$ considers fidelity degradation, e-link age, coherence-budget risk, regeneration cost, and structural routing cost.

4.2.5 Controlled Regeneration Overhead

Frequent regeneration may improve temporal freshness but can consume quantum-memory capacity, elementary-link generation resources, classical coordination time, and Bell-state measurement resources.

The proposed method therefore limits the number of regeneration operations in each time slot:

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\text{max}}. \quad (4.9)$$

Regeneration is prioritized for artificial e-links with high temporal risk and high structural importance.

4.2.6 Reduced Request Rejection

The routing method aims to reduce entanglement-request rejection caused by stale links, insufficient coherence budget, or fidelity-threshold violations.

Let $N_{\text{reject}}(t)$ denote the number of rejected requests at time t . The long-term request-rejection ratio is

$$P_{\text{reject}} = \frac{\sum_{t=1}^T N_{\text{reject}}(t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (4.10)$$

The proposed method seeks to minimize this ratio while respecting memory and regeneration constraints.

4.2.7 Robustness to Dynamic E-Link Conditions

The artificial overlay may change because of successful generation, consumption, expiration, memory failure, or regeneration.

The method adapts route selection whenever the state of a relevant e-link changes. However, unnecessary route recomputation should be avoided.

A route update is triggered only when at least one of the following conditions occurs:

- a selected e-link becomes critical or expired;
- the predicted end-to-end fidelity falls below the required threshold;
- a lower-cost temporally feasible route becomes available;
- an anchor or tracked e-link is regenerated; or
- the current route cannot satisfy the completion-delay constraint.

4.2.8 Scalable Decision Making

The proposed method should remain computationally practical as the number of ESPs increases.

Instead of solving the complete joint optimization problem at every time slot, the framework separates the decision process into

$$\begin{aligned} \text{State Update} &\rightarrow \text{Candidate Construction} \\ &\rightarrow \text{Feasibility Filtering} \rightarrow \text{Cost Based Selection.} \end{aligned} \quad (4.11)$$

This decomposition reduces the number of candidate routes requiring detailed temporal evaluation.

The overall design objective can be summarized as

$$\min \left[\lambda_1 \bar{w} + \lambda_2 P_{\text{reject}} + \lambda_3 C_{\text{regen}} + \lambda_4 \overline{ES} \right], \quad (4.12)$$

subject to

$$\hat{F}_{s,d}^{\text{end}} \geq F_{\min}^{\text{end}}, \quad (4.13)$$

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad (4.14)$$

$$m_i(t) \leq M_i, \quad (4.15)$$

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (4.16)$$

$$\sum_{(i,j)} r_{ij}(t) \leq R_{\max}. \quad (4.17)$$

Here, $\bar{w}$ is the average selected-route cost, C_{regen} is the total regeneration overhead, and $\overline{ES}$ is the average temporal entangling stretch.

4.3 Overview of the Proposed Framework

The proposed framework integrates compact routing, temporal e-link monitoring, fidelity prediction, coherence-budget evaluation, and selective overlay maintenance.

The framework operates in discrete time slots. At the beginning of each time slot, the controller updates the state of every stored artificial e-link and constructs the temporally usable overlay.

The complete processing sequence is represented as

$$G_A(t) \rightarrow \mathcal{L}(t) \rightarrow G_U(t) \rightarrow \mathcal{R}_{s,d}^{\text{cand}}(t) \rightarrow \mathcal{R}_{s,d}^{\text{TF}}(t) \rightarrow R_{s,d}^*(t). \quad (4.18)$$

Here,

- $G_A(t)$ is the currently stored artificial overlay;
- $\mathcal{L}(t)$ is the collection of temporal states of artificial e-links;
- $G_U(t)$ is the temporally usable overlay;
- $\mathcal{R}_{s,d}^{\text{cand}}(t)$ is the candidate-route set;
- $\mathcal{R}_{s,d}^{\text{TF}}(t)$ is the temporally feasible route set; and
- $R_{s,d}^*(t)$ is the selected minimum-cost route.

The framework contains two interacting control processes:

- Reactive routing;

- Proactive overlay maintenance.

Reactive routing is executed when an entanglement request arrives. It constructs candidate compact routes, evaluates their temporal conditions, removes infeasible routes, and selects the lowest-cost temporally feasible route.

Proactive overlay maintenance is executed periodically or when a critical state change occurs. It identifies artificial e-links that should be regenerated, replaced, or discarded before they cause routing failure.

The framework receives the following inputs:

$$\mathcal{I}(t) = [G_P, G_A(t), \mathcal{D}(t), \mathcal{T}(t), \mathcal{M}(t), \mathcal{Q}(t)], \quad (4.19)$$

where

- G_P is the physical quantum network;
- $G_A(t)$ is the current artificial overlay;
- $\mathcal{D}(t)$ is the set of active entanglement requests;
- $\mathcal{T}(t)$ is the compact-routing-table state;
- $\mathcal{M}(t)$ is the quantum-memory state; and
- $\mathcal{Q}(t)$ is the observed or predicted network-quality state.

The output of the framework is

$$\mathcal{O}(t) = [\mathcal{Y}(t), \mathcal{R}^{\text{regen}}(t), \mathcal{E}^{\text{drop}}(t), \mathcal{T}(t+1)], \quad (4.20)$$

where

- $\mathcal{Y}(t)$ is the set of selected routes;
- $\mathcal{R}^{\text{regen}}(t)$ is the regeneration-action set;
- $\mathcal{E}^{\text{drop}}(t)$ is the set of discarded e-links; and
- $\mathcal{T}(t+1)$ is the updated compact-routing-table state.

The proposed framework is divided into six functional modules.

4.3.1 Temporal-State Monitoring Module

This module updates the age, effective fidelity, predicted consumption fidelity, remaining coherence budget, and stale-link state of every stored artificial e-link.

Its output is the temporally usable artificial overlay

$$G_U(t) = (V_2, E_U(t)). \quad (4.21)$$

4.3.2 Compact Candidate-Construction Module

This module constructs a limited number of candidate routes using only the routing information permitted by the Partial-Anchor or Full-Anchor compact-routing scheme.

The candidate-route set generated by this module is represented as

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.22)$$

This restriction preserves compact-routing state requirements and avoids unrestricted global path enumeration.

4.3.3 Temporal-Feasibility Module

This module evaluates each candidate route according to the temporal route-feasibility indicator

$$\chi_{s,d}(R_{s,d}(t)). \quad (4.23)$$

Routes that violate fidelity, coherence-budget, memory, delay, or e-link availability constraints are removed.

The remaining route set is

$$\mathcal{R}_{s,d}^{\text{TF}}(t) \subseteq \mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.24)$$

4.3.4 Time-Dependent Route-Selection Module

For each temporally feasible route, this module calculates the accumulated time-dependent entangling cost:

$$w_t(R) = \sum_{(i,j) \in R} w_t(i,j). \quad (4.25)$$

The selected route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.26)$$

4.3.5 Route-Repair Module

When no directly feasible route exists, this module identifies routes that can become feasible through limited regeneration or purification operations.

The route-repair process is performed only when

$$\widehat{D}_{s,d}^{\text{repair}}(t) \leq D_{\max}^{\text{repair}} \quad (4.27)$$

and

$$C_{s,d}^{\text{repair}}(t) \leq C_{\max}^{\text{repair}}. \quad (4.28)$$

Otherwise, the request is delayed or rejected.

4.3.6 Overlay-Maintenance Module

This module assigns regeneration priority to critical artificial e-links according to temporal risk and structural importance.

The maintenance priority is defined as

$$\Pi_{ij}(t) = \beta_1 C_B(i, j, t) + \beta_2 C_F(i, j, t) + \beta_3 I_{ij}^{\text{struct}} + \beta_4 H_{ij}^{\text{use}}(t), \quad (4.29)$$

where $H_{ij}^{\text{use}}(t)$ represents the recent or predicted utilization level of the artificial e-link.

The weights satisfy

$$\sum_{k=1}^4 \beta_k = 1. \quad (4.30)$$

The highest-priority links are regenerated subject to quantum-memory availability, generation-attempt budget, and control-overhead limitations.

The framework is intentionally modular. Different fidelity models, delay predictors, compact-routing structures, or regeneration policies can be integrated without changing the overall processing sequence.

The proposed method is a model-based routing and control heuristic. It does not require a trained machine-learning model. Any prediction component used in the implementation is derived from measured averages, analytical estimates, or simulation parameters.

4.4 Temporal E-Link State Monitoring

The temporal e-link state-monitoring module continuously evaluates the condition of every artificial e-link stored in the quantum memories of the Entanglement Service Providers.

Its purpose is to distinguish between an artificial e-link that is merely present in the overlay and an e-link that remains sufficiently reliable for future routing and entanglement-swapping operations.

Let the set of currently stored artificial e-links be

$$E_A(t). \quad (4.31)$$

For each artificial e-link $(i, j) \in E_A(t)$, the controller maintains the temporal state record

$$\mathcal{L}_{ij}(t) = [x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, \hat{F}_{ij}^{\text{cons}}(t), B_{ij}(t), H_{ij}^{\text{use}}(t), S_{ij}^{\text{state}}(t)], \quad (4.32)$$

where

- $x_{ij}(t)$ is the current storage and availability indicator;
- $A_{ij}(t)$ is the current e-link age;
- $F_{ij}^{A,0}$ is the initial fidelity after e-link generation;
- $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity;
- $T_{2,ij}$ is the coherence time of the stored entanglement;
- $\hat{F}_{ij}^{\text{cons}}(t)$ is the predicted fidelity at consumption time;
- $B_{ij}(t)$ is the remaining coherence budget;
- $H_{ij}^{\text{use}}(t)$ is the recent or predicted utilization score; and
- $S_{ij}^{\text{state}}(t)$ is the temporal-state classification.

At the end of each time slot, the age of an active e-link is updated as

$$A_{ij}(t+1) = A_{ij}(t) + 1, \quad (4.33)$$

provided that the link remains stored and is neither consumed nor regenerated.

When regeneration succeeds, the age is reset:

$$A_{ij}(t+1) = 0. \quad (4.34)$$

The current effective fidelity is estimated using the selected decoherence model:

$$F_{ij}^{\text{eff}}(t) = f_{\text{dec}} \left(F_{ij}^{A,0}, A_{ij}(t), T_{2,ij} \right). \quad (4.35)$$

The controller also estimates the additional delay before the link is expected to be consumed:

$$\widehat{D}_{ij}^{\text{use}}(t). \quad (4.36)$$

This delay may be estimated from recent queueing observations, route depth, expected swapping order, and classical coordination delay.

The predicted fidelity at consumption time is calculated as

$$\widehat{F}_{ij}^{\text{cons}}(t) = f_{\text{dec}} \left(F_{ij}^{\text{eff}}(t), \widehat{D}_{ij}^{\text{use}}(t), T_{2,ij} \right). \quad (4.37)$$

The remaining coherence budget is updated according to

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t)\Delta - \widehat{D}_{ij}^{\text{use}}(t). \quad (4.38)$$

4.4.1 Temporal Classification and Link Availability

To reduce sensitivity to short-term fluctuations, the utilization score is updated using an exponential moving average:

$$H_{ij}^{\text{use}}(t) = \gamma_H H_{ij}^{\text{use}}(t-1) + (1 - \gamma_H) I_{ij}^{\text{used}}(t), \quad (4.39)$$

where

$$0 \leq \gamma_H < 1 \quad (4.40)$$

and

$$I_{ij}^{\text{used}}(t) = \begin{cases} 1, & \text{if the e-link is selected or consumed at time } t, \\ 0, & \text{otherwise.} \end{cases} \quad (4.41)$$

The temporal state of an artificial e-link is classified according to predicted consumption fidelity and remaining coherence budget.

A link is classified as safe when

$$B_{ij}(t) \geq B_{\text{high}} \quad (4.42)$$

and

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\text{high}}. \quad (4.43)$$

A link is classified as warning when it remains usable but approaches temporal degradation:

$$B_{\text{low}} \leq B_{ij}(t) < B_{\text{high}} \quad (4.44)$$

or

$$F_{\text{min}}^{\text{link}} \leq \hat{F}_{ij}^{\text{cons}}(t) < F_{\text{high}}. \quad (4.45)$$

A link is classified as critical when it is still potentially recoverable but has a high risk of becoming unusable:

$$0 < B_{ij}(t) < B_{\text{low}} \quad (4.46)$$

or

$$\hat{F}_{ij}^{\text{cons}}(t) \approx F_{\text{min}}^{\text{link}}. \quad (4.47)$$

Critical links are not directly preferred for route selection and may trigger proactive regeneration or purification.

A link is classified as expired when

$$B_{ij}(t) \leq 0 \quad (4.48)$$

or

$$x_{ij}(t) = 0. \quad (4.49)$$

The resulting temporal-state classification is

$$S_{ij}^{\text{state}}(t) = \begin{cases} \text{Safe,} & \text{if high-quality conditions are satisfied,} \\ \text{Warning,} & \text{if the link remains usable with degradation risk,} \\ \text{Critical,} & \text{if recovery actions may restore usability,} \\ \text{Expired,} & \text{if the link is unusable.} \end{cases} \quad (4.50)$$

The direct-use indicator is defined as

$$u_{ij}^{\text{direct}}(t) = \begin{cases} 1, & S_{ij}^{\text{state}}(t) \in \{\text{Safe, Warning}\}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.51)$$

The regeneration-candidate indicator is defined as

$$u_{ij}^{\text{regen}}(t) = \begin{cases} 1, & S_{ij}^{\text{state}}(t) = \text{Critical}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.52)$$

Expired links are removed from route construction and may be discarded from memory:

$$u_{ij}^{\text{drop}}(t) = \mathbf{1} [S_{ij}^{\text{state}}(t) = \text{Expired}]. \quad (4.53)$$

The temporally usable edge set is therefore

$$E_U(t) = \{(i, j) \in E_A(t) : u_{ij}^{\text{direct}}(t) = 1\}. \quad (4.54)$$

The temporal-state monitor operates periodically. Let

$$\tau_{\text{mon}} \quad (4.55)$$

denote the monitoring interval.

A shorter monitoring interval improves responsiveness but increases control and state-update overhead. A longer interval reduces overhead but may allow rapidly degrading links to remain incorrectly classified.

Therefore, the monitoring interval is evaluated experimentally under different coherence times, request rates, and overlay sizes.

The output of this module is a synchronized temporal-state table used by candidate-route construction, temporal-feasibility filtering, route selection, and proactive regeneration.

4.5 Candidate Route Construction

The candidate-route construction module generates a limited set of compact-routing paths between a source ESP and a destination ESP. Its purpose is to avoid unrestricted enumeration of all possible paths while preserving the structural properties of the underlying Partial-Anchor or Full-Anchor routing scheme.

For an entanglement request between source ESP v_s and destination ESP v_d , the candidate-route set at time t is denoted by

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.56)$$

Candidate routes are constructed over the currently available compact-overlay information. The module uses the temporally usable overlay

$$G_U(t) = (V_2, E_U(t)), \quad (4.57)$$

where $E_U(t)$ contains artificial e-links classified as safe or warning.

The candidate-generation process first checks whether the source and destination are directly connected by a usable artificial e-link. If

$$(v_s, v_d) \in E_U(t), \quad (4.58)$$

then the direct route

$$R_{s,d}^{\text{dir}}(t) = (v_s, v_d) \quad (4.59)$$

is inserted into the candidate set.

If no direct usable e-link exists, the controller searches through local e-neighbors, anchor ESPs, and tracked long-range links according to the selected compact-routing architecture.

Let

$$\mathcal{N}_U(v_i, t) \quad (4.60)$$

denote the usable e-neighborhood of ESP v_i :

$$\mathcal{N}_U(v_i, t) = \{v_j \in V_2 : (v_i, v_j) \in E_U(t)\}. \quad (4.61)$$

A one-intermediate-node candidate route has the form

$$R_{s,d}^{(1)}(t) = (v_s, v_k, v_d), \quad (4.62)$$

where

$$v_k \in \mathcal{N}_U(v_s, t) \cap \mathcal{N}_U(v_d, t). \quad (4.63)$$

For longer compact routes, the controller follows the routing structure of the selected overlay scheme. In the Partial-Anchor scheme, route construction is restricted by local e-neighborhood information and selected anchor links. In the Full-Anchor scheme, additional tracked long-range links are considered during candidate generation.

4.5.1 Candidate Construction in the Partial-Anchor Scheme

In the Partial-Anchor Scheme, each ESP maintains local e-neighbor information, while a subset of ESPs acts as anchors.

Let

$$\mathcal{T} \subseteq V_2 \quad (4.64)$$

denote the anchor set.

The usable source-anchor set is

$$\mathcal{T}_s^U(t) = \{v_a \in \mathcal{T} : v_a \in \mathcal{N}_U(v_s, t)\}. \quad (4.65)$$

Similarly, the usable destination-anchor set is

$$\mathcal{T}_d^U(t) = \{v_b \in \mathcal{T} : v_b \in \mathcal{N}_U(v_d, t)\}. \quad (4.66)$$

A Partial-Anchor candidate route can be written as

$$R_{s,d}^{\text{PA}}(t) = (v_s, v_a, \dots, v_b, v_d), \quad (4.67)$$

where

$$v_a \in \mathcal{T}_s^U(t) \quad (4.68)$$

and

$$v_b \in \mathcal{T}_d^U(t). \quad (4.69)$$

The internal anchor segment is obtained from the currently usable anchor overlay.

Let

$$G_{\text{anchor}}^U(t) \quad (4.70)$$

denote the usable subgraph formed by anchor ESPs and usable anchor e-links.

A bounded number of low-cost anchor paths is generated between v_a and v_b .

4.5.2 Full-Anchor Candidate Construction

In the Full-Anchor Scheme, every ESP may maintain local e-links and selected tracked long-range e-links.

Let

$$E_{\text{tracked}}^U(t) \quad (4.71)$$

denote the set of currently usable tracked long-range artificial e-links.

The Full-Anchor candidate set is constructed using

$$E_{\text{local}}^{\text{U}}(t) \cup E_{\text{tracked}}^{\text{U}}(t). \quad (4.72)$$

A candidate route is constructed by combining local source segments, tracked long-range segments, and local destination segments.

A general candidate route is represented by

$$R_{s,d}^{\text{FA}}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d), \quad (4.73)$$

where every consecutive node pair corresponds to a usable local or tracked artificial e-link.

4.5.3 Bounded Candidate-Set Size

To control computational overhead, the controller retains at most

$$K_{\text{max}} \quad (4.74)$$

candidate routes for each source–destination request.

Therefore,

$$|\mathcal{R}_{s,d}^{\text{cand}}(t)| \leq K_{\text{max}}. \quad (4.75)$$

Candidate routes are initially ranked using a lightweight preliminary score

$$\tilde{w}_t(R) = \sum_{(i,j) \in R} \tilde{w}_t(i,j), \quad (4.76)$$

where $\tilde{w}_t(i,j)$ includes only readily available local information such as hop count, current fidelity, normalized age, and direct-use state.

A possible preliminary link score is

$$\tilde{w}_t(i,j) = \eta_h + \eta_F (1 - F_{ij}^{\text{eff}}(t)) + \eta_A \frac{A_{ij}(t)}{A_{ij}^{\text{max}}}. \quad (4.77)$$

The coefficients satisfy

$$\eta_h + \eta_F + \eta_A = 1. \quad (4.78)$$

This preliminary score is used only for candidate pruning. Final route selection uses the complete time-dependent entangling-cost metric.

To avoid routing loops, every candidate route must contain unique nodes:

$$v_m \neq v_n, \quad \forall m \neq n. \quad (4.79)$$

A maximum overlay-hop constraint is also applied:

$$|R| - 1 \leq H_{\max}, \quad (4.80)$$

where $H_{\max}$ is the maximum number of artificial e-links allowed in a candidate route.

The final candidate set is

$$\mathcal{R}_{s,d}^{\text{cand}}(t) = \text{TopK}_{K_{\max}} [\tilde{w}_t(R)], \quad (4.81)$$

subject to compact-routing, loop-free, direct-use, and maximum-hop constraints.

The output of this module is a small set of structurally valid candidate routes. These routes are passed to the temporal-feasibility module, where predicted consumption fidelity, remaining coherence budget, delay, memory availability, and end-to-end fidelity are evaluated in detail.

4.5.4 Availability Test

The first test verifies that every required artificial e-link is currently available for direct use:

$$u_{ij}^{\text{direct}}(t) = 1, \quad \forall (i, j) \in E(R). \quad (4.82)$$

The route-level availability indicator is

$$\chi_{\text{avail}}(R, t) = \prod_{(i,j) \in E(R)} u_{ij}^{\text{direct}}(t). \quad (4.83)$$

If

$$\chi_{\text{avail}}(R, t) = 0, \quad (4.84)$$

the route is removed from the directly feasible set.

4.5.5 Predicted Consumption-Time Calculation

Therefore, the temporal feasibility evaluation is performed individually for each e-link based on its predicted consumption time. Therefore, the controller estimates the consumption time of each e-link according to the expected swapping schedule.

Let

$$\hat{\tau}_{ij}^{\text{cons}}(R, t) \quad (4.85)$$

denote the estimated additional time before e-link (i, j) is consumed as part of route R .

A general estimate is

$$\widehat{\tau}_{ij}^{\text{cons}}(R, t) = \widehat{D}_{\text{queue}}(R, t) + \widehat{D}_{\text{coord}}(R, t) + \widehat{D}_{ij}^{\text{sched}}(R, t), \quad (4.86)$$

where

- $\widehat{D}_{\text{queue}}(R, t)$ is the expected request waiting delay;
- $\widehat{D}_{\text{coord}}(R, t)$ is the classical coordination delay; and
- $\widehat{D}_{ij}^{\text{sched}}(R, t)$ is the delay introduced by the selected swapping schedule.

For a simple sequential swapping schedule, the estimated scheduling delay can be approximated as

$$\widehat{D}_{ij}^{\text{sched}}(R, t) = \ell_{ij}(R) D_{\text{swap}}, \quad (4.87)$$

where $\ell_{ij}(R)$ is the number of swapping stages completed before e-link (i, j) is consumed.

For a parallel or nested swapping schedule, this value may be smaller and is determined by the swapping depth of the route.

4.5.6 Link-Level Fidelity Test

The predicted fidelity of e-link (i, j) at its estimated consumption time is

$$\widehat{F}_{ij}^{\text{cons}}(R, t) = f_{\text{dec}} \left(F_{ij}^{\text{eff}}(t), \widehat{\tau}_{ij}^{\text{cons}}(R, t), T_{2,ij} \right). \quad (4.88)$$

The route passes the link-level fidelity test only if

$$\widehat{F}_{ij}^{\text{cons}}(R, t) \geq F_{\min}^{\text{link}}, \quad \forall (i, j) \in E(R). \quad (4.89)$$

The corresponding indicator is

$$\chi_{\text{linkF}}(R, t) = \prod_{(i,j) \in E(R)} \mathbf{1} \left[\widehat{F}_{ij}^{\text{cons}}(R, t) \geq F_{\min}^{\text{link}} \right]. \quad (4.90)$$

4.5.7 End-to-End Fidelity Test

Using the predicted consumption-time fidelities, the controller estimates the end-to-end route fidelity.

Under the Werner-state approximation, the predicted link parameter is

$$\hat{p}_{ij}^W(R, t) = \frac{4\hat{F}_{ij}^{\text{cons}}(R, t) - 1}{3}. \quad (4.91)$$

The predicted end-to-end Werner parameter is

$$\hat{p}_{s,d}^{W,\text{end}}(R, t) = \left(\prod_{(i,j) \in E(R)} \hat{p}_{ij}^W(R, t) \right) \left(\prod_{v_r \in \text{Int}(R)} \lambda_r \right), \quad (4.92)$$

where $\text{Int}(R)$ denotes the set of intermediate ESPs participating in entanglement swapping along route R .

The predicted end-to-end fidelity is

$$\hat{F}_{s,d}^{\text{end}}(R, t) = \frac{1 + 3\hat{p}_{s,d}^{W,\text{end}}(R, t)}{4}. \quad (4.93)$$

The route passes the end-to-end fidelity test when

$$\hat{F}_{s,d}^{\text{end}}(R, t) \geq F_{\min}^{\text{end}}. \quad (4.94)$$

The associated indicator is

$$\chi_{\text{endF}}(R, t) = \mathbf{1} \left[\hat{F}_{s,d}^{\text{end}}(R, t) \geq F_{\min}^{\text{end}} \right]. \quad (4.95)$$

4.5.8 Route-Delay Test

The predicted route-completion delay is denoted by

$$\widehat{D}_{s,d}^{\text{route}}(R, t). \quad (4.96)$$

A general decomposition is

$$\widehat{D}_{s,d}^{\text{route}}(R, t) = \widehat{D}_{\text{queue}}(R, t) + \widehat{D}_{\text{coord}}(R, t) + \widehat{D}_{\text{pur}}(R, t) + \widehat{D}_{\text{swap}}(R, t). \quad (4.97)$$

The estimated delay depends on the route because candidates may differ in swapping operations, purification requirements, and queueing conditions.

The route satisfies the delay constraint only if

$$\widehat{D}_{s,d}^{\text{route}}(R, t) \leq D_{\max}. \quad (4.98)$$

The route-delay indicator is

$$\chi_{\text{delay}}(R, t) = \mathbf{1} \left[\widehat{D}_{s,d}^{\text{route}}(R, t) \leq D_{\max} \right]. \quad (4.99)$$

4.5.9 Memory and E-Bit Availability Test

The number of stored e-bits on each artificial e-link must be sufficient to satisfy the entanglement request.

For a single-pair request, the link-level resource condition is

$$n_{ij}^{\text{ebit}}(t) \geq 1, \quad \forall (i, j) \in E(R). \quad (4.100)$$

For a request requiring $b_{s,d}$ entangled pairs, where $b_{s,d}$ denotes the required number of entangled pairs between source and destination, the condition becomes

$$n_{ij}^{\text{ebit}}(t) \geq b_{s,d}, \quad \forall (i, j) \in E(R). \quad (4.101)$$

The route-level resource indicator is

$$\chi_{\text{mem}}(R, t) = \prod_{(i,j) \in E(R)} \mathbf{1} \left[n_{ij}^{\text{ebit}}(t) \geq b_{s,d} \right]. \quad (4.102)$$

4.5.10 Overall Temporal-Feasibility Decision

The overall temporal-feasibility indicator is defined as

$$\chi_{s,d}(R, t) = \chi_{\text{avail}}(R, t) \chi_{\text{linkF}}(R, t) \chi_{\text{coh}}(R, t) \chi_{\text{endF}}(R, t) \chi_{\text{delay}}(R, t) \chi_{\text{mem}}(R, t). \quad (4.103)$$

A candidate route is directly feasible when

$$\chi_{s,d}(R, t) = 1. \quad (4.104)$$

The temporally feasible route set is therefore

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \left\{ R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \chi_{s,d}(R, t) = 1 \right\}. \quad (4.105)$$

Candidate routes that fail the direct-feasibility test are not immediately discarded if their infeasibility can be corrected through limited regeneration or repair operations.

Such routes are transferred to the route-repair set

$$\mathcal{R}_{s,d}^{\text{repair}}(t). \quad (4.106)$$

A route enters this set only if its predicted repair delay and repair cost satisfy

$$\widehat{D}_{s,d}^{\text{repair}}(R, t) \leq D_{\max}^{\text{repair}} \quad (4.107)$$

and

$$C_{s,d}^{\text{repair}}(R, t) \leq C_{\max}^{\text{repair}}. \quad (4.108)$$

This two-stage filtering process distinguishes among immediately feasible, repairable, and infeasible routes. It prevents the route-selection module from assigning low cost to a path that is topologically available but temporally incapable of satisfying the entanglement request.

4.6 Time-Dependent Route Selection

The time-dependent route-selection module chooses the best route from the set of temporally feasible candidates.

For an entanglement request between source ESP v_s and destination ESP v_d , the feasible candidate set is

$$\mathcal{R}_{s,d}^{\text{TF}}(t). \quad (4.109)$$

For each candidate route

$$R \in \mathcal{R}_{s,d}^{\text{TF}}(t), \quad (4.110)$$

the controller evaluates the complete time-dependent route cost.

The route cost is defined as

$$w_t(R) = \sum_{(i,j) \in E(R)} w_t(i, j) + C_{\text{route}}^{\text{swap}}(R, t) + C_{\text{route}}^{\text{delay}}(R, t), \quad (4.111)$$

where

- $w_t(i, j)$ is the time-dependent cost of artificial e-link (i, j) ;
- $C_{\text{route}}^{\text{swap}}(R, t)$ is the route-level swapping cost; and
- $C_{\text{route}}^{\text{delay}}(R, t)$ is the normalized completion-delay cost.

The link-level time-dependent cost is

$$w_t(i, j) = \alpha_F C_F(i, j, R, t) + \alpha_A C_A(i, j, t) + \alpha_B C_B(i, j, R, t) + \alpha_R C_R(i, j, t) + \alpha_S C_S(i, j). \quad (4.112)$$

The coefficients satisfy

$$\alpha_F + \alpha_A + \alpha_B + \alpha_R + \alpha_S = 1. \quad (4.113)$$

The fidelity cost is defined as

$$C_F(i, j, R, t) = 1 - \widehat{F}_{ij}^{\text{cons}}(R, t). \quad (4.114)$$

This term penalizes e-links that are expected to have low fidelity when they are consumed.

The age cost is

$$C_A(i, j, t) = \min \left\{ \frac{A_{ij}(t)}{A_{ij}^{\max}}, 1 \right\}. \quad (4.115)$$

The normalized coherence-budget cost is defined as

$$C_B(i, j, R, t) = \frac{1}{\widetilde{B}_{ij}(R, t) + \varepsilon}, \quad (4.116)$$

where

$$\widetilde{B}_{ij}(R, t) = \frac{B_{ij}(R, t)}{T_{2,ij}}. \quad (4.117)$$

Since only temporally feasible routes reach this module, links with non-positive coherence budget are already removed before route selection:

$$\widetilde{B}_{ij}(R, t) > 0. \quad (4.118)$$

The normalized regeneration-risk cost is

$$C_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\max}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\max}^{\text{regen}}}, \quad (4.119)$$

where

$$\lambda_D + \lambda_C = 1. \quad (4.120)$$

The structural routing cost is represented by

$$C_S(i, j). \quad (4.121)$$

This component distinguishes local e-neighbor links, anchor links, and tracked long-range links according to their resource and maintenance requirements.

4.6.1 Route-Level Cost Components

The swapping cost of route R is defined as

$$C_{\text{route}}^{\text{swap}}(R, t) = \alpha_Q \sum_{v_r \in \text{Int}(R)} (1 - q_r), \quad (4.122)$$

where q_r is the swapping-success probability at intermediate ESP v_r .

The normalized route-delay cost is

$$C_{\text{route}}^{\text{delay}}(R, t) = \alpha_D \frac{\widehat{D}_{s,d}^{\text{route}}(R, t)}{D_{\max}}. \quad (4.123)$$

The selected directly feasible route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.124)$$

When multiple routes have the same minimum cost, the controller applies the following tie-breaking order:

1. select the route with the highest predicted end-to-end fidelity;
2. select the route with the largest minimum coherence budget;
3. select the route with the smallest predicted completion delay;
4. select the route with the fewest artificial e-links.

The minimum route-level coherence budget is

$$B_{\min}(R, t) = \min_{(i,j) \in E(R)} B_{ij}(R, t). \quad (4.125)$$

The route-selection decision can therefore be represented lexicographically as

$$\min \left[w_t(R), -\widehat{F}_{s,d}^{\text{end}}(R, t), -B_{\min}(R, t), \widehat{D}_{s,d}^{\text{route}}(R, t), |E(R)| \right]. \quad (4.126)$$

After selecting a route, the required e-bits are temporarily reserved before swapping begins.

For a request requiring $b_{s,d}$ entangled pairs, the reservation variable is

$$z_{ij}^{s,d}(t) = \begin{cases} b_{s,d}, & \text{if } (i,j) \text{ belongs to the selected route,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.127)$$

The reservation constraint is

$$\sum_{(s,d) \in \mathcal{D}(t)} z_{ij}^{s,d}(t) \leq n_{ij}^{\text{ebit}}(t). \quad (4.128)$$

This reservation prevents the same stored e-bit from being assigned to multiple requests.

After successful end-to-end swapping, the consumed e-bits are removed:

$$n_{ij}^{\text{ebit}}(t+1) = n_{ij}^{\text{ebit}}(t) - z_{ij}^{s,d}(t). \quad (4.129)$$

If a swapping operation fails, all e-bits already consumed by the failed route are treated as lost.

The route outcome is represented by

$$Y_{s,d}(t) = \begin{cases} 1, & \text{if end-to-end entanglement is established,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.130)$$

The realized route success probability is estimated as

$$P_{\text{succ}}(R, t) = \left(\prod_{(i,j) \in E(R)} p_{ij}^{\text{avail}}(t) \right) \left(\prod_{v_r \in \text{Int}(R)} q_r \right), \quad (4.131)$$

where $p_{ij}^{\text{avail}}(t)$ represents the probability that artificial e-link (i, j) remains usable until its scheduled consumption time.

A reliability-aware extension of the route cost is

$$w_t^{\text{rel}}(R) = w_t(R) + \alpha_P [-\log P_{\text{succ}}(R, t)]. \quad (4.132)$$

This logarithmic transformation converts the multiplicative success probability into an additive routing penalty.

The route selected under the reliability-aware metric is

$$R_{s,d}^{*,\text{rel}}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t^{\text{rel}}(R). \quad (4.133)$$

The reliability-aware metric provides an optional extension of the basic temporal entangling-cost metric. Its effectiveness will be evaluated through comparative experiments.

When

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \quad (4.134)$$

the module does not select a directly feasible route.

Instead, it forwards the request and the repairable candidate set

$$\mathcal{R}_{s,d}^{\text{repair}}(t) \quad (4.135)$$

to the proactive regeneration and route-repair process described in the following section.

The normalized fidelity risk is defined as

$$\hat{C}_F(i, j, t) = \min \left\{ \frac{F_{\text{high}} - \hat{F}_{ij}^{\text{cons}}(t)}{F_{\text{high}} - F_{\text{min}}^{\text{link}} + \varepsilon}, 1 \right\}. \quad (4.136)$$

The predicted demand relevance may be estimated from the number of active or expected routes that can use the e-link:

$$Q_{ij}^{\text{demand}}(t) = \frac{N_{ij}^{\text{demand}}(t)}{\max_{(u,v)} N_{uv}^{\text{demand}}(t) + \varepsilon}. \quad (4.137)$$

The normalized regeneration cost is

$$\hat{C}_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\text{max}}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\text{max}}^{\text{regen}}}, \quad (4.138)$$

where

$$\lambda_D + \lambda_C = 1. \quad (4.139)$$

A link becomes a proactive-regeneration candidate when

$$\Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\text{min}} \quad (4.140)$$

and sufficient regeneration resources are available.

The candidate set is

$$\mathcal{E}_{\text{regen}}^{\text{cand}}(t) = \left\{ (i, j) \in \mathcal{E}_{\text{crit}}(t) : \Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\text{min}} \right\}. \quad (4.141)$$

The regeneration-decision variable is

$$r_{ij}(t) \in \{0, 1\}. \quad (4.142)$$

It is defined as

$$r_{ij}(t) = \begin{cases} 1, & \text{if e-link } (i, j) \text{ is selected for regeneration,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.143)$$

The controller selects at most $R_{\max}$ highest-priority regeneration candidates:

$$\mathcal{E}_{\text{regen}}^*(t) = \text{TopK} \left(\mathcal{E}_{\text{regen}}^{\text{cand}}(t), R_{\max} \right). \quad (4.144)$$

The selected regeneration actions must satisfy the regeneration budget constraint:

$$|\mathcal{E}_{\text{regen}}^*(t)| \leq R_{\max}. \quad (4.145)$$

Let $m_i(t)$ denote the currently occupied quantum-memory capacity at ESP v_i . The regeneration decision must satisfy

$$m_i(t) + \sum_{j:(i,j) \in \mathcal{E}_{\text{regen}}^*(t)} r_{ij}(t) \leq M_i, \quad \forall v_i \in V_2. \quad (4.146)$$

They must also satisfy the physical-generation capacity constraint:

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) g_{ij}^{\text{req}} \leq G_{\max}, \quad (4.147)$$

where g_{ij}^{req} represents the expected number of elementary-link generation attempts required for regenerating artificial e-link (i, j) , and $G_{\max}$ denotes the available generation-attempt budget.

For an artificial e-link generated through physical path P_{ij}^{gen} , the expected regeneration delay is

$$D_{ij}^{\text{regen}} = D^{\text{phys}}(P_{ij}^{\text{gen}}) + D_{ij}^{\text{swap}} + D_{ij}^{\text{coord}}. \quad (4.148)$$

When purification is required, the regeneration delay becomes

$$D_{ij}^{\text{regen,pur}} = D_{ij}^{\text{regen}} + D_{ij}^{\text{pur}}. \quad (4.149)$$

The regeneration process may follow either a replace-after-success strategy or a discard-before-generation strategy.

Under the replace-after-success strategy, the old artificial e-link remains available until the new e-link is successfully generated. After successful regeneration,

$$F_{ij}^{\text{eff}}(t+1) = F_{ij}^{\text{A},0}, \quad (4.150)$$

$$A_{ij}(t+1) = 0, \quad (4.151)$$

and

$$B_{ij}(t+1) = T_{2,ij} - \widehat{D}_{ij}^{\text{use}}(t+1). \quad (4.152)$$

This strategy reduces temporary disconnection risk but requires additional memory capacity because both old and newly generated e-links may coexist during the transition period.

Under the discard-before-generation strategy, the old e-link is released before regeneration. This reduces memory consumption but temporarily removes the link from the usable overlay.

The replacement-policy variable is defined as

$$\rho_{ij}^{\text{replace}} \in \{0, 1\}, \quad (4.153)$$

where

$$\rho_{ij}^{\text{replace}} = 1 \quad (4.154)$$

indicates that the replace-after-success strategy is selected.

The policy is determined according to available memory capacity:

$$\rho_{ij}^{\text{replace}} = \mathbf{1} [M_i - m_i(t) \geq 1 \wedge M_j - m_j(t) \geq 1]. \quad (4.155)$$

Regeneration is considered successful only when the required elementary-link generation, entanglement swapping, and coordination operations are completed successfully.

Let

$$Z_{ij}^{\text{regen}}(t) \in \{0, 1\} \quad (4.156)$$

denote the regeneration outcome of artificial e-link (i, j) at time slot t .

The regeneration success probability is approximated as

$$P[Z_{ij}^{\text{regen}}(t) = 1] = P_{ij}^{\text{regen}} = P_{\text{succ}}(P_{ij}^{\text{gen}}). \quad (4.157)$$

If regeneration fails, the e-link remains in its previous state under the replace-after-success strategy. Under the discard-before-generation strategy, the e-link remains unavailable until a subsequent regeneration attempt succeeds.

Reactive route repair uses the same regeneration mechanism but assigns additional priority according to the blocked request.

For a repairable route R , the repair priority of a critical e-link is increased by

$$\Pi_{ij}^{\text{repair}}(R, t) = \Pi_{ij}^{\text{regen}}(t) + \beta_{\text{req}} I_{ij}^R, \quad (4.158)$$

where

$$I_{ij}^R = \mathbf{1}[(i, j) \in \mathcal{E}_{\text{crit}}(R)]. \quad (4.159)$$

This request-specific term provides higher priority to links whose regeneration can restore a currently infeasible route.

A regeneration action is approved only when its expected benefit exceeds its expected cost.

The expected regeneration utility is defined as

$$U_{ij}^{\text{regen}}(t) = P_{ij}^{\text{regen}} G_{ij}^{\text{future}}(t) - \kappa_D D_{ij}^{\text{regen}} - \kappa_C C_{ij}^{\text{regen}}, \quad (4.160)$$

where $G_{ij}^{\text{future}}(t)$ represents the predicted future routing benefit obtained by refreshing artificial e-link (i, j) .

Regeneration is executed only if

$$U_{ij}^{\text{regen}}(t) > 0. \quad (4.161)$$

This condition prevents unnecessary regeneration of low-impact e-links whose expected routing benefit does not justify the required generation resources and delay.

The proposed proactive-regeneration mechanism therefore balances temporal freshness, fidelity preservation, structural importance, future demand, memory availability, physical-generation capacity, and regeneration overhead.

4.7 Overlay Maintenance Strategy

The overlay-maintenance strategy updates the artificial entanglement overlay according to temporal degradation, routing demand, memory availability, and compact-routing constraints.

Its purpose is not to keep every artificial e-link permanently fresh. Instead, the strategy maintains a bounded set of useful e-links whose expected routing benefit justifies their generation and storage cost.

At time slot t , the stored artificial overlay is

$$G_A(t) = (V_2, E_A(t)). \quad (4.162)$$

The overlay is updated through four principal actions:

$$\text{Retain,} \quad \text{Regenerate,} \quad \text{Replace,} \quad \text{Discard.} \quad (4.163)$$

For each artificial e-link (i, j) , the maintenance action is denoted by

$$a_{ij}^{\text{maint}}(t) \in \{\text{retain, regenerate, replace, discard}\}. \quad (4.164)$$

4.7.1 Retention Decision

An e-link is retained without regeneration when it remains temporally reliable and sufficiently useful.

The retention indicator is

$$k_{ij}^{\text{retain}}(t) = \mathbf{1} \left[S_{ij}^{\text{state}}(t) \in \{\text{Safe, Warning}\} \right]. \quad (4.165)$$

A warning-state link may be retained only if its predicted consumption fidelity and remaining coherence budget satisfy the minimum operational margins:

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}} \quad (4.166)$$

and

$$B_{ij}(t) \geq \delta_{\text{safe}}. \quad (4.167)$$

4.7.2 Regeneration Decision

A critical e-link is considered for regeneration when its expected future routing benefit is high and its regeneration cost is acceptable.

The regeneration decision follows the previously defined variable:

$$r_{ij}(t) = \mathbf{1} \left[\Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\min} \right], \quad (4.168)$$

subject to memory, physical-generation, and per-slot regeneration constraints.

The total number of regeneration actions satisfies

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\max}. \quad (4.169)$$

4.7.3 Replacement Decision

Replacement changes the endpoint pair or physical realization of an artificial e-link when repeated regeneration becomes inefficient or when the existing link has limited routing value.

Let

$$h_{ij}^{\text{fail}}(t) \quad (4.170)$$

denote the recent number of failed regeneration attempts for e-link (i, j) . The link becomes a replacement candidate when

$$h_{ij}^{\text{fail}}(t) \geq H_{\text{max}}^{\text{fail}}, \quad (4.171)$$

or when

$$U_{ij}^{\text{regen}}(t) \leq 0. \quad (4.172)$$

A replacement candidate may also be identified when its structural importance is low while another non-stored e-link provides higher predicted benefit.

Let

$$\mathcal{E}_{\text{new}}^{\text{cand}}(t) \quad (4.173)$$

denote the set of candidate artificial e-links that are not currently stored. For a candidate new e-link (u, v) , the predicted placement utility is defined as

$$U_{uv}^{\text{place}}(t) = \theta_D Q_{uv}^{\text{demand}}(t) + \theta_S I_{uv}^{\text{struct}} + \theta_C G_{uv}^{\text{connect}}(t) - \theta_R \hat{C}_R(u, v, t). \quad (4.174)$$

The weights satisfy

$$\theta_D + \theta_S + \theta_C + \theta_R = 1. \quad (4.175)$$

A stored e-link (i, j) may be replaced by candidate (u, v) when

$$U_{uv}^{\text{place}}(t) - U_{ij}^{\text{retain}}(t) \geq \Delta U_{\text{min}}, \quad (4.176)$$

where $U_{ij}^{\text{retain}}(t)$ represents the predicted utility of keeping the existing artificial e-link without replacement.

The threshold ΔU_{min} prevents frequent overlay changes caused by small utility fluctuations.

4.7.4 Discard Decision

An artificial e-link is discarded when it becomes expired, repeatedly unused, structurally redundant, or too expensive to maintain.

The discard indicator is

$$d_{ij}(t) \in \{0, 1\}. \quad (4.177)$$

It is defined as

$$d_{ij}(t) = \mathbf{1} \left[S_{ij}^{\text{state}}(t) = \text{Expired} \right]. \quad (4.178)$$

A non-expired link may also be discarded when the following redundancy conditions are simultaneously satisfied:

$$H_{ij}^{\text{use}}(t) < H_{\min}^{\text{use}}, \quad (4.179)$$

$$I_{ij}^{\text{struct}} < I_{\min}^{\text{struct}}, \quad (4.180)$$

and

$$U_{ij}^{\text{regen}}(t) \leq 0. \quad (4.181)$$

An e-link is retained if discarding it would violate an overlay-connectivity requirement.

4.7.5 Compact-Overlay Constraints

For every ESP v_i , the number of maintained routing-table entries is bounded by

$$|\mathcal{T}_i(t)| \leq T_i^{\max}, \quad (4.182)$$

where

$$T_i^{\max} = c_{\text{tab}} \sqrt{n_e} \log n_e. \quad (4.183)$$

This constraint preserves the compact-routing property by preventing each ESP from maintaining complete global routing information.

In the Partial-Anchor Scheme, the required local and anchor connectivity structures must be preserved:

$$E_{\text{local}}^{\text{req}} \subseteq E_A^{\text{PA}}(t). \quad (4.184)$$

In addition, the anchor overlay subgraph should remain connected whenever sufficient physical entanglement resources are available.

In the Full-Anchor Scheme, each ESP maintains a bounded tracked-link set:

$$|\mathcal{L}_i^{\text{track}}(t)| \leq L_{\text{max}}^{\text{track}}. \quad (4.185)$$

New tracked links are admitted only when available routing-table capacity exists or when an existing tracked link is removed.

4.7.6 Memory-Aware Maintenance

The overlay-maintenance strategy must satisfy the quantum-memory constraint

$$\sum_{j:(i,j) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i, \quad \forall v_i \in V_2. \quad (4.186)$$

When memory capacity becomes insufficient, the controller releases artificial e-links according to a discard-priority score.

The discard-priority score is defined as

$$\Pi_{ij}^{\text{drop}}(t) = \phi_1 (1 - H_{ij}^{\text{use}}(t)) + \phi_2 (1 - I_{ij}^{\text{struct}}) + \phi_3 \hat{C}_R(i, j, t) + \phi_4 I_{ij}^{\text{expired}}(t), \quad (4.187)$$

where

$$I_{ij}^{\text{expired}}(t) = \mathbf{1} [S_{ij}^{\text{state}}(t) = \text{Expired}]. \quad (4.188)$$

Links with higher discard priority are released first while preserving required compact-overlay connectivity.

4.7.7 Maintenance Triggering

Overlay maintenance may be executed periodically or through event-driven triggers.

Periodic maintenance is performed every

$$\tau_{\text{maint}} \quad (4.189)$$

time slots.

Event-driven maintenance is triggered when one of the following conditions occurs:

$$S_{ij}^{\text{state}}(t) = \text{Critical} \quad (4.190)$$

for a structurally important e-link,

$$S_{ij}^{\text{state}}(t) = \text{Expired}, \quad (4.191)$$

$$m_i(t) \geq M_i, \quad (4.192)$$

or

$$P_{\text{disc}}(t) \geq P_{\text{disc}}^{\text{max}}. \quad (4.193)$$

To prevent excessive overlay reconfiguration, a minimum holding interval

$$\tau_{\text{hold}} \quad (4.194)$$

is enforced between two structural changes affecting the same routing-table entry.

4.7.8 Overlay-State Update

Let

$$E_{\text{retain}}(t), \quad E_{\text{regen}}^{\text{succ}}(t), \quad E_{\text{new}}^{\text{succ}}(t), \quad E_{\text{drop}}(t) \quad (4.195)$$

denote the sets of retained, successfully regenerated, newly established, and discarded artificial e-links, respectively.

The updated artificial overlay edge set is defined as

$$E_A(t+1) = \left(E_{\text{retain}}(t) \cup E_{\text{regen}}^{\text{succ}}(t) \cup E_{\text{new}}^{\text{succ}}(t) \right) \setminus E_{\text{drop}}(t). \quad (4.196)$$

The corresponding routing tables are updated only after successful entanglement generation or regeneration is confirmed through classical heralding information.

Therefore, a routing entry is not advertised as usable until the corresponding artificial e-link has been successfully established and its initial temporal state has been recorded.

The proposed maintenance strategy balances temporal reliability and resource efficiency by preserving useful links, selectively regenerating critical links, replacing inefficient links, and removing stale or low-value links while maintaining the compact structure of the artificial overlay.

4.8 Proposed Routing Algorithm

The proposed routing algorithm integrates temporal-state monitoring, compact candidate construction, temporal-feasibility filtering, time-dependent route selection, and selective artificial e-link regeneration.

The algorithm is executed at each time slot t and processes the active entanglement-request set

$$\mathcal{D}(t). \quad (4.197)$$

The main inputs are

$$G_P, \quad G_A(t), \quad \mathcal{D}(t), \quad \mathcal{T}(t), \quad \mathcal{M}(t), \quad (4.198)$$

together with fidelity, coherence, delay, memory, and regeneration parameters.

The main outputs are

$$\mathcal{Y}(t), \quad \mathcal{E}_{\text{regen}}^*(t), \quad \mathcal{E}_{\text{drop}}(t), \quad G_A(t+1), \quad (4.199)$$

where $\mathcal{Y}(t)$ contains selected routes and their execution outcomes.

The complete procedure is described as follows.

4.8.1 Algorithmic Procedure

1. Update temporal e-link states.

For every artificial e-link $(i, j) \in E_A(t)$, update

$$A_{ij}(t), \quad F_{ij}^{\text{eff}}(t), \quad \widehat{F}_{ij}^{\text{cons}}(t), \quad B_{ij}(t), \quad H_{ij}^{\text{use}}(t). \quad (4.200)$$

Each e-link is classified as safe, warning, critical, or expired according to the temporal-state classification rules.

2. Construct the temporally usable overlay.

The usable edge set is constructed as

$$E_U(t) = \left\{ (i, j) \in E_A(t) : S_{ij}^{\text{state}}(t) \in \{\text{Safe}, \text{Warning}\} \right\}. \quad (4.201)$$

The temporally usable overlay is then defined as

$$G_U(t) = (V_2, E_U(t)). \quad (4.202)$$

3. Process each entanglement request.

For every request

$$(s, d) \in \mathcal{D}(t), \quad (4.203)$$

the controller generates a compact candidate-route set

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.204)$$

The candidate set contains no more than $K_{\max}$ routes.

4. Evaluate direct temporal feasibility.

For every candidate route R , calculate

$$\chi_{\text{avail}}(R, t), \quad \chi_{\text{linkF}}(R, t), \quad \chi_{\text{coh}}(R, t), \quad (4.205)$$

and

$$\chi_{\text{endF}}(R, t), \quad \chi_{\text{delay}}(R, t), \quad \chi_{\text{mem}}(R, t). \quad (4.206)$$

The overall feasibility indicator is

$$\begin{aligned} \chi_{s,d}(R, t) &= \chi_{\text{avail}}(R, t) \chi_{\text{linkF}}(R, t) \chi_{\text{coh}}(R, t) \\ &\quad \times \chi_{\text{endF}}(R, t) \chi_{\text{delay}}(R, t) \chi_{\text{mem}}(R, t). \end{aligned} \quad (4.207)$$

5. Build the directly feasible route set.

The temporally feasible route set is defined as

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \left\{ R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \chi_{s,d}(R, t) = 1 \right\}. \quad (4.208)$$

6. Select a directly feasible route.

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) \neq \emptyset, \quad (4.209)$$

the controller calculates the complete time-dependent cost

$$W_t(R) \quad (4.210)$$

for every feasible route.

The selected route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} W_t(R). \quad (4.211)$$

The required e-bits are reserved and the planned swapping operations are executed.

7. Construct the repairable route set.

If no route is directly feasible, candidates with a bounded number of regenerable critical links are considered.

The repairable route set is defined as

$$\mathcal{R}_{s,d}^{\text{repair}}(t) = \left\{ R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \widehat{D}_{s,d}^{\text{repair}}(R, t) \leq D_{\max}^{\text{repair}}, \right. \\ \left. C_{s,d}^{\text{repair}}(R, t) \leq C_{\max}^{\text{repair}} \right\}. \quad (4.212)$$

8. Select a repairable route.

If

$$\mathcal{R}_{s,d}^{\text{repair}}(t) \neq \emptyset, \quad (4.213)$$

the repair-aware cost is calculated as

$$W_t^{\text{repair}}(R) = W_t(R) + \rho_D \widehat{D}_{s,d}^{\text{repair}}(R, t) + \rho_C C_{s,d}^{\text{repair}}(R, t). \quad (4.214)$$

The selected repairable route is

$$R_{s,d}^{\text{repair},*}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{repair}}(t)} W_t^{\text{repair}}(R). \quad (4.215)$$

The critical e-links of this route are added to the request-specific regeneration queue.

9. Reject or postpone infeasible requests.

If both

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset \quad (4.216)$$

and

$$\mathcal{R}_{s,d}^{\text{repair}}(t) = \emptyset, \quad (4.217)$$

the request is classified as temporarily infeasible.

Depending on the request deadline, the controller either postpones or rejects the request.

10. Calculate proactive regeneration priorities.

For every critical artificial e-link, calculate

$$\Pi_{ij}^{\text{regen}}(t). \quad (4.218)$$

Request-specific critical links receive an additional repair-priority term.

11. Select regeneration actions.

Regeneration candidates are ranked according to priority and expected utility.

The selected regeneration set is

$$\mathcal{E}_{\text{regen}}^*(t), \quad (4.219)$$

subject to

$$|\mathcal{E}_{\text{regen}}^*(t)| \leq R_{\text{max}}, \quad (4.220)$$

memory constraints, and physical-generation capacity constraints.

12. Execute overlay-maintenance actions.

Useful links are retained, selected critical links are regenerated, inefficient links are replaced, and expired or low-value links are discarded.

13. Update the artificial overlay.

After receiving generation and swapping outcomes, update

$$E_A(t+1) = \left(E_{\text{retain}}(t) \cup E_{\text{regen}}^{\text{succ}}(t) \cup E_{\text{new}}^{\text{succ}}(t) \right) \setminus E_{\text{drop}}(t). \quad (4.221)$$

14. Update compact routing tables.

A routing-table entry is added only after the corresponding artificial e-link is successfully established and confirmed through classical communication.

Entries associated with consumed, discarded, or permanently unavailable e-links are removed.

4.8.2 Algorithm Characteristics

The proposed algorithm is heuristic rather than globally optimal. It does not solve the complete joint optimization problem involving route selection, regeneration scheduling, and overlay maintenance at every time slot.

Instead, the computational cost is reduced through

- compact routing-state restriction;
- bounded candidate-route generation;
- early temporal-feasibility filtering;
- additive time-dependent route-cost evaluation;
- bounded regeneration operations; and
- threshold-based overlay maintenance.

For a fixed network state, parameter configuration, candidate ordering, and tie-breaking rule, the algorithm produces deterministic routing decisions.

Randomness appears only from physical entanglement generation processes, decoherence behavior, and probabilistic swapping operations.

The performance of the proposed method is evaluated by comparison with static compact routing, fidelity-only routing, age-only routing, shortest-path routing, and regeneration-disabled variants.

4.9 Computational Complexity Analysis

The computational complexity of the proposed method consists of temporal-state monitoring, candidate-route construction, temporal-feasibility evaluation, route-cost calculation, and regeneration-priority computation.

Let

$$n = |V_2| \tag{4.222}$$

denote the number of Entanglement Service Providers (ESPs), and let

$$m_A(t) = |E_A(t)| \quad (4.223)$$

denote the number of stored artificial e-links at time slot t .

The number of active entanglement requests at time t is

$$D_t = |\mathcal{D}(t)|. \quad (4.224)$$

The maximum number of candidate routes generated for each request is denoted by

$$K_{\max}, \quad (4.225)$$

while the maximum number of artificial e-links contained in one candidate route is denoted by

$$H_{\max}. \quad (4.226)$$

4.9.1 Temporal-State Monitoring Complexity

The temporal-state monitoring module updates the age, effective fidelity, predicted consumption fidelity, remaining coherence budget, utilization score, and state classification of every stored artificial e-link.

Since each update requires a constant number of operations, the monitoring complexity is

$$O(m_A(t)). \quad (4.227)$$

The memory requirement for storing temporal-state information is also

$$O(m_A(t)), \quad (4.228)$$

because one temporal-state record is maintained for each artificial e-link.

4.9.2 Candidate-Route Construction Complexity

Candidate-route construction is performed on the temporally usable compact overlay rather than the complete physical network.

When Dijkstra's algorithm or a bounded K -shortest-path method is applied to the usable overlay, the shortest-path computation using adjacency lists and a binary heap has complexity

$$O(m_U(t) \log n), \quad (4.229)$$

where

$$m_U(t) = |E_U(t)|. \quad (4.230)$$

For generating at most $K_{\max}$ candidate routes, a conservative upper bound is

$$O(K_{\max} m_U(t) \log n). \quad (4.231)$$

In practice, the computational cost is lower because candidate generation is restricted by compact routing tables, anchor structures, tracked long-range links, and maximum-hop constraints.

If every ESP maintains at most

$$T_{\max} = O(\sqrt{n} \log n) \quad (4.232)$$

routing entries, the local search space remains significantly smaller than unrestricted global path enumeration.

4.9.3 Temporal-Feasibility Filtering Complexity

For each candidate route, the temporal-feasibility module evaluates every artificial e-link contained in the route.

Since a candidate route contains at most $H_{\max}$ artificial e-links, the availability, fidelity, coherence-budget, and resource-availability checks require

$$O(H_{\max}) \quad (4.233)$$

computational complexity per candidate route.

The end-to-end fidelity estimation also requires a product operation over at most $H_{\max}$ link parameters and swapping-retention factors. Therefore, its complexity is

$$O(H_{\max}). \quad (4.234)$$

For at most $K_{\max}$ candidate routes, the total feasibility-filtering complexity for one request is

$$O(K_{\max} H_{\max}). \quad (4.235)$$

For all entanglement requests in time slot t , the total complexity becomes

$$O(D_t K_{\max} H_{\max}). \quad (4.236)$$

4.9.4 Time-Dependent Route-Selection Complexity

After feasibility filtering, let

$$K_{\text{TF}} \leq K_{\text{max}} \quad (4.237)$$

denote the number of temporally feasible candidate routes.

The complete time-dependent cost evaluation of one route requires examining at most H_{max} artificial e-links. Therefore, evaluating all feasible routes requires

$$O(K_{\text{TF}}H_{\text{max}}) \quad (4.238)$$

computational complexity per request.

Selecting the minimum-cost route requires

$$O(K_{\text{TF}}) \quad (4.239)$$

additional comparisons.

Because

$$K_{\text{TF}} \leq K_{\text{max}}, \quad (4.240)$$

the overall route-selection complexity is bounded by

$$O(K_{\text{max}}H_{\text{max}}). \quad (4.241)$$

4.9.5 Route-Repair Complexity

When no directly feasible route is available, the algorithm evaluates the repairable candidate-route set.

For each candidate route, the controller identifies critical artificial e-links and estimates the corresponding repair delay and regeneration cost.

Since each route contains at most H_{max} artificial e-links, the repair evaluation requires

$$O(H_{\text{max}}) \quad (4.242)$$

computational complexity per candidate route.

Therefore, the total repair-evaluation complexity per request is

$$O(K_{\text{max}}H_{\text{max}}). \quad (4.243)$$

This repair procedure is activated only when the directly feasible route set is empty, and therefore does not introduce additional overhead during normal routing operation.

4.9.6 Regeneration-Priority Complexity

The proactive regeneration module evaluates all currently critical artificial e-links.

Let

$$m_{\text{crit}}(t) = |\mathcal{E}_{\text{crit}}(t)| \quad (4.244)$$

denote the number of critical e-links at time t .

The calculation of regeneration-priority and utility scores requires constant computation per critical e-link. Therefore, the scoring complexity is

$$O(m_{\text{crit}}(t)). \quad (4.245)$$

Selecting the highest-priority R_{max} regeneration candidates through complete sorting requires

$$O(m_{\text{crit}}(t) \log m_{\text{crit}}(t)). \quad (4.246)$$

Alternatively, maintaining a bounded priority queue with maximum size R_{max} reduces the selection complexity to

$$O(m_{\text{crit}}(t) \log R_{\text{max}}). \quad (4.247)$$

Since

$$R_{\text{max}} \ll m_{\text{crit}}(t), \quad (4.248)$$

the bounded-priority-queue implementation is preferred for large-scale quantum networks.

4.9.7 Overlay-Maintenance Complexity

The retain, regenerate, replace, and discard decisions are evaluated over the currently stored artificial e-link set.

Therefore, the basic maintenance-state evaluation requires

$$O(m_A(t)). \quad (4.249)$$

If replacement candidates are selected from a bounded candidate set of size

$$m_{\text{new}}(t), \quad (4.250)$$

the ranking complexity becomes

$$O(m_{\text{new}}(t) \log m_{\text{new}}(t)). \quad (4.251)$$

In the proposed implementation, replacement evaluation is restricted to a bounded candidate set instead of considering all possible ESP pairs. This limitation preserves the scalability of the compact-overlay architecture.

4.9.8 Total Per-Slot Complexity

Combining temporal-state monitoring, candidate construction, feasibility filtering, route selection, and regeneration operations, the overall computational complexity in time slot t is

$$O(m_A(t) + D_t K_{\max} m_U(t) \log n + D_t K_{\max} H_{\max} + m_{\text{crit}}(t) \log R_{\max}). \quad (4.252)$$

Generating candidate routes from existing compact routing structures avoids a complete graph search for each request and reduces the route-construction cost.

In this case, the dominant computational complexity becomes

$$O(m_A(t) + D_t K_{\max} H_{\max} + m_{\text{crit}}(t) \log R_{\max}). \quad (4.253)$$

With bounded $K_{\max}$ and $H_{\max}$, the processing complexity scales approximately linearly with the number of active requests.

4.9.9 Space Complexity

The artificial-overlay graph requires

$$O(n + m_A(t)) \quad (4.254)$$

storage space.

The temporal-state information maintained for each artificial e-link requires additional

$$O(m_A(t)) \quad (4.255)$$

storage.

The temporary storage required for candidate routes of all active requests is bounded by

$$O(D_t K_{\max} H_{\max}). \quad (4.256)$$

Therefore, the overall space complexity is

$$O(n + m_A(t) + D_t K_{\max} H_{\max}). \quad (4.257)$$

The compact-routing-table constraint prevents each ESP from maintaining complete global routing information. Instead of requiring $O(n)$ routing state, the proposed architecture maintains the target per-ESP routing-state complexity of

$$O(\sqrt{n} \log n). \quad (4.258)$$

Therefore, the proposed algorithm sacrifices global optimality in exchange for bounded candidate evaluation, compact routing-state requirements, and scalable temporal-state processing.

4.10 Chapter Summary

This chapter presented the proposed temporal-state-aware compact routing and overlay-maintenance framework for quantum-native entanglement networks.

The proposed method extends compact quantum routing by incorporating the temporal evolution of stored artificial e-links into routing decisions. The main objectives were to improve temporal reliability, preserve entanglement fidelity, maintain compact routing-state requirements, control regeneration overhead, reduce request rejection, and achieve scalable decision making.

The proposed framework consists of two interacting processes: reactive routing and proactive overlay maintenance. Reactive routing handles incoming entanglement requests by constructing a bounded candidate-route set, evaluating temporal feasibility, and selecting the minimum-cost feasible route. Proactive maintenance continuously evaluates stored artificial e-links and determines whether they should be retained, regenerated, replaced, or discarded.

A temporal-state monitoring mechanism was introduced to update e-link age, effective fidelity, predicted consumption fidelity, remaining coherence budget, utilization history, and temporal classification. Based on these factors, artificial e-links are classified into safe, warning, critical, and expired states.

The candidate-route construction mechanism preserves compact routing by limiting route discovery to local e-neighbors, anchor information, and tracked long-range links. The number of candidate routes and maximum route length are explicitly bounded to avoid unrestricted path exploration.

The temporal-feasibility filtering module evaluates candidate routes using link availability, predicted fidelity, coherence budget, end-to-end fidelity, completion delay, and e-bit resource availability. This process separates directly feasible routes from repairable and infeasible routes.

A time-dependent route-selection metric was developed by combining temporal e-link quality, fidelity degradation, age, coherence risk, regeneration cost, swapping reliability, and route delay. Additional reliability-aware extensions were introduced using predicted route-success probability.

The proactive regeneration mechanism prioritizes critical artificial e-links according to temporal risk, structural importance, utilization history, predicted demand, and regeneration overhead. Regeneration decisions are constrained by quantum-memory capacity, physical-generation capability, and per-slot regeneration limits.

The overlay-maintenance strategy provides a controlled mechanism for preserving useful links, refreshing degraded links, replacing inefficient links, and removing obsolete links while maintaining compact routing constraints.

The complete routing procedure was summarized as a model-based heuristic algorithm. It does not attempt to solve the globally optimal joint routing and maintenance problem at every time slot. Instead, it achieves practical scalability through bounded candidate generation, temporal filtering, and limited regeneration operations.

Finally, the computational complexity analysis demonstrated that the proposed framework depends mainly on the number of stored artificial e-links, active requests, bounded candidate routes, maximum route length, and critical-link population. With fixed candidate and route-length limits, the request-processing complexity scales approximately linearly with the number of active requests.

The next chapter presents the simulation environment, implementation details, parameter configuration, baseline algorithms, evaluation metrics, and experimental scenarios used to validate the proposed method.

Chapter 5

Simulation Methodology and Experimental Setup

5.1 Introduction

This chapter describes the corrected simulation methodology used to evaluate the proposed temporal-state-aware quantum-routing method. The implementation follows the system model and routing constraints defined in Chapters 3 and 4. In particular, the simulator explicitly represents physical links, stored entanglement links, memory decoherence, predicted consumption-time fidelity, remaining coherence budget, request-level quality-of-service requirements, probabilistic entanglement generation, swapping outcomes, and budget-limited regeneration.

All reported experiments use matched random inputs across the compared methods. For a given seed, the methods receive the same physical topology, initial link parameters, traffic requests, entanglement-generation outcomes, and request-service random numbers. This paired design reduces confounding variation and permits seed-wise statistical comparison.

5.2 Simulation Environment

The simulator was implemented in Python 3.13.13. Network topology construction and path computation were implemented using NetworkX. NumPy and pandas were used for numerical processing and tabular analysis, Matplotlib was used for figure generation, and SciPy was used for paired statistical tests.

The implementation was divided into the following modules:

- `topology.py`: spatial physical-network generation;

- `physical_links.py`: physical and quantum-link parameters;
- `overlay.py`: physical and artificial entanglement overlay;
- `temporal_state.py`: age, decoherence, and coherence-budget updates;
- `routing.py`: composite routing cost and route-feasibility checks;
- `traffic.py`: Poisson traffic and heterogeneous request classes;
- `entanglement_generation.py`: continuous probabilistic ebit generation;
- `regeneration.py`: budget-limited probabilistic regeneration;
- `request_service.py`: request outcomes and resource consumption;
- `metrics.py`: end-to-end fidelity and performance metrics;
- `baselines.py`: matched benchmark-routing implementations.

The source code and experiment runners were maintained under Git version control. The corrected implementation was developed on the `thesis-aligned-simulation` branch so that the preliminary prototype remained preserved.

5.3 Spatial Physical-Network Model

5.3.1 Node Placement

For a network containing N nodes, each node was placed uniformly in a two-dimensional $50 \text{ km} \times 50 \text{ km}$ region. If node i has coordinate (x_i, y_i) , the physical distance between nodes i and j is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (5.1)$$

Unlike the preliminary prototype, link distance was not sampled independently after topology generation. The Euclidean distance determined by the node coordinates was preserved throughout the simulation.

5.3.2 Connectivity and Local Links

A minimum spanning tree over the complete distance-weighted graph was first constructed to guarantee physical connectivity. Each node was then connected to its three nearest neighbors. Duplicate undirected edges were automatically removed. Consequently, every generated physical network was connected while remaining sparse.

No artificial quantum shortcut was inserted directly into the physical graph. Physical connectivity and stored-entanglement connectivity were maintained as separate layers.

5.4 Physical-Link Parameters

For every physical edge (i, j) , the elementary entanglement-generation probability was modeled as

$$p_{ij}^{\text{gen}} = p_0 \exp(-\alpha d_{ij}), \quad (5.2)$$

where $p_0 = 0.90$ and $\alpha = 0.02 \text{ km}^{-1}$. The expected generation delay was

$$D_{ij}^{\text{gen}} = \frac{\tau_{\text{attempt}}}{\max(p_{ij}^{\text{gen}}, \varepsilon)}, \quad (5.3)$$

where $\tau_{\text{attempt}} = 0.01 \text{ s}$ and $\varepsilon = 10^{-9}$.

The initial elementary-link fidelity was sampled uniformly from $[0.90, 0.99]$. Unless coherence time was the controlled independent variable, the memory coherence time was sampled uniformly from $[0.5, 5.0] \text{ s}$. The default link-memory capacity was ten ebits.

The swapping-success probability was set to 0.95, the operational error rate to 0.01, the per-swap delay to 0.002 s, and the classical coordination delay to 0.005 s.

5.5 Entanglement Overlay Construction

5.5.1 Physical Overlay Edges

Each physical edge was copied into the entanglement overlay with a valid two-node physical path $[i, j]$. A physical overlay edge initially stored five ebits. Therefore, with a memory capacity of ten, its initial free-memory value was five.

5.5.2 Artificial Entanglement Shortcuts

Candidate artificial edges were selected only between node pairs that were not directly connected in the physical graph. For each candidate, the underlying minimum-distance physical path was computed. Thus, every artificial edge was backed by a valid sequence of physical edges.

The number of artificial shortcuts was

$$M_A = \min(|\mathcal{C}|, \text{round}(0.35N)), \quad (5.4)$$

where $\mathcal{C}$ is the candidate set. Candidates with larger physical hop count and distance were considered first.

For a physical path containing h elementary edges, the artificial-link generation probability was

$$p_{ij}^A = \left(\prod_{\ell=1}^h p_{\ell}^{\text{gen}} \right) p_{\text{swap}}^{h-1}. \quad (5.5)$$

The artificial-link delay included the elementary generation delays, swapping delays, and classical coordination delay. Its coherence time and memory capacity were bounded by the minimum values along the supporting physical path.

The end-to-end initial fidelity was obtained through Werner-state visibility composition. For elementary fidelity F_{ℓ} , the corresponding visibility is

$$v_{\ell} = \max \left(0, \min \left(1, \frac{4F_{\ell} - 1}{3} \right) \right). \quad (5.6)$$

The artificial-link fidelity was calculated as

$$F_{ij}^{A,0} = \frac{1}{4} + \frac{3}{4} \left(\prod_{\ell=1}^h v_{\ell} \right) (1 - e_{\text{op}})^{h-1}. \quad (5.7)$$

Each artificial edge initially stored one ebit, leaving nine free memory units under the default capacity. Resource conservation was enforced as

$$n_{ij}^{\text{ebit}} + m_{ij}^{\text{free}} = m_{ij}^{\text{cap}}. \quad (5.8)$$

5.6 Temporal-State Evolution

The slot duration was $\Delta = 0.1$ s. At each slot, the ebit age was increased and effective fidelity was updated using the Werner-state memory-decay model

$$F_{ij}^{\text{eff}}(t) = \frac{1}{4} + \left(F_{ij}^0 - \frac{1}{4}\right) \exp\left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}}\right). \quad (5.9)$$

The fidelity predicted at the expected consumption time was

$$\hat{F}_{ij}^{\text{cons}}(t) = \frac{1}{4} + \left(F_{ij}^{\text{eff}}(t) - \frac{1}{4}\right) \exp\left(-\frac{\widehat{D}_{ij}^{\text{use}}(t)}{T_{2,ij}}\right). \quad (5.10)$$

The remaining coherence budget was

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t)\Delta - \widehat{D}_{ij}^{\text{use}}(t). \quad (5.11)$$

The default minimum link fidelity was 0.70. A link was unusable when its predicted consumption fidelity fell below this threshold, its coherence budget was non-positive, or it contained no stored ebit. Free memory was not treated as a service-feasibility constraint because zero free memory means that the memory is full, not that its stored ebits are unavailable.

The normalized safe and warning coherence-budget thresholds were 0.30 and 0.10, respectively.

5.7 Traffic and Request Model

5.7.1 Poisson Arrivals

The number of requests arriving in a slot was sampled from a Poisson distribution. Unless otherwise stated, the mean arrival rate was one request per slot. Source and destination nodes were sampled uniformly without replacement.

5.7.2 Request Classes

Three service classes were used:

Table 5.1: Entanglement-request service classes.

Service class	Probability	Required fidelity	Deadline
Standard	0.50	0.80	5 slots
High fidelity	0.30	0.85	4 slots
Urgent	0.20	0.75	2 slots

Every request demanded one end-to-end entangled pair. The deadline in seconds was obtained by multiplying the slot deadline by Δ .

5.8 Entanglement Generation, Service, and Regeneration

5.8.1 Continuous Ebit Generation

At every slot, each overlay edge with free memory performed at most one generation attempt. The attempt succeeded according to the edge-specific generation probability. A successful attempt added one stored ebit, reduced free memory by one, reset the ebit age, and restored the initial fidelity and coherence budget.

5.8.2 Request Service

A request was accepted only when all of the following conditions held:

1. a usable route existed;
2. every route edge contained the demanded stored ebit;
3. every route edge met the minimum link-fidelity constraint;
4. route-level predicted fidelity met the request requirement;
5. expected completion delay did not exceed the deadline;
6. the probabilistic swapping operation succeeded.

Following successful service, one stored ebit was consumed from every overlay edge on the route and the corresponding memory was released. Therefore, finding a path alone was not counted as request acceptance.

5.8.3 Budget-Limited Regeneration

Warning, critical, and expired links were eligible for regeneration. Candidate priority combined temporal urgency, fidelity deficit, coherence-budget risk, structural importance, blocked-request count, success probability, and expected generation delay. The priority was proportional to expected regeneration utility rather than urgency alone.

Each regeneration attempt succeeded probabilistically. A successful attempt reset age, restored fidelity, created a fresh stored ebit, and recorded the regeneration delay. Attempt, success, and failure counts were collected separately.

5.9 Routing Methods

Three methods were compared in the main baseline experiment:

1. **Shortest Path:** minimum-hop routing over currently usable overlay edges;
2. **Fidelity-Based:** routing that minimizes the negative logarithm of composed Werner-state visibility;
3. **Proposed:** the temporal-state-aware composite-cost method with route-level fidelity and delay feasibility checks and proactive regeneration.

All methods used identical link-availability conditions and matched stochastic inputs. Independent graph copies were used so that one method’s resource consumption did not modify another method’s state.

The proposed composite cost used the following weights:

Table 5.2: Composite routing-cost weights.

Cost component	Weight
Predicted-fidelity risk	0.30
Age risk	0.15
Coherence-budget risk	0.25
Regeneration risk	0.15
Structural risk	0.15

At most the 20 lowest-cost simple candidate routes were checked. This bound prevented uncontrolled path enumeration in denser networks while preserving alternative-route feasibility evaluation.

5.10 Experimental Scenarios

The corrected evaluation included six experiments:

1. **Baseline comparison:** Shortest Path, Fidelity-Based, and Proposed routing over 20 seeds, 20 nodes, and 100 slots.
2. **Multi-seed comparison:** paired Proposed-versus-Static evaluation over 20 seeds.
3. **Ablation study:** Static, Adaptive-Only, and Proposed variants over 20 seeds.

4. **Coherence-time analysis:** controlled $T_2 \in \{0.5, 1, 2, 5, 10, 20\}$ s, with 20 seeds per setting.
5. **Regeneration-budget analysis:** budgets $\{0, 1, 3, 5, 10\}$ attempts per slot, with 20 seeds per setting.
6. **Network scalability:** network sizes $\{6, 20, 40, 60, 100\}$, with 20 seeds per size.

The ablation, coherence-time, regeneration-budget, multi-seed, and scalability experiments used 100 slots per run. The default network size for the multi-seed, ablation, coherence-time, and regeneration-budget experiments was 50 nodes.

5.11 Evaluation Metrics

The principal metrics were:

- request acceptance ratio;
- accepted and failed request counts;
- average accepted-route fidelity;
- average accepted-route completion delay;
- average accepted-route hop count;
- regeneration attempts, successes, and failures;
- physical and overlay edge counts;
- total runtime and runtime per request.

Fidelity, delay, and hop metrics were averaged over accepted requests. When combining multiple seeds, weighted means used accepted-request counts as weights to avoid bias from seeds with zero accepted requests.

5.12 Statistical Methodology

Every reported configuration was evaluated using 20 independent seeds. Mean acceptance ratios were reported with two-sided 95% confidence intervals. Because methods received matched inputs for each seed, paired two-sided t -tests were used for method comparisons. The significance threshold was $p < 0.05$, while results with $p < 0.001$ were treated as highly significant.

5.13 Implementation Validation

Eight automated tests verified the core scientific and resource invariants:

1. physical topology connectivity;
2. preservation of spatial link distance;
3. validity of artificial-link physical paths;
4. agreement with the Werner-state decay equation;
5. decreasing fidelity with additional swapping hops;
6. request-service memory and ebit conservation;
7. generation-capacity conservation;
8. reproducibility of seeded traffic.

All eight tests passed before the full experiments were executed.

5.14 Chapter Summary

This chapter presented the corrected and reproducible simulation methodology. The evaluation separates physical and artificial connectivity, applies consumption-time temporal feasibility, models stochastic generation and regeneration, consumes resources after successful service, and compares methods using matched inputs and paired statistics. The resulting experimental findings are presented in Chapter 6.

Chapter 6

Results and Discussion

6.1 Introduction

This chapter presents the results obtained from the corrected thesis-aligned simulation described in Chapter 5. The reported values were generated after separating the physical network from the artificial entanglement overlay, implementing valid underlying physical paths, applying Werner-state memory decoherence, enforcing consumption-time fidelity and deadline constraints, modeling probabilistic entanglement generation and swapping, consuming stored ebits after successful service, and introducing budget-limited probabilistic regeneration.

Every experimental configuration was evaluated using 20 independent random seeds. Methods compared under the same configuration received matched topologies, traffic requests, generation outcomes, and service random numbers. The reported confidence intervals are two-sided 95% confidence intervals. Paired two-sided t -tests were used where methods were compared seed by seed.

6.2 Implementation Validation

Before running the full experiments, eight automated tests were executed to validate the principal scientific and resource invariants. These tests verified that:

1. every spatial physical topology was connected;
2. topology-derived physical distances were preserved;
3. every artificial e-link was supported by a valid physical path;

4. temporal fidelity followed the Werner-state decay equation;
5. end-to-end fidelity decreased with additional swapping hops;
6. successful service conserved stored ebits and memory capacity;
7. probabilistic ebit generation conserved memory capacity;
8. seeded traffic generation was reproducible.

All eight tests passed. Small integration tests also confirmed that request constraints, route feasibility, regeneration outcomes, resource consumption, and metric collection operated together without runtime errors.

6.3 Baseline Routing Comparison

6.3.1 Request Acceptance

Table 6.1 compares the Shortest Path, Fidelity-Based, and Proposed methods. Across the 20 matched seeds, every method received 1,779 requests in total.

Table 6.1: Corrected baseline comparison over 20 matched seeds.

Method	Acceptance	95% CI	Weighted fidelity	Delay (s)
Shortest Path	0.0404	[0.0303, 0.0505]	0.8501	0.0207
Fidelity-Based	0.0416	[0.0321, 0.0510]	0.8560	0.0229
Proposed	0.1052	[0.0911, 0.1192]	0.8694	0.0191

The Proposed method achieved a mean request-acceptance ratio of 0.1052, compared with 0.0404 for Shortest Path and 0.0416 for Fidelity-Based routing. Its relative acceptance improvement was therefore

$$\frac{0.1052 - 0.0404}{0.0404} \times 100 \approx 160.4\% \quad (6.1)$$

over Shortest Path, and

$$\frac{0.1052 - 0.0416}{0.0416} \times 100 \approx 152.9\% \quad (6.2)$$

over Fidelity-Based routing.

The Proposed method accepted 187 requests, whereas Shortest Path and Fidelity-Based routing accepted 72 and 74 requests, respectively. The absolute acceptance ratios remain moderate because acceptance required simultaneous route availability, sufficient stored ebits, link-level temporal feasibility,

request-level end-to-end fidelity, deadline satisfaction, and successful swapping. Therefore, the results should not be interpreted as ordinary classical path-connectivity ratios.

Figure 6.1 presents the mean acceptance ratios with their 95% confidence intervals.

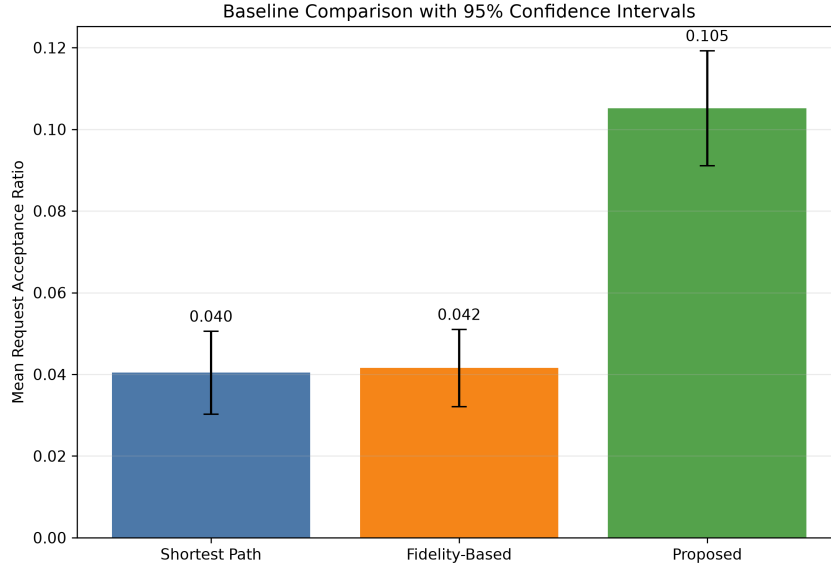


Figure 6.1: Corrected baseline request-acceptance comparison with 95% confidence intervals.

6.3.2 Fidelity, Delay, and Route Length

The Proposed method also achieved the highest request-weighted end-to-end fidelity, 0.8694. The corresponding values were 0.8501 for Shortest Path and 0.8560 for Fidelity-Based routing.

The request-weighted completion delay of the Proposed method was 0.0191 s. This value was lower than the Fidelity-Based delay of 0.0229 s and slightly lower than the Shortest-Path delay of 0.0207 s. The result indicates that the proposed method did not obtain higher acceptance by selecting arbitrarily long routes. Instead, its consumption-time prediction and coherence-budget filtering avoided routes whose expected completion time would make their stored entanglement unusable.

The request-weighted mean hop counts were 1.444, 1.595, and 1.380 for Shortest Path, Fidelity-Based, and Proposed routing, respectively. The Proposed method therefore achieved the smallest accepted-route hop count in the corrected experiment.

6.3.3 Statistical Significance

Paired tests were conducted on the seed-level acceptance ratios. The comparison between Proposed and Shortest Path produced

$$t(19) = 10.622, \quad p = 1.97 \times 10^{-9}. \quad (6.3)$$

The comparison between Proposed and Fidelity-Based routing produced

$$t(19) = 11.138, \quad p = 9.03 \times 10^{-10}. \quad (6.4)$$

Both results satisfy $p < 0.001$. Consequently, the observed acceptance improvements are highly statistically significant under the evaluated configuration.

6.4 Multi-Seed Proposed-versus-Static Comparison

The separate multi-seed experiment compared the complete Proposed method with Static shortest-path routing under matched traffic and stochastic outcomes.

Table 6.2: Multi-seed Proposed-versus-Static comparison.

Method	Mean acceptance	Weighted fidelity	Weighted delay (s)
Static	0.0152	0.8752	0.0154
Proposed	0.0286	0.8800	0.0149

The Proposed method increased acceptance from 0.0152 to 0.0286, an improvement of approximately 88.5%. It also produced slightly higher weighted fidelity and slightly lower weighted delay. These results are consistent with the ablation experiment presented in the following section.

6.5 Ablation Study

The ablation study separated the contribution of adaptive temporal routing and regeneration. The evaluated variants were:

- **Static:** usable-link shortest-path routing without regeneration;
- **Adaptive-Only:** temporal composite routing without regeneration;

- **Proposed:** temporal composite routing with probabilistic regeneration.

Table 6.3: Corrected ablation results over 20 matched seeds.

Variant	Acceptance	95% CI	Weighted fidelity	Delay (s)
Static	0.0152	[0.0100, 0.0204]	0.8752	0.0154
Adaptive-Only	0.0163	[0.0110, 0.0216]	0.8719	0.0168
Proposed	0.0286	[0.0224, 0.0348]	0.8800	0.0149

Adaptive-Only routing produced a small improvement over Static routing. However, the complete Proposed method achieved a substantially larger acceptance ratio. Relative to Static routing, its improvement was approximately 88.5%. Relative to Adaptive-Only routing, its improvement was approximately 75.6%.

Figure 6.2 shows the seed-level means and confidence intervals.

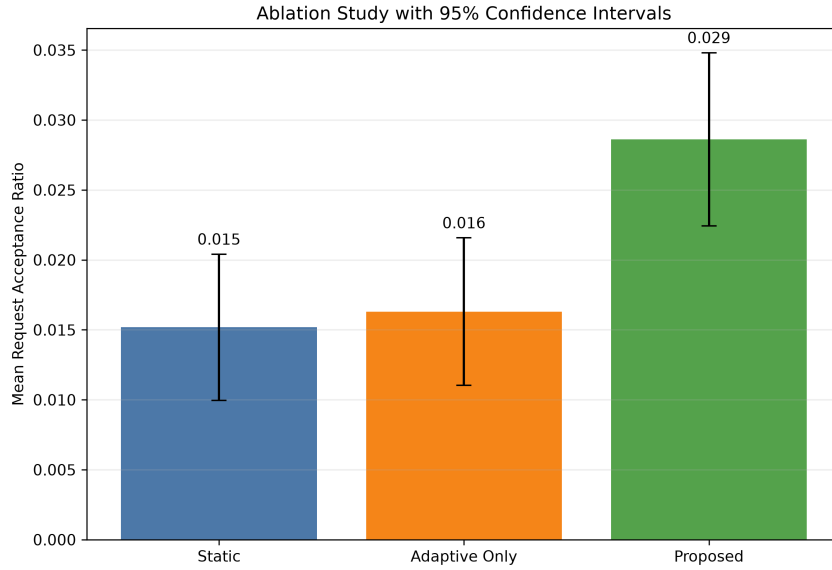


Figure 6.2: Corrected ablation study with 95% confidence intervals.

The paired comparison between Proposed and Static produced

$$t(19) = 4.215, \quad p = 0.000469. \quad (6.5)$$

The paired comparison between Proposed and Adaptive-Only produced

$$t(19) = 4.242, \quad p = 0.000441. \quad (6.6)$$

Both comparisons satisfy $p < 0.001$. The experiment therefore provides statistical evidence that the complete method, particularly its regeneration component, contributed materially to request acceptance.

6.6 Impact of Memory Coherence Time

Memory coherence time directly determines how long stored entanglement remains useful. Table 6.4 summarizes the results.

Table 6.4: Effect of memory coherence time.

T_2 (s)	Acceptance	95% CI	Weighted fidelity	Mean regeneration successes
0.5	0.0101	[0.0056, 0.0146]	0.8346	83.75
1.0	0.0186	[0.0123, 0.0248]	0.8659	80.05
2.0	0.0225	[0.0153, 0.0298]	0.8797	75.20
5.0	0.0472	[0.0350, 0.0594]	0.8717	65.65
10.0	0.0731	[0.0585, 0.0876]	0.8636	50.80
20.0	0.0922	[0.0786, 0.1058]	0.8670	19.80

Increasing T_2 from 0.5 s to 20 s increased the mean acceptance ratio from 0.0101 to 0.0922. This corresponds to an increase of more than nine times.

The number of successful regenerations decreased as coherence time increased. Short-coherence memories caused links to become stale frequently, creating strong regeneration demand. Longer coherence times preserved existing ebits, reduced temporal disconnections, and decreased the need for regeneration.

Figure 6.3 shows the monotonic acceptance trend.

The weighted fidelity remained between approximately 0.835 and 0.880. This confirms that the increase in acceptance did not result from relaxing the request-fidelity requirements.

6.7 Impact of Regeneration Budget

The regeneration-budget experiment evaluated maximum budgets of 0, 1, 3, 5, and 10 attempts per slot.

Acceptance increased strongly as the budget increased from zero to five. However, the improvement from budget five to budget ten was only

$$0.0815 - 0.0776 = 0.0039, \quad (6.7)$$

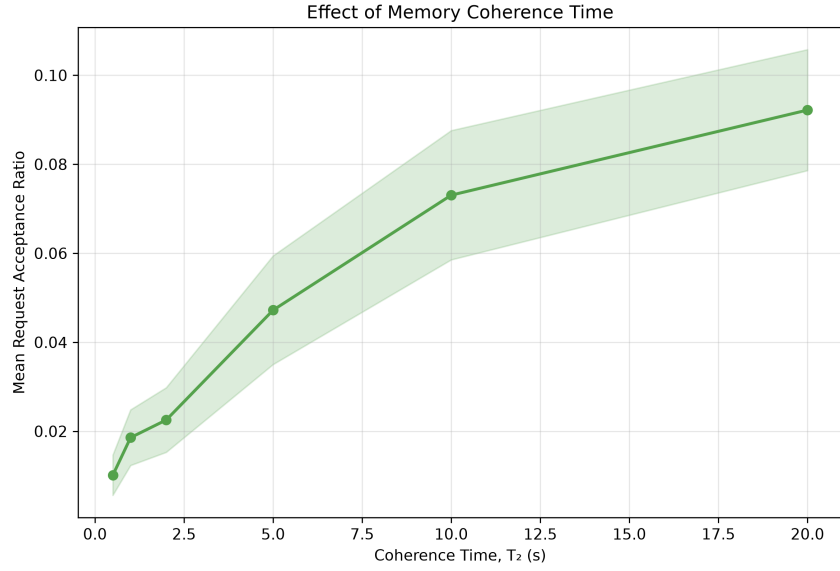


Figure 6.3: Effect of memory coherence time on request acceptance.

Table 6.5: Regeneration-budget sensitivity.

Case	Budget	Acceptance	95% CI	Weighted fidelity	Mean successes
Disabled	0	0.0163	[0.0110, 0.0216]	0.8719	0.00
Low	1	0.0281	[0.0221, 0.0342]	0.8801	80.40
Medium	3	0.0534	[0.0424, 0.0644]	0.8729	227.15
High	5	0.0776	[0.0638, 0.0914]	0.8673	361.80
Maximum	10	0.0815	[0.0713, 0.0917]	0.8750	430.45

or 0.39 percentage point. Therefore, budget five represents a practical efficiency point in the evaluated configuration. Doubling the maximum budget from five to ten generated additional successful regenerations but produced only a small acceptance improvement.

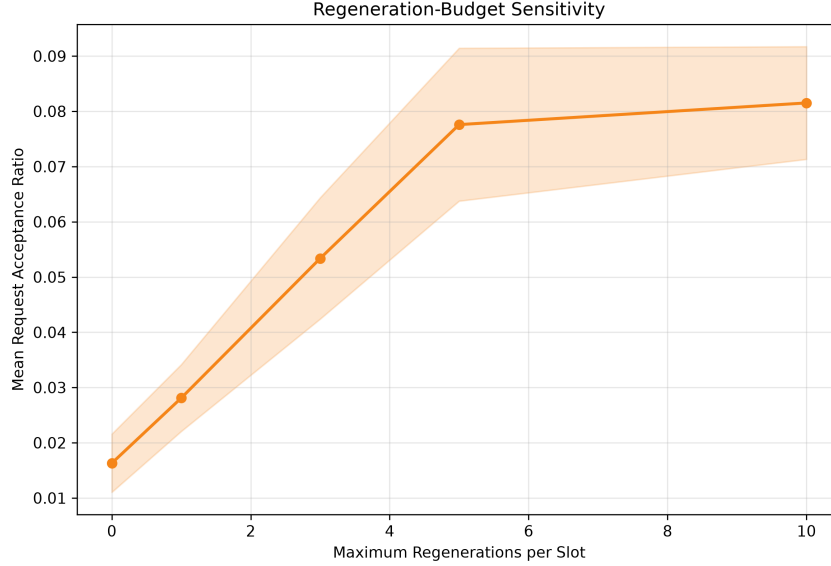


Figure 6.4: Request acceptance under different regeneration budgets.

The weighted end-to-end fidelity remained close to 0.87 for all settings. Regeneration therefore increased temporal availability without producing a major fidelity penalty among accepted requests.

6.8 Network-Size Scalability

The scalability experiment evaluated networks containing 6, 20, 40, 60, and 100 nodes. Each size was evaluated with 20 independent seeds.

Table 6.6: Corrected network-size scalability results.

Nodes	Physical edges	Acceptance	Weighted fidelity	Runtime (s)	ms/request
6	11.4	0.7566	0.8743	0.304	3.423
20	38.6	0.2237	0.8686	1.124	12.644
40	77.1	0.1073	0.8733	2.081	23.401
60	115.0	0.0574	0.8619	2.172	24.406
100	191.3	0.0219	0.8723	2.727	30.674

Request acceptance decreased as network size increased. Larger networks generally required longer physical and overlay routes, creating more opportunities for generation failure, fidelity degradation, temporal expiration, resource depletion, and swapping failure.

Despite the reduction in acceptance, the request-weighted fidelity remained between 0.862 and 0.874. This result demonstrates that the feasibility filter continued to preserve the fidelity of accepted requests even in larger networks.

Mean execution time increased from 0.304 s for six nodes to 2.727 s for 100 nodes. Mean runtime per request increased from 3.423 ms to 30.674 ms.

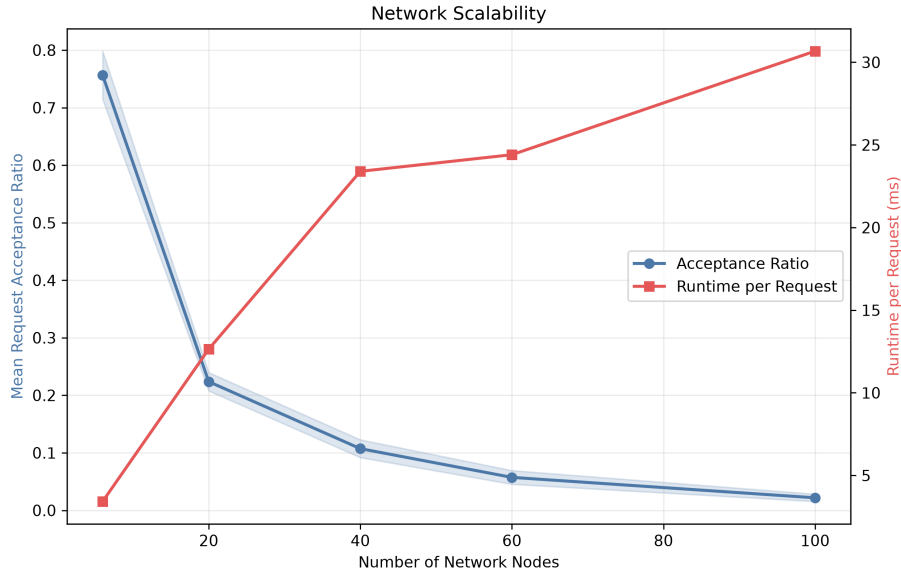


Figure 6.5: Acceptance and runtime scalability with increasing network size.

The route search examined at most the 20 lowest-cost simple candidate routes. This bound prevented uncontrolled path enumeration in denser networks. The runtime results indicate that the implementation remained computationally manageable at 100 nodes, although further optimization would be required for substantially larger or higher-load networks.

6.9 Discussion of the Main Findings

The corrected experiments support five principal findings.

First, the Proposed method achieved substantially higher request acceptance than both baseline methods while maintaining the highest weighted

fidelity. The paired tests indicate that these improvements were highly significant.

Second, the benefit did not arise solely from fidelity-based path selection. Fidelity-Based routing only slightly improved acceptance over Shortest Path. The complete method benefited from combining predicted fidelity, entanglement age, coherence budget, structural risk, route-level feasibility, and regeneration.

Third, regeneration was essential under short coherence times. The ablation study showed that Adaptive-Only routing provided only a small improvement, whereas the complete method produced a much larger and statistically significant gain.

Fourth, coherence time and regeneration budget interacted strongly. Short coherence times created high regeneration demand, while longer coherence times reduced this demand. Increasing regeneration budget improved acceptance, but the gain saturated beyond five attempts per slot in the evaluated setting.

Fifth, the feasibility mechanism preserved accepted-route fidelity across network sizes. Acceptance decreased with network size, but weighted fidelity remained approximately stable. The method therefore rejected temporally unsafe requests rather than accepting them with poor fidelity.

6.10 Limitations

The corrected evaluation has several limitations.

- The simulator is a discrete-time abstract network simulator rather than a hardware-level quantum-network emulator.
- Physical-link parameters are based on controlled analytical ranges rather than measurements from a specific deployed platform.
- Purification delay was configured as zero and purification operations were not modeled explicitly.
- Requests demanded one end-to-end entangled pair; multi-pair requests require further evaluation.
- Node placement was uniform in a square region and did not include geographic obstacles or heterogeneous repeater hardware.
- Traffic was spatially uniform and did not include hotspot or application-correlated demand.

- Only the experiments reported in this chapter were executed under the corrected lifecycle. Earlier prototype-only scenarios were excluded from the final claims.
- Although 20 seeds support paired statistical analysis, more seeds and alternative parameter regimes would improve external validity.

These limitations restrict the scope of the conclusions. The results support the effectiveness of the proposed method under the stated simulation model; they do not establish universal superiority for every quantum-network architecture.

6.11 Chapter Summary

This chapter presented the corrected experimental results. The Proposed method achieved a mean baseline acceptance ratio of 0.1052, compared with 0.0404 for Shortest Path and 0.0416 for Fidelity-Based routing. Its weighted fidelity was 0.8694, and both paired baseline comparisons produced $p < 0.001$.

The ablation study demonstrated a statistically significant contribution from the complete adaptive-routing and regeneration mechanism. Coherence-time and regeneration-budget experiments showed the expected temporal-availability trends, including diminishing returns beyond a regeneration budget of five. The scalability experiment showed increasing computation time and decreasing acceptance with network size, while accepted-route fidelity remained stable.

Overall, the corrected evidence supports the conclusion that temporal-state prediction, coherence-budget-aware feasibility, and targeted regeneration can improve fidelity-preserving entanglement distribution under finite memory coherence and stochastic link generation.

Chapter 7

Conclusion and Future Work

7.1 Summary of Research

This thesis investigated the problem of reliable entanglement distribution in dynamic quantum networks where quantum links experience temporal degradation, fidelity reduction, and limited resource availability.

Unlike traditional static routing approaches that treat quantum links as unchanging resources, this work proposed a Temporal-State-Aware Routing framework that considers the evolution of entanglement states over time. The proposed framework integrates temporal state evaluation, adaptive route selection, coherence-budget management, and regeneration mechanisms to maintain reliable communication paths.

A simulation framework was developed to evaluate the proposed method under different network conditions. The evaluation considered baseline comparison, multi-seed experiments, scalability analysis, coherence-time variation, traffic-load variation, and additional sensitivity studies.

The experimental results demonstrated that considering temporal link conditions provides more reliable routing decisions compared with static routing approaches. The proposed framework improves the capability of quantum networks to maintain usable entanglement paths under dynamic operating conditions.

7.2 Main Contributions

The main contributions of this thesis are summarized as follows:

- A temporal-state-aware quantum routing framework was proposed for fidelity-preserving entanglement distribution in dynamic quantum net-

works. The framework considers the time-dependent evolution of entanglement resources instead of relying only on static topology information.

- A routing strategy was developed that incorporates temporal link conditions, including link aging, fidelity degradation, and remaining coherence budget, into route selection decisions.
- A regeneration-assisted resource management mechanism was introduced to recover degraded entanglement resources and improve long-term route availability.
- A simulator was implemented to evaluate routing under different network sizes, coherence times, traffic conditions, and random seeds.
- A comprehensive evaluation framework was established using metrics including request success ratio, fidelity performance, temporal disconnection ratio, completion delay, regeneration overhead, and quantum-memory utilization.

7.3 Limitations

Although the proposed framework improves routing reliability in simulation, several limitations remain.

First, the current evaluation is based on simulated quantum network environments. Real quantum communication systems may introduce additional hardware-dependent imperfections, noise sources, and operational constraints.

Second, the current model uses a simplified temporal fidelity degradation process. More detailed physical noise models, including hardware-specific errors and complex quantum channel behaviors, can be incorporated in future work.

Third, the regeneration mechanism assumes that recovery resources are available when required. Practical quantum networks may face limitations in entanglement generation capability, memory availability, and operational cost.

These limitations indicate several possible directions for extending the proposed framework toward more realistic quantum network implementations.

7.4 Future Research Directions

Several research directions can be explored based on this work.

First, future studies can incorporate more realistic quantum noise models and hardware parameters to improve the physical accuracy of the simulation framework.

Second, machine learning and reinforcement learning approaches can be investigated for adaptive routing decisions in larger and more complex quantum networks.

Third, future work can consider distributed routing architectures where multiple quantum network nodes collaboratively exchange temporal state information.

Finally, the proposed framework can be validated using quantum network simulation platforms and experimental quantum communication testbeds to evaluate its practical applicability.

7.5 Conclusion

This thesis developed a temporal-state-aware framework for reliable entanglement distribution in dynamic quantum networks. It combines temporal resource tracking, adaptive routing, and regeneration as an alternative to static routing.

The simulation results confirm the importance of temporal awareness and resource recovery mechanisms for maintaining stable quantum communication paths. The findings demonstrate that future quantum networks require routing strategies capable of adapting to continuously changing entanglement states.

Overall, this work contributes toward the development of reliable and resource-efficient routing techniques for future large-scale quantum network infrastructures.

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